K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDoubleDoor / demo_99 / s288 / L

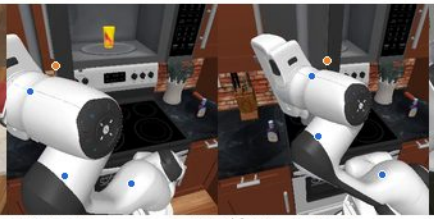


Query now Actual future +32 steps

Object: actual 11.0 cm; forecast 6.6 cm

Min point movement: 24.0 / 14.0 px (actual / forecast)

CloseSingleDoor / demo_651 / s160 / L



Query now Actual future +32 steps

Object: actual 12.8 cm; forecast 2.6 cm

Min point movement: 24.6 / 8.0 px (actual / forecast)

CloseSingleDoor / demo_972 / s176 / L



Query now Actual future +32 steps

Object: actual 12.9 cm; forecast 11.8 cm

Min point movement: 25.5 / 25.8 px (actual / forecast)

CloseDoubleDoor / demo_1717 / s304 / L

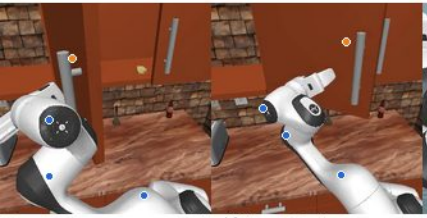


Query now Actual future +32 steps

Object: actual 21.9 cm; forecast 19.8 cm

Min point movement: 14.5 / 4.8 px (actual / forecast)

CloseDoubleDoor / demo_1930 / s336 / L



Query now Actual future +32 steps

Object: actual 24.6 cm; forecast 29.7 cm

Min point movement: 7.4 / 29.6 px (actual / forecast)

CloseSingleDoor / demo_535 / s160 / L



Query now Actual future +32 steps

Object: actual 9.9 cm; forecast 5.9 cm

Min point movement: 20.2 / 11.3 px (actual / forecast)

CloseSingleDoor / demo_535 / s176 / L



Query now Actual future +19 steps

Object: actual 24.0 cm; forecast 25.9 cm

Min point movement: 30.4 / 40.6 px (actual / forecast)

CloseSingleDoor / demo_1645 / s176 / L

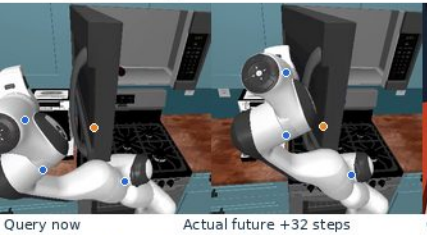


Query now Actual future +32 steps

Object: actual 4.1 cm; forecast 9.5 cm

Min point movement: 5.6 / 8.0 px (actual / forecast)

CloseSingleDoor / demo_1829 / s192 / L



Query now Actual future +32 steps

Object: actual 5.3 cm; forecast 12.9 cm

Min point movement: 9.9 / 25.6 px (actual / forecast)

CloseSingleDoor / demo_695 / s192 / L



Query now Actual future +32 steps

Object: actual 25.6 cm; forecast 26.6 cm

Min point movement: 24.9 / 27.4 px (actual / forecast)

CloseDoubleDoor / demo_1906 / s288 / L



Query now Actual future +32 steps

Object: actual 3.7 cm; forecast 4.3 cm

Min point movement: 8.1 / 7.8 px (actual / forecast)

CloseSingleDoor / demo_288 / s176 / L

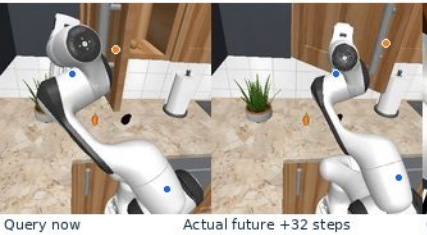


Query now Actual future +32 steps

Object: actual 7.6 cm; forecast 3.5 cm

Min point movement: 14.3 / 7.0 px (actual / forecast)

CloseSingleDoor / demo_288 / s192 / L



Query now Actual future +32 steps

Object: actual 24.4 cm; forecast 18.5 cm

Min point movement: 5.0 / 7.6 px (actual / forecast)

CloseSingleDoor / demo_610 / s176 / L



Query now Actual future +32 steps

Object: actual 34.7 cm; forecast 33.1 cm

Min point movement: 21.3 / 15.7 px (actual / forecast)

CloseSingleDoor / demo_2151 / s160 / L



Query now Actual future +32 steps

Object: actual 14.1 cm; forecast 9.2 cm

Min point movement: 3.6 / 6.4 px (actual / forecast)

CloseSingleDoor / demo_2151 / s176 / L



Query now Actual future +32 steps

Object: actual 28.0 cm; forecast 29.6 cm

Min point movement: 44.6 / 45.8 px (actual / forecast)

CloseSingleDoor / demo_258 / s176 / L



Query now Actual future +32 steps

Object: actual 10.4 cm; forecast 5.2 cm

Min point movement: 13.9 / 8.8 px (actual / forecast)

CloseSingleDoor / demo_258 / s192 / L

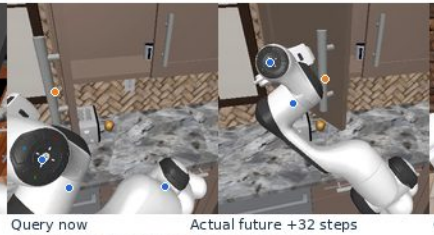


Query now Actual future +32 steps

Object: actual 23.8 cm; forecast 20.4 cm

Min point movement: 23.8 / 20.2 px (actual / forecast)

CloseDoubleDoor / demo_1717 / s288 / L

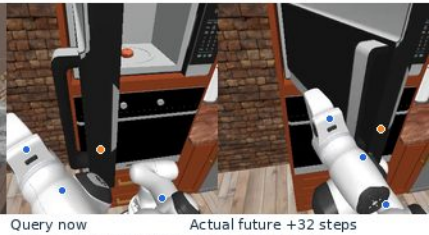


Query now Actual future +32 steps

Object: actual 17.6 cm; forecast 6.7 cm

Min point movement: 15.7 / 14.4 px (actual / forecast)

CloseSingleDoor / demo_609 / s176 / L



Query now Actual future +32 steps

Object: actual 17.5 cm; forecast 10.0 cm

Min point movement: 9.1 / 3.9 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseSingleDoor / demo_96 / s144 / R

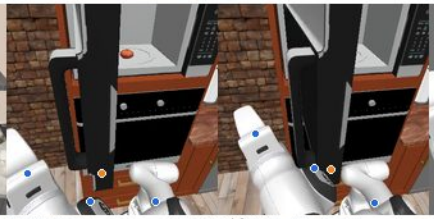


Query now Actual future +32 steps

Object: actual 4.7 cm; forecast 3.2 cm

Min point movement: 9.5 / 6.0 px (actual / forecast)

CloseSingleDoor / demo_609 / s160 / L

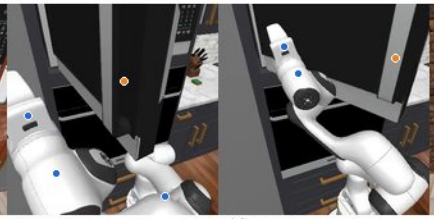


Query now Actual future +32 steps

Object: actual 4.2 cm; forecast 4.4 cm

Min point movement: 5.3 / 9.1 px (actual / forecast)

CloseSingleDoor / demo_1932 / s160 / L



Query now Actual future +32 steps

Object: actual 18.8 cm; forecast 3.9 cm

Min point movement: 13.3 / 8.1 px (actual / forecast)

CloseSingleDoor / demo_609 / s192 / L



Query now Actual future +31 steps

Object: actual 29.2 cm; forecast 28.3 cm

Min point movement: 3.6 / 4.1 px (actual / forecast)

CloseDoubleDoor / demo_2827 / s480 / L



Query now Actual future +29 steps

Object: actual 22.7 cm; forecast 11.2 cm

Min point movement: 10.7 / 9.1 px (actual / forecast)

CloseSingleDoor / demo_96 / s160 / R



Query now Actual future +32 steps

Object: actual 24.3 cm; forecast 21.2 cm

Min point movement: 30.6 / 23.9 px (actual / forecast)

CloseSingleDoor / demo_2758 / s176 / L

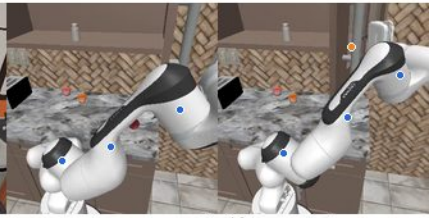


Query now Actual future +32 steps

Object: actual 14.0 cm; forecast 8.8 cm

Min point movement: 16.4 / 13.4 px (actual / forecast)

CloseDoubleDoor / demo_2238 / s144 / R



Query now Actual future +32 steps

Object: actual 10.5 cm; forecast 15.4 cm

Min point movement: 7.8 / 18.1 px (actual / forecast)

CloseSingleDoor / demo_1682 / s208 / L



Query now Actual future +24 steps

Object: actual 22.5 cm; forecast 20.3 cm

Min point movement: 32.9 / 17.4 px (actual / forecast)

CloseDoubleDoor / demo_126 / s320 / L



Query now Actual future +32 steps

Object: actual 14.4 cm; forecast 9.7 cm

Min point movement: 16.9 / 7.0 px (actual / forecast)

CloseSingleDoor / demo_1892 / s176 / L

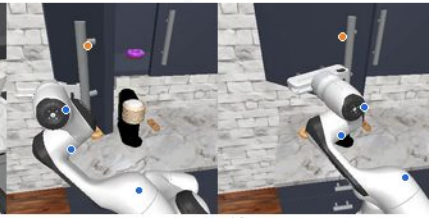


Query now Actual future +32 steps

Object: actual 15.1 cm; forecast 20.1 cm

Min point movement: 21.2 / 25.1 px (actual / forecast)

CloseDoubleDoor / demo_987 / s384 / L

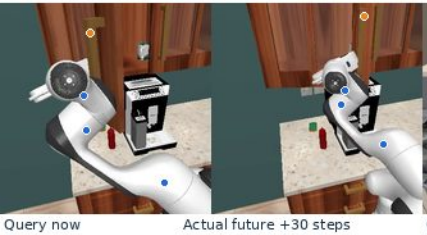


Query now Actual future +32 steps

Object: actual 19.1 cm; forecast 30.3 cm

Min point movement: 27.4 / 30.7 px (actual / forecast)

CloseDoubleDoor / demo_341 / s496 / L

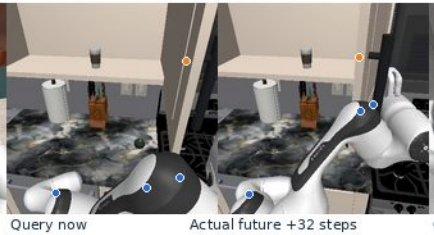


Query now Actual future +30 steps

Object: actual 39.1 cm; forecast 31.6 cm

Min point movement: 26.8 / 20.3 px (actual / forecast)

CloseDoubleDoor / demo_1175 / s128 / R



Query now Actual future +32 steps

Object: actual 8.7 cm; forecast 11.0 cm

Min point movement: 10.8 / 10.1 px (actual / forecast)

CloseSingleDoor / demo_24 / s208 / L

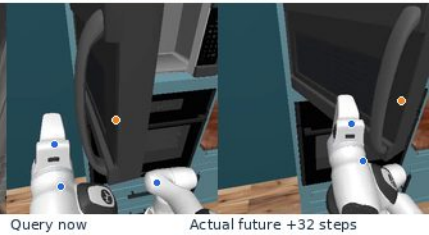


Query now Actual future +32 steps

Object: actual 27.3 cm; forecast 32.8 cm

Min point movement: 4.6 / 6.4 px (actual / forecast)

CloseSingleDoor / demo_1838 / s192 / L



Query now Actual future +32 steps

Object: actual 18.2 cm; forecast 6.1 cm

Min point movement: 3.8 / 3.9 px (actual / forecast)

CloseSingleDoor / demo_611 / s144 / R

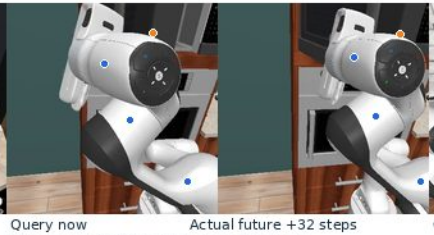


Query now Actual future +32 steps

Object: actual 23.4 cm; forecast 19.6 cm

Min point movement: 37.1 / 38.3 px (actual / forecast)

CloseSingleDoor / demo_2592 / s176 / L

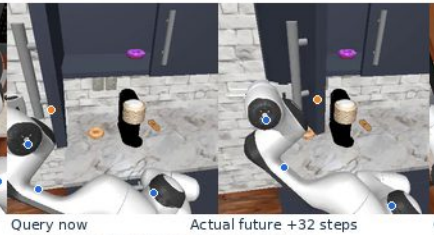


Query now Actual future +32 steps

Object: actual 9.1 cm; forecast 8.5 cm

Min point movement: 12.7 / 14.7 px (actual / forecast)

CloseDoubleDoor / demo_987 / s352 / L

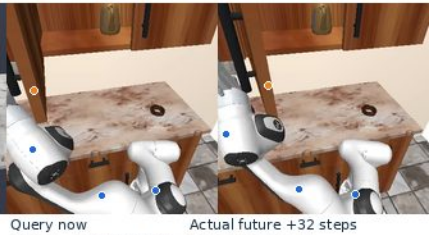


Query now Actual future +32 steps

Object: actual 14.8 cm; forecast 3.9 cm

Min point movement: 17.7 / 9.8 px (actual / forecast)

CloseDoubleDoor / demo_2731 / s320 / L



Query now Actual future +32 steps

Object: actual 5.7 cm; forecast 19.7 cm

Min point movement: 7.6 / 19.9 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseSingleDoor / demo_1172 / s176 / L

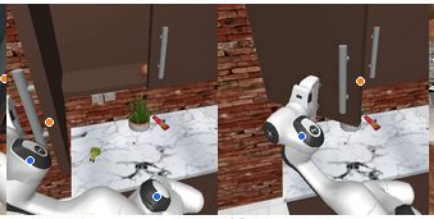


Query now Actual future +29 steps

Object: actual 27.1 cm; forecast 22.2 cm

Min point movement: 20.7 / 17.4 px (actual / forecast)

CloseDoubleDoor / demo_842 / s352 / L



Query now Actual future +32 steps

Object: actual 36.2 cm; forecast 40.5 cm

Min point movement: 10.9 / 17.0 px (actual / forecast)

CloseDoubleDoor / demo_443 / s144 / R

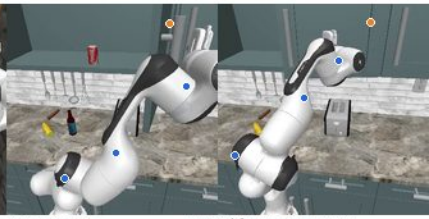


Query now Actual future +31 steps

Object: actual 5.6 cm; forecast 6.2 cm

Min point movement: 6.4 / 15.1 px (actual / forecast)

CloseDoubleDoor / demo_254 / s144 / R

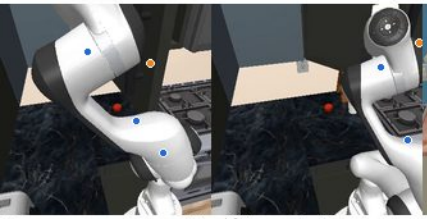


Query now Actual future +31 steps

Object: actual 8.0 cm; forecast 6.9 cm

Min point movement: 6.7 / 6.4 px (actual / forecast)

CloseSingleDoor / demo_466 / s224 / L

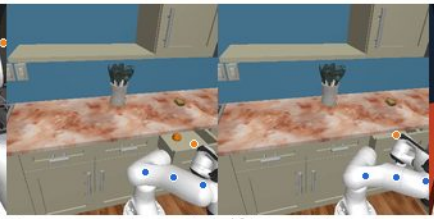


Query now Actual future +21 steps

Object: actual 35.7 cm; forecast 31.8 cm

Min point movement: 29.0 / 29.3 px (actual / forecast)

CloseDrawer / demo_183 / s160 / L



Query now Actual future +32 steps

Object: actual 13.1 cm; forecast 14.1 cm

Min point movement: 5.9 / 7.2 px (actual / forecast)

CloseSingleDoor / demo_695 / s176 / L

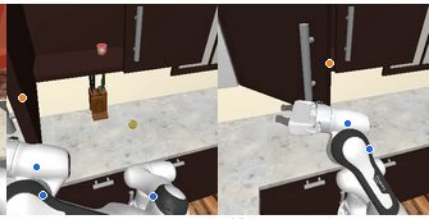


Query now Actual future +32 steps

Object: actual 7.6 cm; forecast 6.2 cm

Min point movement: 15.5 / 11.9 px (actual / forecast)

CloseDoubleDoor / demo_1987 / s352 / L



Query now Actual future +32 steps

Object: actual 30.7 cm; forecast 30.5 cm

Min point movement: 12.5 / 14.1 px (actual / forecast)

CloseSingleDoor / demo_1859 / s192 / L



Query now Actual future +32 steps

Object: actual 27.9 cm; forecast 24.8 cm

Min point movement: 24.5 / 21.6 px (actual / forecast)

CloseDrawer / demo_2877 / s144 / L



Query now Actual future +32 steps

Object: actual 13.8 cm; forecast 13.2 cm

Min point movement: 6.5 / 9.4 px (actual / forecast)

CloseDrawer / demo_2877 / s192 / L

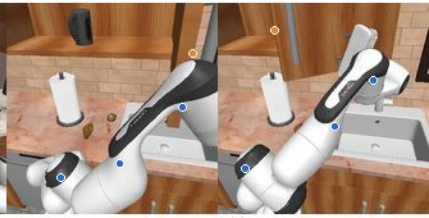


Query now Actual future +28 steps

Object: actual 11.6 cm; forecast 9.4 cm

Min point movement: 3.4 / 4.9 px (actual / forecast)

CloseDoubleDoor / demo_1499 / s160 / R

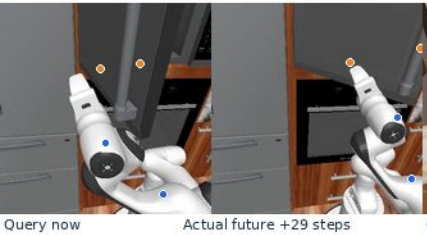


Query now Actual future +32 steps

Object: actual 43.2 cm; forecast 49.7 cm

Min point movement: 19.5 / 14.9 px (actual / forecast)

CloseSingleDoor / demo_258 / s208 / L

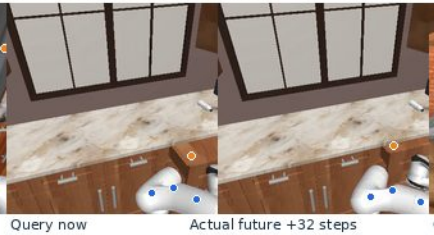


Query now Actual future +29 steps

Object: actual 33.4 cm; forecast 35.7 cm

Min point movement: 25.4 / 23.6 px (actual / forecast)

CloseDrawer / demo_2877 / s160 / L

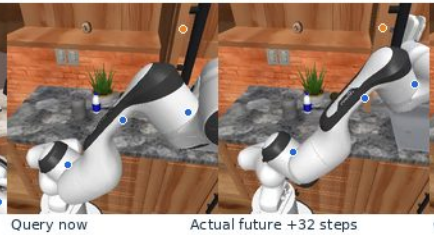


Query now Actual future +32 steps

Object: actual 12.2 cm; forecast 14.1 cm

Min point movement: 5.2 / 9.0 px (actual / forecast)

CloseDoubleDoor / demo_1569 / s144 / R

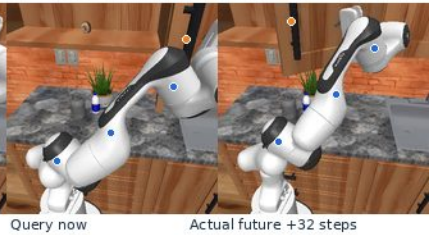


Query now Actual future +32 steps

Object: actual 2.5 cm; forecast 16.4 cm

Min point movement: 7.0 / 17.2 px (actual / forecast)

CloseDoubleDoor / demo_1569 / s160 / R

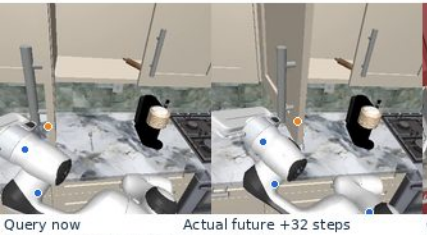


Query now Actual future +32 steps

Object: actual 35.3 cm; forecast 26.1 cm

Min point movement: 13.2 / 8.4 px (actual / forecast)

CloseDoubleDoor / demo_2121 / s320 / L

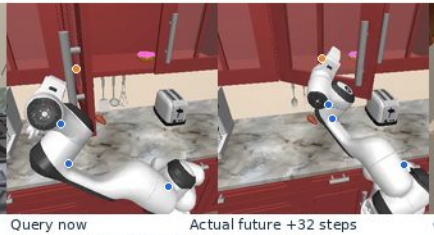


Query now Actual future +32 steps

Object: actual 11.6 cm; forecast 8.8 cm

Min point movement: 12.3 / 11.8 px (actual / forecast)

CloseDoubleDoor / demo_2123 / s304 / L

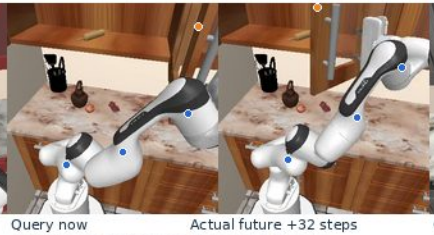


Query now Actual future +32 steps

Object: actual 18.6 cm; forecast 17.6 cm

Min point movement: 19.6 / 6.0 px (actual / forecast)

CloseDoubleDoor / demo_1087 / s160 / R



Query now Actual future +32 steps

Object: actual 27.6 cm; forecast 37.7 cm

Min point movement: 9.9 / 10.4 px (actual / forecast)

CloseSingleDoor / demo_96 / s176 / R



Query now Actual future +29 steps

Object: actual 47.0 cm; forecast 39.9 cm

Min point movement: 19.1 / 15.6 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

OpenDoubleDoor / demo_1262 / s368 / R

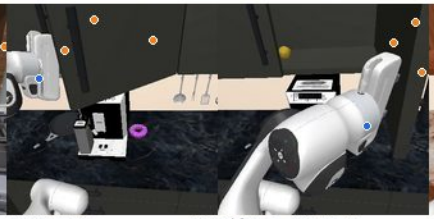


Query now Actual future +32 steps

Object: actual 42.6 cm; forecast 36.9 cm

Min point movement: 19.3 / 20.4 px (actual / forecast)

OpenDoubleDoor / demo_1176 / s416 / R



Query now Actual future +31 steps

Object: actual 40.6 cm; forecast 37.4 cm

Min point movement: 18.4 / 17.0 px (actual / forecast)

CloseDoubleDoor / demo_2731 / s160 / R

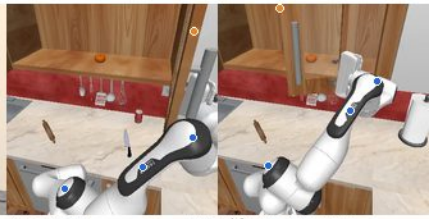


Query now Actual future +32 steps

Object: actual 38.6 cm; forecast 51.6 cm

Min point movement: 6.5 / 6.9 px (actual / forecast)

CloseDoubleDoor / demo_99 / s160 / R



Query now Actual future +21 steps

Object: actual 66.2 cm; forecast 59.2 cm

Min point movement: 26.2 / 22.4 px (actual / forecast)

CloseDoubleDoor / demo_1906 / s160 / R

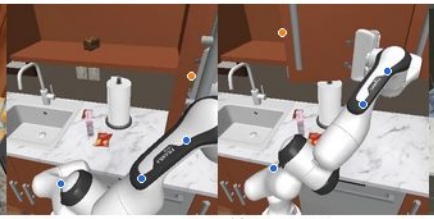


Query now Actual future +24 steps

Object: actual 60.8 cm; forecast 59.6 cm

Min point movement: 11.2 / 17.9 px (actual / forecast)

CloseDoubleDoor / demo_1020 / s176 / R

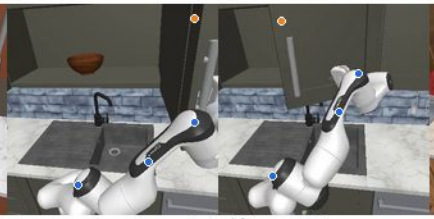


Query now Actual future +29 steps

Object: actual 54.2 cm; forecast 59.1 cm

Min point movement: 10.4 / 11.3 px (actual / forecast)

CloseDoubleDoor / demo_2216 / s176 / R

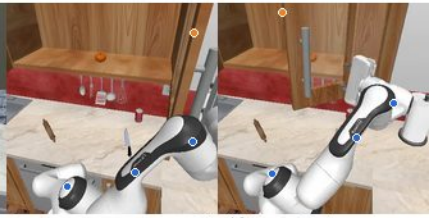


Query now Actual future +32 steps

Object: actual 47.0 cm; forecast 44.9 cm

Min point movement: 11.2 / 11.8 px (actual / forecast)

CloseDoubleDoor / demo_99 / s144 / R



Query now Actual future +32 steps

Object: actual 38.4 cm; forecast 37.1 cm

Min point movement: 8.6 / 8.3 px (actual / forecast)

CloseSingleDoor / demo_821 / s208 / L

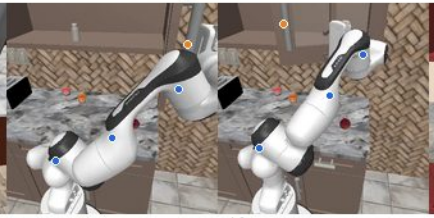


Query now Actual future +32 steps

Object: actual 39.9 cm; forecast 40.4 cm

Min point movement: 6.8 / 13.5 px (actual / forecast)

CloseDoubleDoor / demo_2238 / s160 / R

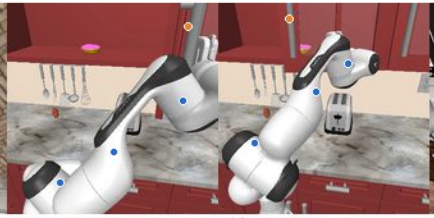


Query now Actual future +32 steps

Object: actual 46.4 cm; forecast 47.1 cm

Min point movement: 9.6 / 14.7 px (actual / forecast)

CloseDoubleDoor / demo_2123 / s144 / R



Query now Actual future +27 steps

Object: actual 55.8 cm; forecast 53.0 cm

Min point movement: 37.0 / 30.5 px (actual / forecast)

CloseSingleDoor / demo_1060 / s240 / R

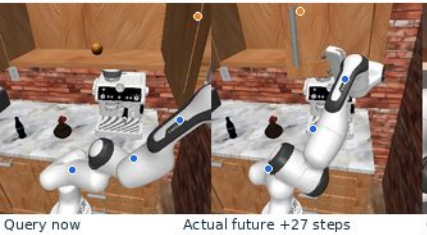


Query now Actual future +32 steps

Object: actual 50.3 cm; forecast 59.5 cm

Min point movement: 6.3 / 5.6 px (actual / forecast)

CloseDoubleDoor / demo_2128 / s192 / R

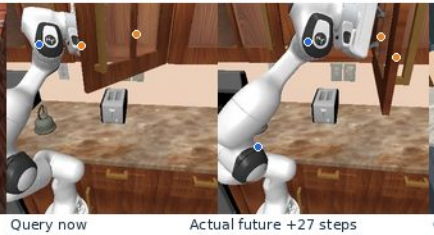


Query now Actual future +27 steps

Object: actual 55.1 cm; forecast 51.7 cm

Min point movement: 11.3 / 14.6 px (actual / forecast)

OpenDoubleDoor / demo_503 / s368 / R

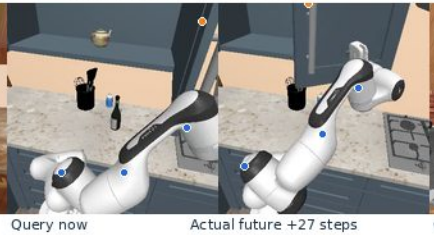


Query now Actual future +27 steps

Object: actual 32.5 cm; forecast 20.1 cm

Min point movement: 15.4 / 16.6 px (actual / forecast)

CloseDoubleDoor / demo_2470 / s160 / R



Query now Actual future +27 steps

Object: actual 46.4 cm; forecast 43.7 cm

Min point movement: 21.8 / 26.4 px (actual / forecast)

OpenDoubleDoor / demo_176 / s752 / L

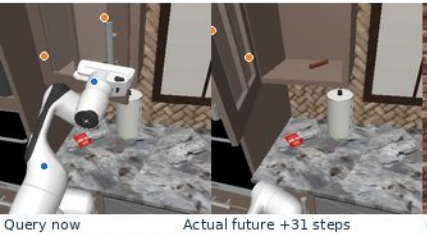


Query now Actual future +32 steps

Object: actual 28.6 cm; forecast 21.9 cm

Min point movement: 7.4 / 4.5 px (actual / forecast)

OpenSingleDoor / demo_959 / s160 / R



Query now Actual future +31 steps

Object: actual 34.7 cm; forecast 31.3 cm

Min point movement: 4.4 / 3.6 px (actual / forecast)

OpenSingleDoor / demo_1154 / s272 / L

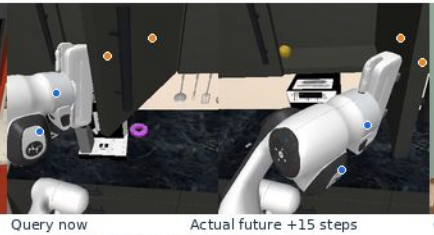


Query now Actual future +19 steps

Object: actual 19.8 cm; forecast 22.6 cm

Min point movement: 7.3 / 9.2 px (actual / forecast)

OpenDoubleDoor / demo_1176 / s432 / R

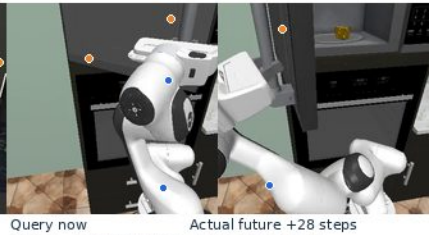


Query now Actual future +15 steps

Object: actual 28.8 cm; forecast 31.8 cm

Min point movement: 31.9 / 36.6 px (actual / forecast)

OpenSingleDoor / demo_49 / s208 / L



Query now Actual future +28 steps

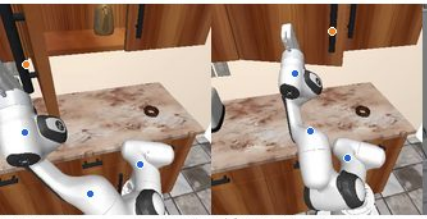
Object: actual 28.5 cm; forecast 30.8 cm

Min point movement: 12.7 / 15.4 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDoubleDoor / demo_2731 / s352 / L



Query now Actual future +32 steps

Object: actual 41.8 cm; forecast 55.8 cm

Min point movement: 4.4 / 9.5 px (actual / forecast)

OpenSingleDoor / demo_156 / s272 / L

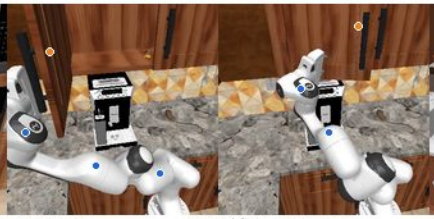


Query now Actual future +27 steps

Object: actual 28.4 cm; forecast 29.8 cm

Min point movement: 9.1 / 10.5 px (actual / forecast)

CloseDoubleDoor / demo_1906 / s352 / L

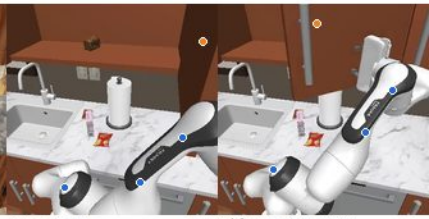


Query now Actual future +32 steps

Object: actual 49.6 cm; forecast 47.1 cm

Min point movement: 12.3 / 14.4 px (actual / forecast)

CloseDoubleDoor / demo_1020 / s160 / R



Query now Actual future +32 steps

Object: actual 32.2 cm; forecast 36.7 cm

Min point movement: 10.7 / 9.1 px (actual / forecast)

CloseDoubleDoor / demo_2872 / s304 / L



Query now Actual future +32 steps

Object: actual 25.9 cm; forecast 4.6 cm

Min point movement: 3.8 / 14.0 px (actual / forecast)

CloseSingleDoor / demo_1682 / s224 / L



Query now Actual future +32 steps

Object: actual 30.5 cm; forecast 35.8 cm

Min point movement: 31.5 / 31.9 px (actual / forecast)

OpenSingleDoor / demo_32 / s224 / L



Query now Actual future +19 steps

Object: actual 25.3 cm; forecast 27.9 cm

Min point movement: 10.3 / 12.2 px (actual / forecast)

CloseDoubleDoor / demo_2316 / s352 / L

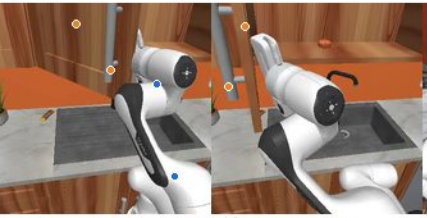


Query now Actual future +32 steps

Object: actual 38.3 cm; forecast 33.3 cm

Min point movement: 18.4 / 3.4 px (actual / forecast)

OpenDoubleDoor / demo_562 / s976 / L

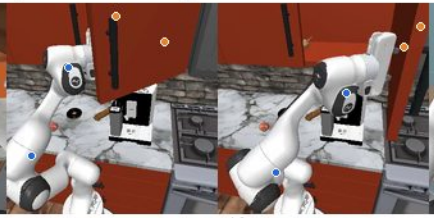


Query now Actual future +24 steps

Object: actual 25.3 cm; forecast 32.9 cm

Min point movement: 22.5 / 22.4 px (actual / forecast)

OpenDoubleDoor / demo_737 / s352 / R



Query now Actual future +23 steps

Object: actual 37.7 cm; forecast 32.1 cm

Min point movement: 20.4 / 17.1 px (actual / forecast)

OpenDoubleDoor / demo_1189 / s384 / R



Query now Actual future +23 steps

Object: actual 24.3 cm; forecast 28.8 cm

Min point movement: 13.0 / 14.1 px (actual / forecast)

CloseSingleDoor / demo_651 / s176 / L

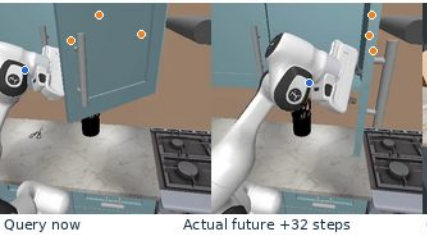


Query now Actual future +32 steps

Object: actual 22.2 cm; forecast 14.8 cm

Min point movement: 23.1 / 10.5 px (actual / forecast)

OpenDoubleDoor / demo_1189 / s368 / R

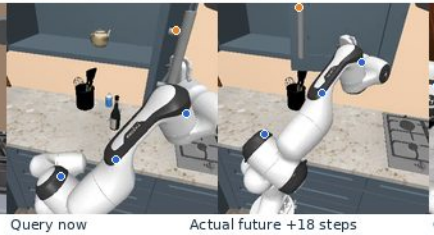


Query now Actual future +32 steps

Object: actual 29.5 cm; forecast 30.3 cm

Min point movement: 9.5 / 9.6 px (actual / forecast)

CloseDoubleDoor / demo_2470 / s176 / R



Query now Actual future +18 steps

Object: actual 53.9 cm; forecast 55.1 cm

Min point movement: 47.3 / 38.6 px (actual / forecast)

OpenDoubleDoor / demo_1130 / s608 / R

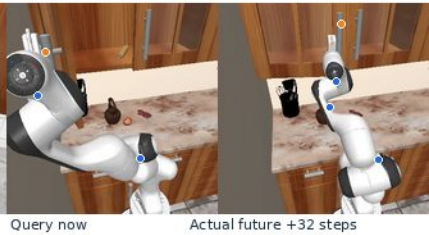


Query now Actual future +26 steps

Object: actual 28.8 cm; forecast 32.3 cm

Min point movement: 9.3 / 13.9 px (actual / forecast)

CloseDoubleDoor / demo_1087 / s336 / L



Query now Actual future +32 steps

Object: actual 46.5 cm; forecast 54.4 cm

Min point movement: 23.4 / 29.5 px (actual / forecast)

CloseSingleDoor / demo_2451 / s352 / R

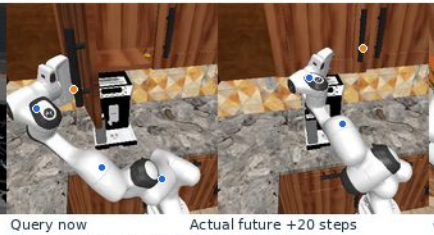


Query now Actual future +32 steps

Object: actual 32.2 cm; forecast 28.2 cm

Min point movement: 6.4 / 5.2 px (actual / forecast)

CloseDoubleDoor / demo_1906 / s368 / L

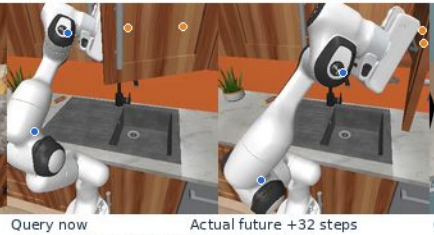


Query now Actual future +20 steps

Object: actual 53.7 cm; forecast 52.0 cm

Min point movement: 3.0 / 16.7 px (actual / forecast)

OpenDoubleDoor / demo_562 / s320 / R

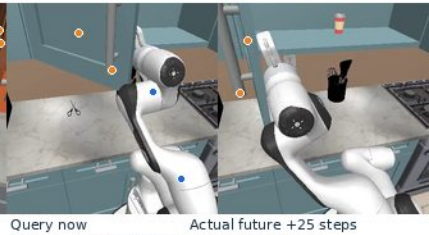


Query now Actual future +32 steps

Object: actual 31.9 cm; forecast 29.6 cm

Min point movement: 17.3 / 12.7 px (actual / forecast)

OpenDoubleDoor / demo_1189 / s848 / L



Query now Actual future +25 steps

Object: actual 25.1 cm; forecast 26.6 cm

Min point movement: 25.9 / 20.7 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseSingleDoor / demo_1645 / s208 / L

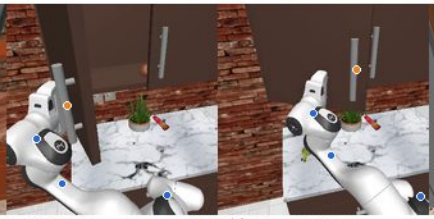


Query now Actual future +32 steps

Object: actual 41.1 cm; forecast 36.4 cm

Min point movement: 11.4 / 16.5 px (actual / forecast)

CloseDoubleDoor / demo_842 / s368 / L



Query now Actual future +32 steps

Object: actual 34.8 cm; forecast 49.0 cm

Min point movement: 26.2 / 24.0 px (actual / forecast)

OpenSingleDoor / demo_1105 / s176 / R

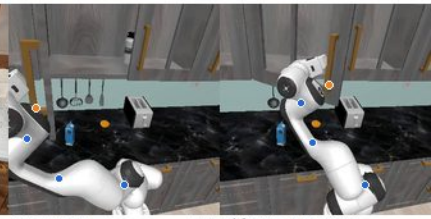


Query now Actual future +32 steps

Object: actual 27.9 cm; forecast 30.7 cm

Min point movement: 5.6 / 7.2 px (actual / forecast)

CloseDoubleDoor / demo_1729 / s320 / L

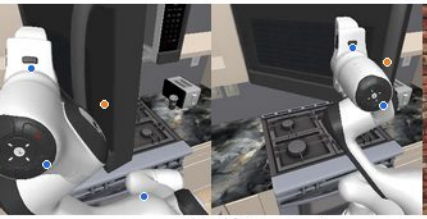


Query now Actual future +32 steps

Object: actual 35.1 cm; forecast 30.9 cm

Min point movement: 18.1 / 10.8 px (actual / forecast)

CloseSingleDoor / demo_1881 / s208 / L



Query now Actual future +32 steps

Object: actual 30.8 cm; forecast 33.0 cm

Min point movement: 34.2 / 34.0 px (actual / forecast)

OpenSingleDoor / demo_1154 / s256 / L



Query now Actual future +32 steps

Object: actual 28.5 cm; forecast 28.6 cm

Min point movement: 7.0 / 7.0 px (actual / forecast)

CloseDoubleDoor / demo_1399 / s352 / L



Query now Actual future +32 steps

Object: actual 38.8 cm; forecast 36.2 cm

Min point movement: 21.8 / 25.7 px (actual / forecast)

CloseSingleDoor / demo_2131 / s240 / L



Query now Actual future +32 steps

Object: actual 36.9 cm; forecast 41.3 cm

Min point movement: 17.5 / 24.0 px (actual / forecast)

CloseSingleDoor / demo_1789 / s192 / L



Query now Actual future +32 steps

Object: actual 25.2 cm; forecast 19.2 cm

Min point movement: 11.7 / 8.4 px (actual / forecast)

CloseSingleDoor / demo_1892 / s192 / L



Query now Actual future +31 steps

Object: actual 42.4 cm; forecast 40.6 cm

Min point movement: 37.7 / 29.9 px (actual / forecast)

CloseDoubleDoor / demo_2128 / s384 / L



Query now Actual future +32 steps

Object: actual 36.9 cm; forecast 37.2 cm

Min point movement: 4.4 / 3.6 px (actual / forecast)

CloseDoubleDoor / demo_876 / s160 / R



Query now Actual future +32 steps

Object: actual 26.7 cm; forecast 17.6 cm

Min point movement: 7.8 / 3.8 px (actual / forecast)

CloseSingleDoor / demo_24 / s224 / L

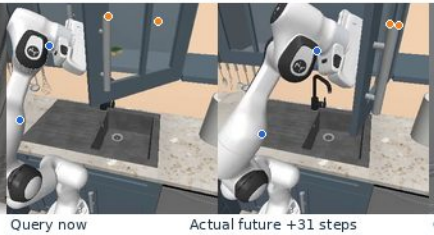


Query now Actual future +32 steps

Object: actual 33.4 cm; forecast 36.0 cm

Min point movement: 13.0 / 10.5 px (actual / forecast)

OpenDoubleDoor / demo_1392 / s448 / R

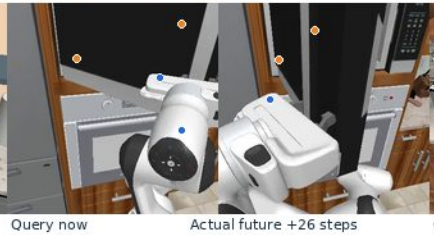


Query now Actual future +31 steps

Object: actual 23.3 cm; forecast 25.0 cm

Min point movement: 13.4 / 13.9 px (actual / forecast)

OpenSingleDoor / demo_1115 / s208 / L



Query now Actual future +26 steps

Object: actual 23.8 cm; forecast 28.7 cm

Min point movement: 5.3 / 7.6 px (actual / forecast)

CloseDoubleDoor / demo_443 / s160 / R



Query now Actual future +23 steps

Object: actual 39.3 cm; forecast 31.4 cm

Min point movement: 14.0 / 12.2 px (actual / forecast)

OpenSingleDoor / demo_2015 / s224 / L



Query now Actual future +28 steps

Object: actual 25.4 cm; forecast 24.3 cm

Min point movement: 9.4 / 7.9 px (actual / forecast)

OpenSingleDoor / demo_2753 / s176 / L

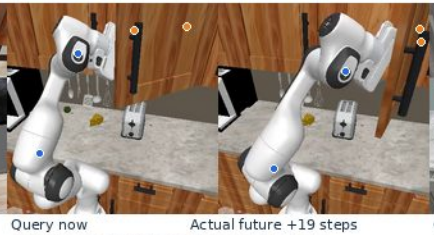


Query now Actual future +32 steps

Object: actual 31.2 cm; forecast 31.0 cm

Min point movement: 10.3 / 10.7 px (actual / forecast)

OpenDoubleDoor / demo_456 / s368 / R



Query now Actual future +19 steps

Object: actual 38.3 cm; forecast 34.5 cm

Min point movement: 18.1 / 17.8 px (actual / forecast)

OpenSingleDoor / demo_32 / s192 / R



Query now Actual future +32 steps

Object: actual 25.5 cm; forecast 26.1 cm

Min point movement: 10.1 / 10.6 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

OpenDoubleDoor / demo_296 / s336 / R



Query now Actual future +27 steps

Object: actual 30.0 cm; forecast 31.1 cm

Min point movement: 17.4 / 19.8 px (actual / forecast)

OpenDoubleDoor / demo_1088 / s336 / R



Query now Actual future +32 steps

Object: actual 24.0 cm; forecast 16.4 cm

Min point movement: 9.3 / 7.3 px (actual / forecast)

OpenSingleDoor / demo_32 / s208 / L

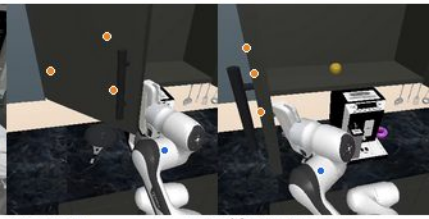


Query now Actual future +32 steps

Object: actual 29.8 cm; forecast 29.5 cm

Min point movement: 10.4 / 10.4 px (actual / forecast)

OpenDoubleDoor / demo_1176 / s848 / L

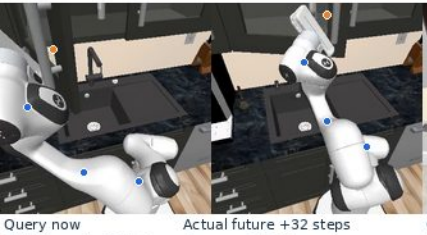


Query now Actual future +23 steps

Object: actual 24.1 cm; forecast 26.5 cm

Min point movement: 5.1 / 3.4 px (actual / forecast)

CloseDoubleDoor / demo_1584 / s352 / L

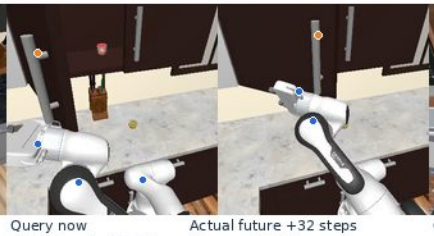


Query now Actual future +32 steps

Object: actual 33.9 cm; forecast 43.4 cm

Min point movement: 22.7 / 24.7 px (actual / forecast)

CloseDoubleDoor / demo_1987 / s368 / L

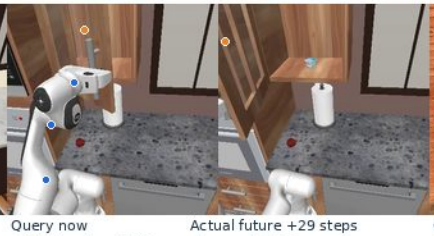


Query now Actual future +32 steps

Object: actual 23.2 cm; forecast 35.5 cm

Min point movement: 3.4 / 5.4 px (actual / forecast)

OpenSingleDoor / demo_1017 / s160 / R

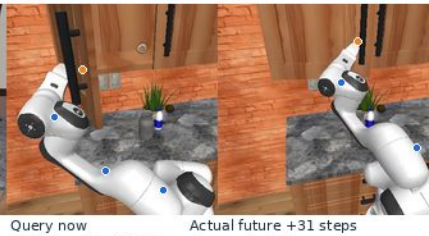


Query now Actual future +29 steps

Object: actual 25.3 cm; forecast 26.6 cm

Min point movement: 11.7 / 28.4 px (actual / forecast)

CloseDoubleDoor / demo_1569 / s352 / L

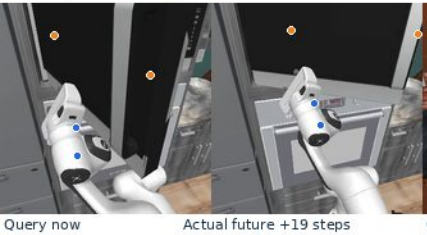


Query now Actual future +31 steps

Object: actual 43.6 cm; forecast 43.5 cm

Min point movement: 31.5 / 34.4 px (actual / forecast)

CloseSingleDoor / demo_2519 / s192 / L



Query now Actual future +19 steps

Object: actual 61.2 cm; forecast 41.6 cm

Min point movement: 18.7 / 19.4 px (actual / forecast)

CloseSingleDoor / demo_651 / s192 / L



Query now Actual future +32 steps

Object: actual 24.5 cm; forecast 17.0 cm

Min point movement: 12.0 / 10.1 px (actual / forecast)

CloseSingleDoor / demo_1829 / s240 / L

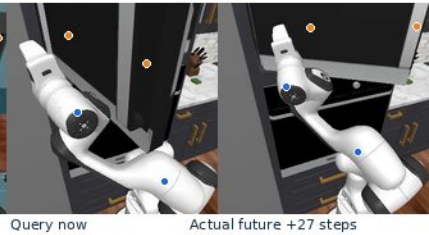


Query now Actual future +31 steps

Object: actual 43.3 cm; forecast 40.6 cm

Min point movement: 35.5 / 31.7 px (actual / forecast)

CloseSingleDoor / demo_1932 / s176 / L

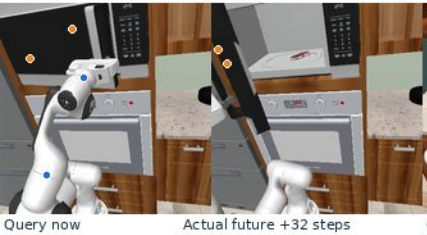


Query now Actual future +27 steps

Object: actual 50.1 cm; forecast 12.5 cm

Min point movement: 17.2 / 8.6 px (actual / forecast)

OpenSingleDoor / demo_1115 / s176 / R

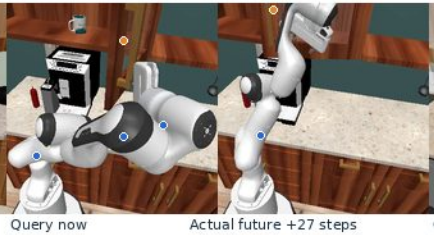


Query now Actual future +32 steps

Object: actual 27.7 cm; forecast 27.4 cm

Min point movement: 9.1 / 9.2 px (actual / forecast)

CloseDoubleDoor / demo_341 / s144 / R



Query now Actual future +27 steps

Object: actual 51.6 cm; forecast 37.8 cm

Min point movement: 18.8 / 7.9 px (actual / forecast)

CloseSingleDoor / demo_611 / s160 / R

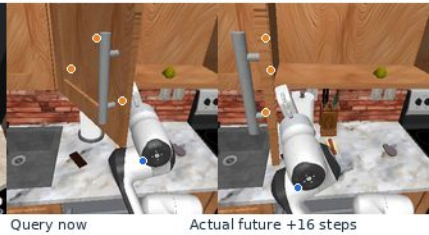


Query now Actual future +32 steps

Object: actual 33.6 cm; forecast 37.9 cm

Min point movement: 21.1 / 20.5 px (actual / forecast)

OpenDoubleDoor / demo_176 / s768 / L



Query now Actual future +16 steps

Object: actual 17.4 cm; forecast 25.1 cm

Min point movement: 7.2 / 10.2 px (actual / forecast)

OpenSingleDoor / demo_641 / s160 / R



Query now Actual future +24 steps

Object: actual 31.6 cm; forecast 27.7 cm

Min point movement: 12.4 / 10.3 px (actual / forecast)

CloseSingleDoor / demo_1859 / s208 / L



Query now Actual future +32 steps

Object: actual 29.5 cm; forecast 30.7 cm

Min point movement: 17.7 / 18.6 px (actual / forecast)

OpenSingleDoor / demo_935 / s192 / R

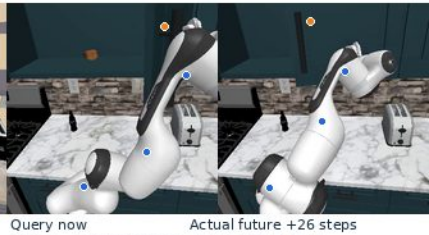


Query now Actual future +28 steps

Object: actual 30.5 cm; forecast 28.7 cm

Min point movement: 11.0 / 11.3 px (actual / forecast)

CloseDoubleDoor / demo_2593 / s176 / R



Query now Actual future +26 steps

Object: actual 44.4 cm; forecast 20.4 cm

Min point movement: 17.9 / 11.5 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDoubleDoor / demo_876 / s192 / L



Query now Actual future +17 steps

Object: actual 33.8 cm; forecast 36.3 cm

Min point movement: 6.6 / 9.5 px (actual / forecast)

CloseSingleDoor / demo_535 / s192 / L

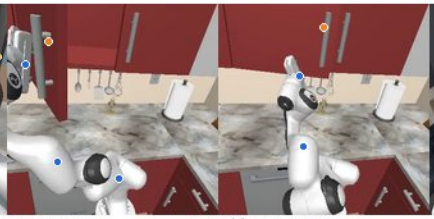


Query now Actual future +32 steps

Object: actual 32.4 cm; forecast 35.5 cm

Min point movement: 27.0 / 27.5 px (actual / forecast)

CloseDoubleDoor / demo_1683 / s384 / L



Query now Actual future +32 steps

Object: actual 40.8 cm; forecast 44.9 cm

Min point movement: 10.5 / 7.2 px (actual / forecast)

OpenSingleDoor / demo_1549 / s176 / R



Query now Actual future +32 steps

Object: actual 27.0 cm; forecast 27.6 cm

Min point movement: 12.6 / 12.6 px (actual / forecast)

CloseSingleDoor / demo_1645 / s224 / L

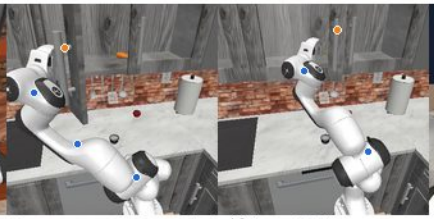


Query now Actual future +23 steps

Object: actual 61.0 cm; forecast 60.4 cm

Min point movement: 8.4 / 32.8 px (actual / forecast)

CloseDoubleDoor / demo_1895 / s336 / L

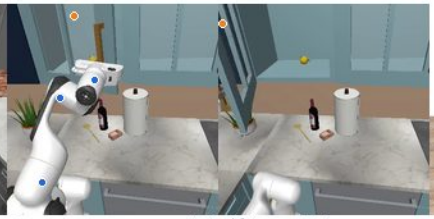


Query now Actual future +23 steps

Object: actual 47.5 cm; forecast 50.9 cm

Min point movement: 24.9 / 26.3 px (actual / forecast)

OpenSingleDoor / demo_191 / s144 / R



Query now Actual future +32 steps

Object: actual 26.0 cm; forecast 23.6 cm

Min point movement: 22.6 / 21.9 px (actual / forecast)

CloseSingleDoor / demo_2591 / s160 / L



Query now Actual future +25 steps

Object: actual 28.7 cm; forecast 26.4 cm

Min point movement: 17.6 / 15.0 px (actual / forecast)

CloseSingleDoor / demo_1182 / s224 / L



Query now Actual future +29 steps

Object: actual 43.2 cm; forecast 38.6 cm

Min point movement: 28.6 / 26.4 px (actual / forecast)

OpenDoubleDoor / demo_456 / s928 / L



Query now Actual future +14 steps

Object: actual 14.7 cm; forecast 11.1 cm

Min point movement: 10.9 / 8.5 px (actual / forecast)

OpenDoubleDoor / demo_1176 / s400 / R

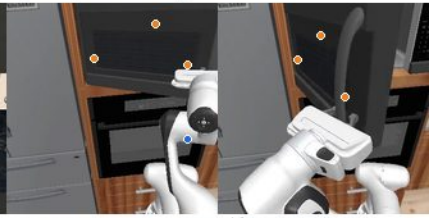


Query now Actual future +32 steps

Object: actual 25.1 cm; forecast 33.2 cm

Min point movement: 6.8 / 3.3 px (actual / forecast)

OpenSingleDoor / demo_156 / s256 / L

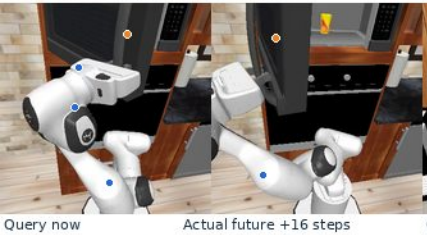


Query now Actual future +32 steps

Object: actual 31.8 cm; forecast 31.1 cm

Min point movement: 4.7 / 4.0 px (actual / forecast)

OpenSingleDoor / demo_2753 / s192 / L



Query now Actual future +16 steps

Object: actual 17.8 cm; forecast 20.7 cm

Min point movement: 21.7 / 40.8 px (actual / forecast)

OpenDoubleDoor / demo_456 / s912 / L

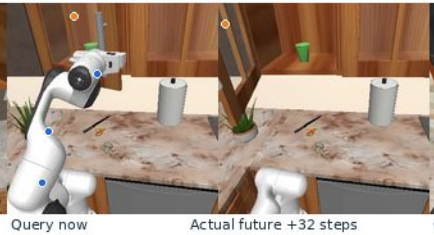


Query now Actual future +32 steps

Object: actual 21.5 cm; forecast 15.6 cm

Min point movement: 24.4 / 24.2 px (actual / forecast)

OpenSingleDoor / demo_2350 / s176 / R

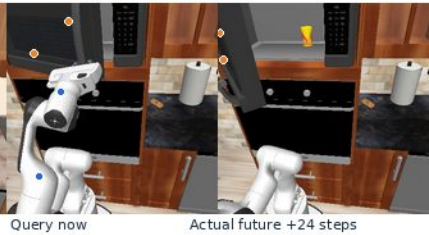


Query now Actual future +32 steps

Object: actual 24.2 cm; forecast 21.8 cm

Min point movement: 18.4 / 19.5 px (actual / forecast)

OpenSingleDoor / demo_2753 / s160 / R



Query now Actual future +24 steps

Object: actual 31.3 cm; forecast 30.6 cm

Min point movement: 17.0 / 17.0 px (actual / forecast)

OpenSingleDoor / demo_49 / s176 / R

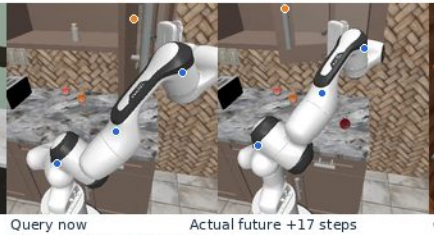


Query now Actual future +32 steps

Object: actual 27.6 cm; forecast 28.6 cm

Min point movement: 10.0 / 8.2 px (actual / forecast)

CloseDoubleDoor / demo_2238 / s176 / R

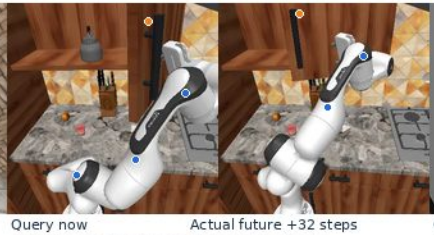


Query now Actual future +17 steps

Object: actual 51.5 cm; forecast 52.9 cm

Min point movement: 32.5 / 28.6 px (actual / forecast)

CloseDoubleDoor / demo_1029 / s192 / R

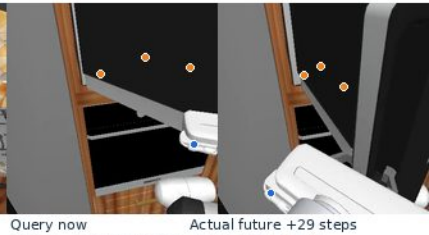


Query now Actual future +32 steps

Object: actual 36.1 cm; forecast 42.4 cm

Min point movement: 11.4 / 18.3 px (actual / forecast)

OpenSingleDoor / demo_1033 / s192 / L



Query now Actual future +29 steps

Object: actual 17.0 cm; forecast 20.7 cm

Min point movement: 5.0 / 8.8 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDoubleDoor / demo_2827 / s336 / L



Query now Actual future +31 steps

Object: actual 20.8 cm; forecast 31.0 cm

Min point movement: 12.2 / 26.5 px (actual / forecast)

CloseSingleDoor / demo_972 / s208 / L



Query now Actual future +32 steps

Object: actual 34.9 cm; forecast 34.3 cm

Min point movement: 26.6 / 23.0 px (actual / forecast)

CloseSingleDoor / demo_821 / s224 / L

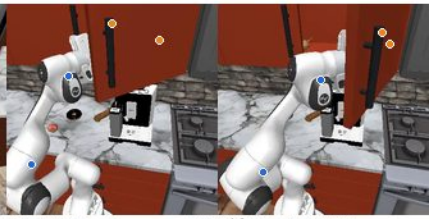


Query now Actual future +28 steps

Object: actual 48.4 cm; forecast 51.0 cm

Min point movement: 22.4 / 19.8 px (actual / forecast)

OpenDoubleDoor / demo_737 / s336 / R



Query now Actual future +32 steps

Object: actual 19.8 cm; forecast 19.8 cm

Min point movement: 11.4 / 11.3 px (actual / forecast)

CloseDoubleDoor / demo_2216 / s352 / L

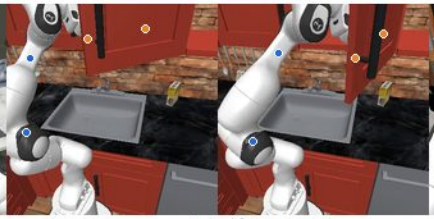


Query now Actual future +16 steps

Object: actual 46.7 cm; forecast 51.7 cm

Min point movement: 31.7 / 36.0 px (actual / forecast)

OpenDoubleDoor / demo_332 / s384 / R



Query now Actual future +32 steps

Object: actual 26.3 cm; forecast 24.6 cm

Min point movement: 11.2 / 10.5 px (actual / forecast)

OpenSingleDoor / demo_321 / s176 / R

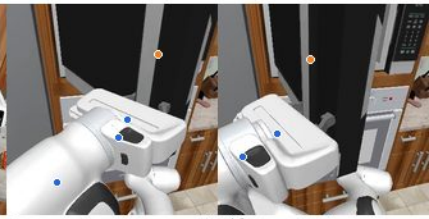


Query now Actual future +32 steps

Object: actual 20.6 cm; forecast 23.7 cm

Min point movement: 9.4 / 11.0 px (actual / forecast)

OpenSingleDoor / demo_1115 / s224 / L



Query now Actual future +10 steps

Object: actual 9.7 cm; forecast 15.7 cm

Min point movement: 27.9 / 31.8 px (actual / forecast)

OpenSingleDoor / demo_2989 / s144 / R

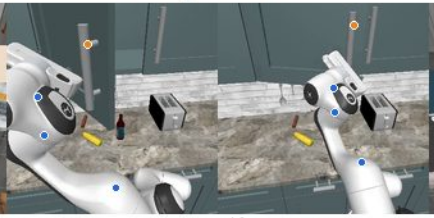


Query now Actual future +18 steps

Object: actual 38.7 cm; forecast 40.2 cm

Min point movement: 15.8 / 12.9 px (actual / forecast)

CloseDoubleDoor / demo_254 / s336 / L

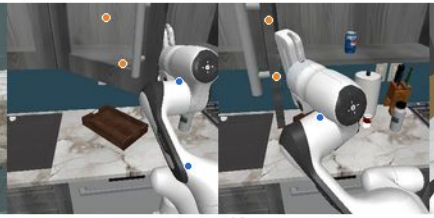


Query now Actual future +32 steps

Object: actual 30.4 cm; forecast 22.9 cm

Min point movement: 26.2 / 24.3 px (actual / forecast)

OpenDoubleDoor / demo_296 / s896 / L



Query now Actual future +31 steps

Object: actual 21.5 cm; forecast 20.9 cm

Min point movement: 27.7 / 28.1 px (actual / forecast)

OpenSingleDoor / demo_1049 / s160 / R

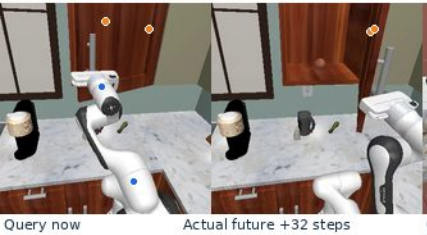


Query now Actual future +32 steps

Object: actual 25.3 cm; forecast 25.5 cm

Min point movement: 11.9 / 11.8 px (actual / forecast)

OpenSingleDoor / demo_255 / s240 / L

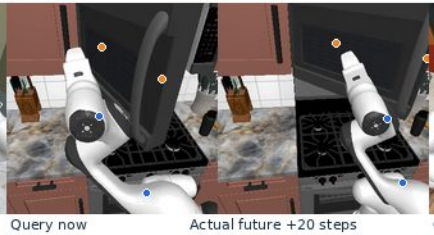


Query now Actual future +32 steps

Object: actual 26.7 cm; forecast 21.6 cm

Min point movement: 6.4 / 4.1 px (actual / forecast)

CloseSingleDoor / demo_2151 / s192 / L

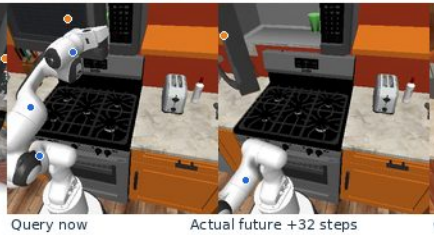


Query now Actual future +20 steps

Object: actual 45.6 cm; forecast 39.5 cm

Min point movement: 20.3 / 22.1 px (actual / forecast)

OpenSingleDoor / demo_2015 / s192 / R

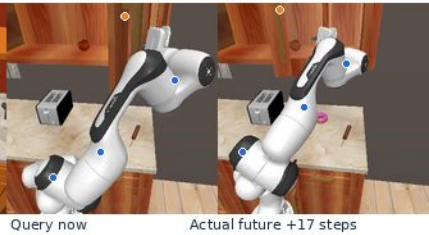


Query now Actual future +32 steps

Object: actual 26.5 cm; forecast 24.2 cm

Min point movement: 15.0 / 13.5 px (actual / forecast)

CloseDoubleDoor / demo_1606 / s160 / R



Query now Actual future +17 steps

Object: actual 48.4 cm; forecast 46.5 cm

Min point movement: 34.4 / 35.3 px (actual / forecast)

CloseSingleDoor / demo_1680 / s192 / L



Query now Actual future +32 steps

Object: actual 28.4 cm; forecast 33.0 cm

Min point movement: 13.0 / 15.7 px (actual / forecast)

CloseDoubleDoor / demo_876 / s176 / R



Query now Actual future +27 steps

Object: actual 39.3 cm; forecast 34.0 cm

Min point movement: 33.9 / 19.0 px (actual / forecast)

OpenSingleDoor / demo_1419 / s160 / R

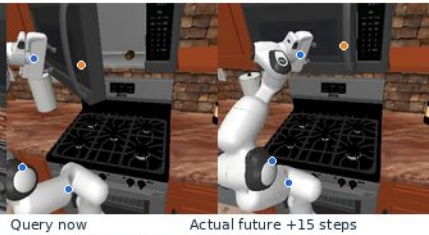


Query now Actual future +32 steps

Object: actual 23.3 cm; forecast 25.3 cm

Min point movement: 8.1 / 9.9 px (actual / forecast)

CloseSingleDoor / demo_1645 / s240 / R



Query now Actual future +15 steps

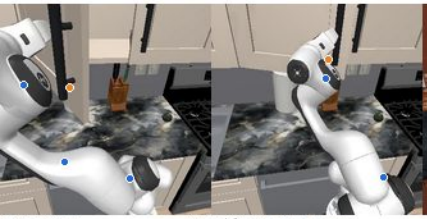
Object: actual 32.0 cm; forecast 35.9 cm

Min point movement: 6.4 / 5.6 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDoubleDoor / demo_1175 / s352 / L

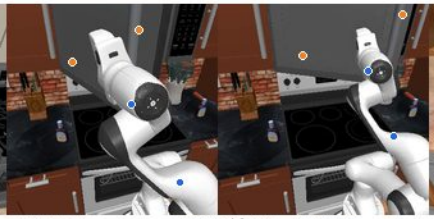


Query now Actual future +32 steps

Object: actual 30.7 cm; forecast 31.6 cm

Min point movement: 25.6 / 30.6 px (actual / forecast)

CloseSingleDoor / demo_651 / s208 / L

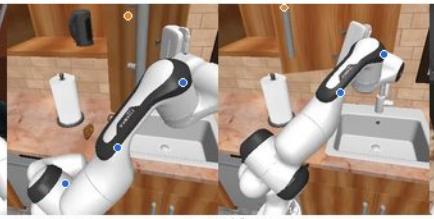


Query now Actual future +32 steps

Object: actual 36.2 cm; forecast 40.5 cm

Min point movement: 12.0 / 12.3 px (actual / forecast)

CloseDoubleDoor / demo_1499 / s176 / R



Query now Actual future +24 steps

Object: actual 43.0 cm; forecast 44.0 cm

Min point movement: 21.5 / 30.8 px (actual / forecast)

PnPMicrowaveToCounter / demo_463 / s160 / R



Query now Actual future +7 steps

Object: actual 31.7 cm; forecast 33.7 cm

Min point movement: 9.3 / 9.7 px (actual / forecast)

CloseSingleDoor / demo_2334 / s192 / L



Query now Actual future +32 steps

Object: actual 25.4 cm; forecast 27.2 cm

Min point movement: 20.3 / 15.2 px (actual / forecast)

OpenSingleDoor / demo_49 / s192 / L



Query now Actual future +32 steps

Object: actual 27.2 cm; forecast 26.8 cm

Min point movement: 11.5 / 12.3 px (actual / forecast)

OpenDoubleDoor / demo_293 / s400 / R

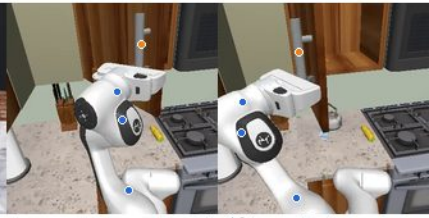


Query now Actual future +32 steps

Object: actual 18.3 cm; forecast 17.6 cm

Min point movement: 8.7 / 9.5 px (actual / forecast)

OpenSingleDoor / demo_2131 / s192 / L



Query now Actual future +12 steps

Object: actual 15.7 cm; forecast 24.5 cm

Min point movement: 26.3 / 48.3 px (actual / forecast)

OpenSingleDoor / demo_1845 / s240 / L

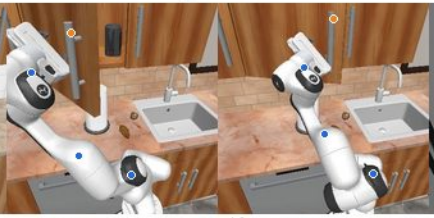


Query now Actual future +24 steps

Object: actual 18.5 cm; forecast 22.6 cm

Min point movement: 32.6 / 41.1 px (actual / forecast)

CloseDoubleDoor / demo_1499 / s368 / L

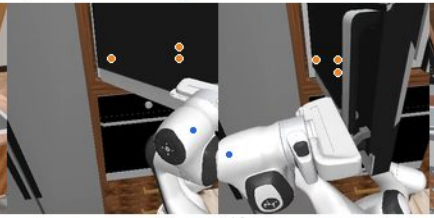


Query now Actual future +32 steps

Object: actual 28.7 cm; forecast 43.3 cm

Min point movement: 18.6 / 16.9 px (actual / forecast)

OpenSingleDoor / demo_1105 / s208 / L

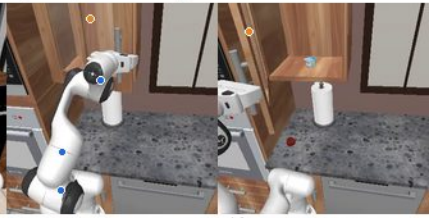


Query now Actual future +26 steps

Object: actual 18.7 cm; forecast 22.6 cm

Min point movement: 4.1 / 5.7 px (actual / forecast)

OpenSingleDoor / demo_1017 / s144 / R



Query now Actual future +32 steps

Object: actual 25.1 cm; forecast 23.9 cm

Min point movement: 13.0 / 14.3 px (actual / forecast)

CloseDoubleDoor / demo_126 / s352 / L



Query now Actual future +23 steps

Object: actual 44.1 cm; forecast 41.2 cm

Min point movement: 32.4 / 30.5 px (actual / forecast)

OpenSingleDoor / demo_1227 / s144 / R



Query now Actual future +32 steps

Object: actual 25.6 cm; forecast 22.9 cm

Min point movement: 24.9 / 25.6 px (actual / forecast)

OpenDoubleDoor / demo_1130 / s928 / L

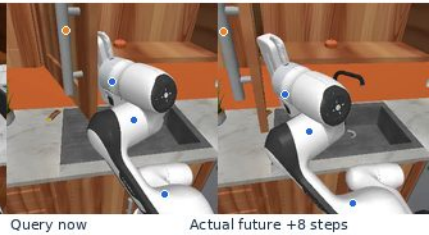


Query now Actual future +30 steps

Object: actual 15.5 cm; forecast 14.4 cm

Min point movement: 16.7 / 14.4 px (actual / forecast)

OpenDoubleDoor / demo_562 / s992 / L



Query now Actual future +8 steps

Object: actual 10.3 cm; forecast 26.2 cm

Min point movement: 15.3 / 31.1 px (actual / forecast)

CloseDoubleDoor / demo_1029 / s208 / L

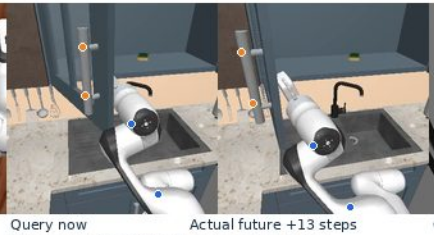


Query now Actual future +25 steps

Object: actual 28.3 cm; forecast 27.3 cm

Min point movement: 5.9 / 3.7 px (actual / forecast)

OpenDoubleDoor / demo_1392 / s880 / L

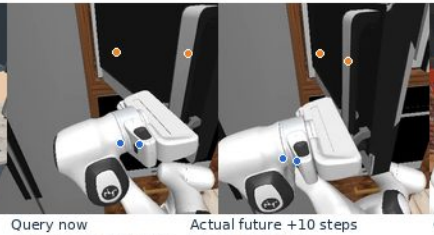


Query now Actual future +13 steps

Object: actual 12.0 cm; forecast 12.6 cm

Min point movement: 13.9 / 13.8 px (actual / forecast)

OpenSingleDoor / demo_1105 / s224 / L

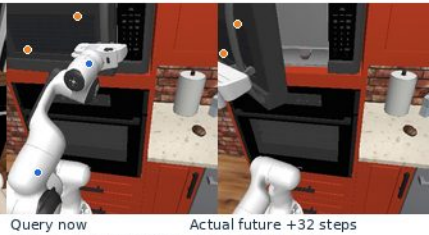


Query now Actual future +10 steps

Object: actual 9.2 cm; forecast 13.9 cm

Min point movement: 4.3 / 7.2 px (actual / forecast)

OpenSingleDoor / demo_1154 / s224 / R



Query now Actual future +32 steps

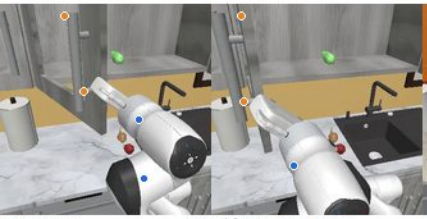
Object: actual 24.2 cm; forecast 26.9 cm

Min point movement: 9.7 / 10.3 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

OpenDoubleDoor / demo_1225 / s736 / L

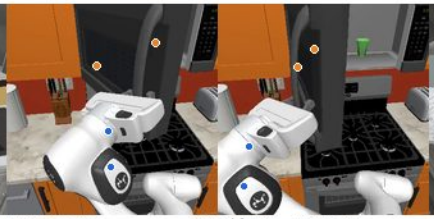


Query now Actual future +15 steps

Object: actual 13.2 cm; forecast 12.0 cm

Min point movement: 20.5 / 18.1 px (actual / forecast)

OpenSingleDoor / demo_2015 / s240 / L

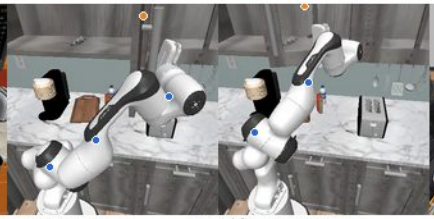


Query now Actual future +12 steps

Object: actual 12.8 cm; forecast 18.4 cm

Min point movement: 6.3 / 8.8 px (actual / forecast)

CloseDoubleDoor / demo_928 / s160 / R



Query now Actual future +25 steps

Object: actual 37.9 cm; forecast 33.3 cm

Min point movement: 30.9 / 25.1 px (actual / forecast)

OpenSingleDoor / demo_1381 / s288 / R



Query now Actual future +32 steps

Object: actual 24.7 cm; forecast 29.6 cm

Min point movement: 25.4 / 22.7 px (actual / forecast)

OpenSingleDoor / demo_2131 / s160 / R



Query now Actual future +32 steps

Object: actual 23.1 cm; forecast 21.3 cm

Min point movement: 21.6 / 22.6 px (actual / forecast)

OpenSingleDoor / demo_1845 / s224 / L

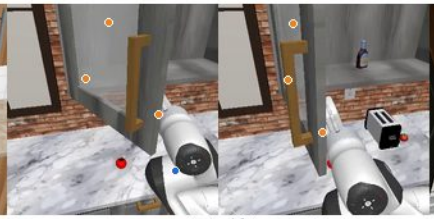


Query now Actual future +32 steps

Object: actual 22.4 cm; forecast 23.8 cm

Min point movement: 10.3 / 11.8 px (actual / forecast)

OpenDoubleDoor / demo_293 / s832 / L



Query now Actual future +20 steps

Object: actual 14.0 cm; forecast 9.2 cm

Min point movement: 5.6 / 4.7 px (actual / forecast)

OpenSingleDoor / demo_271 / s192 / L

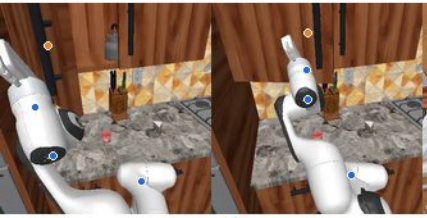


Query now Actual future +20 steps

Object: actual 22.1 cm; forecast 28.5 cm

Min point movement: 5.4 / 7.7 px (actual / forecast)

CloseDoubleDoor / demo_1029 / s400 / L



Query now Actual future +32 steps

Object: actual 25.0 cm; forecast 35.6 cm

Min point movement: 4.2 / 4.5 px (actual / forecast)

CloseSingleDoor / demo_2131 / s256 / L

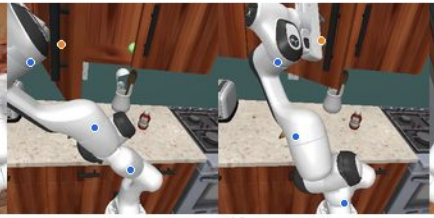


Query now Actual future +29 steps

Object: actual 38.5 cm; forecast 39.2 cm

Min point movement: 30.1 / 31.8 px (actual / forecast)

CloseDoubleDoor / demo_1617 / s352 / L



Query now Actual future +32 steps

Object: actual 21.6 cm; forecast 16.4 cm

Min point movement: 18.2 / 22.1 px (actual / forecast)

OpenSingleDoor / demo_1026 / s160 / R

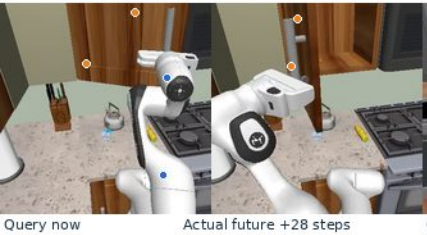


Query now Actual future +32 steps

Object: actual 21.9 cm; forecast 25.3 cm

Min point movement: 18.3 / 15.5 px (actual / forecast)

OpenSingleDoor / demo_2131 / s176 / L

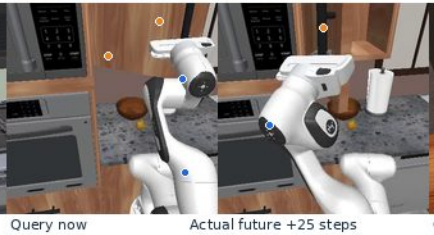


Query now Actual future +28 steps

Object: actual 23.7 cm; forecast 22.6 cm

Min point movement: 3.9 / 4.0 px (actual / forecast)

OpenSingleDoor / demo_498 / s208 / L



Query now Actual future +25 steps

Object: actual 23.4 cm; forecast 25.5 cm

Min point movement: 4.6 / 7.1 px (actual / forecast)

OpenSingleDoor / demo_2350 / s192 / L

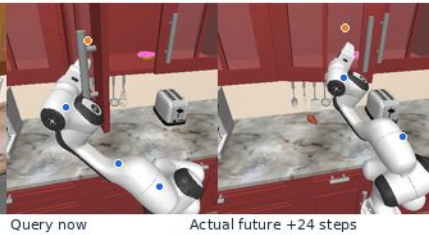


Query now Actual future +23 steps

Object: actual 22.8 cm; forecast 24.4 cm

Min point movement: 4.7 / 6.4 px (actual / forecast)

CloseDoubleDoor / demo_2123 / s320 / L

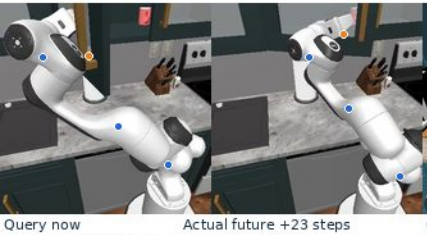


Query now Actual future +24 steps

Object: actual 39.5 cm; forecast 37.9 cm

Min point movement: 21.6 / 31.2 px (actual / forecast)

CloseDoubleDoor / demo_275 / s336 / L

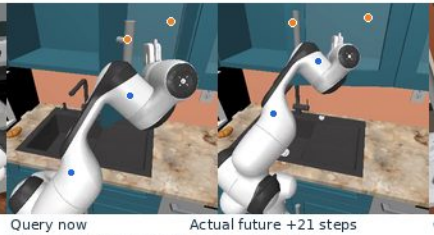


Query now Actual future +23 steps

Object: actual 41.2 cm; forecast 38.7 cm

Min point movement: 15.2 / 17.5 px (actual / forecast)

CloseDoubleDoor / demo_1112 / s176 / R

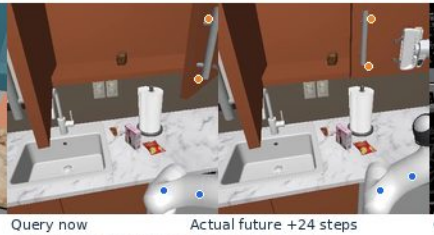


Query now Actual future +21 steps

Object: actual 43.6 cm; forecast 33.9 cm

Min point movement: 9.2 / 9.9 px (actual / forecast)

CloseDoubleDoor / demo_1020 / s192 / L

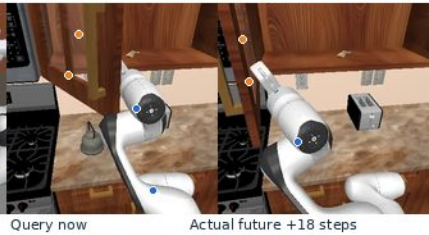


Query now Actual future +24 steps

Object: actual 25.6 cm; forecast 24.6 cm

Min point movement: 4.0 / 4.4 px (actual / forecast)

OpenDoubleDoor / demo_503 / s720 / L



Query now Actual future +18 steps

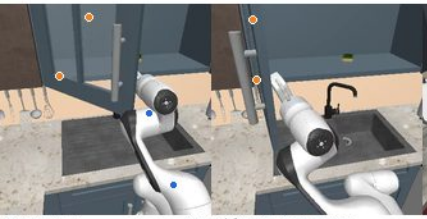
Object: actual 13.8 cm; forecast 12.5 cm

Min point movement: 14.8 / 14.0 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

OpenDoubleDoor / demo_1392 / s864 / L



Query now Actual future +29 steps

Object: actual 16.7 cm; forecast 10.8 cm

Min point movement: 8.8 / 5.0 px (actual / forecast)

OpenSingleDoor / demo_461 / s160 / R



Query now Actual future +32 steps

Object: actual 25.9 cm; forecast 28.4 cm

Min point movement: 4.8 / 20.2 px (actual / forecast)

OpenSingleDoor / demo_461 / s192 / L

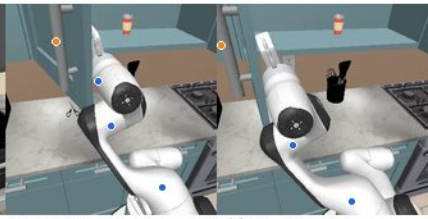


Query now Actual future +14 steps

Object: actual 17.4 cm; forecast 21.1 cm

Min point movement: 6.6 / 8.5 px (actual / forecast)

OpenDoubleDoor / demo_1189 / s864 / L

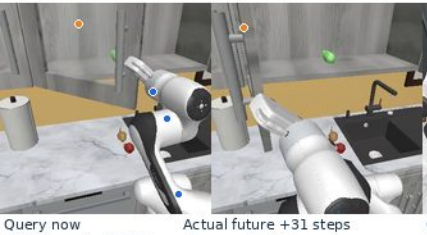


Query now Actual future +9 steps

Object: actual 9.5 cm; forecast 20.6 cm

Min point movement: 15.8 / 28.4 px (actual / forecast)

OpenDoubleDoor / demo_1225 / s720 / L

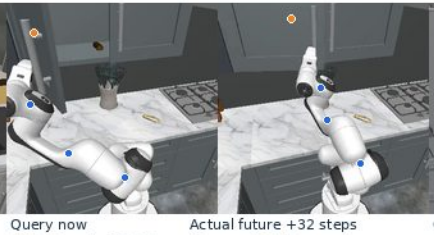


Query now Actual future +31 steps

Object: actual 17.2 cm; forecast 14.0 cm

Min point movement: 10.7 / 10.9 px (actual / forecast)

CloseDoubleDoor / demo_1480 / s336 / L

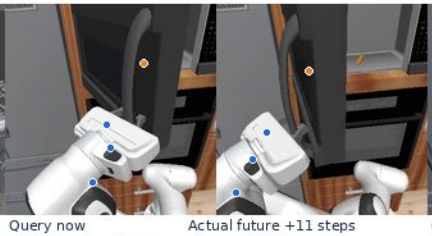


Query now Actual future +32 steps

Object: actual 30.8 cm; forecast 31.7 cm

Min point movement: 17.1 / 23.5 px (actual / forecast)

OpenSingleDoor / demo_156 / s288 / L

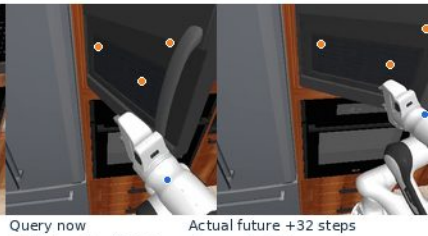


Query now Actual future +11 steps

Object: actual 10.8 cm; forecast 14.1 cm

Min point movement: 28.4 / 38.1 px (actual / forecast)

CloseSingleDoor / demo_1172 / s192 / L

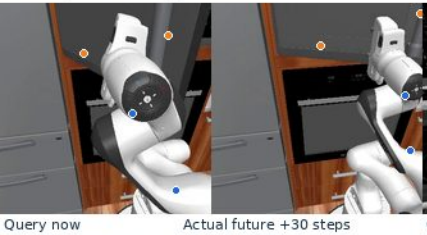


Query now Actual future +32 steps

Object: actual 26.0 cm; forecast 30.1 cm

Min point movement: 7.2 / 7.6 px (actual / forecast)

CloseSingleDoor / demo_1682 / s240 / L



Query now Actual future +30 steps

Object: actual 35.2 cm; forecast 38.8 cm

Min point movement: 16.6 / 15.5 px (actual / forecast)

OpenDoubleDoor / demo_1130 / s944 / L

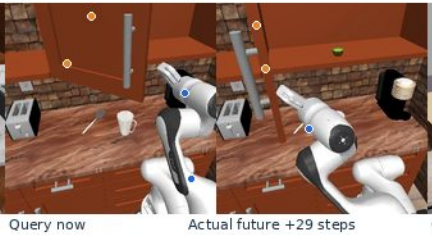


Query now Actual future +14 steps

Object: actual 10.1 cm; forecast 17.0 cm

Min point movement: 15.9 / 30.5 px (actual / forecast)

OpenDoubleDoor / demo_1088 / s720 / L



Query now Actual future +29 steps

Object: actual 17.7 cm; forecast 14.1 cm

Min point movement: 8.4 / 4.0 px (actual / forecast)

OpenSingleDoor / demo_935 / s224 / L

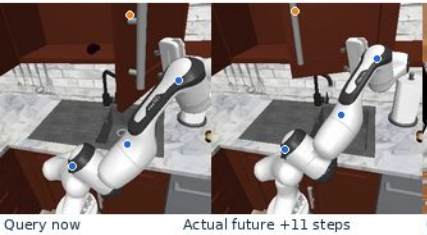


Query now Actual future +26 steps

Object: actual 19.0 cm; forecast 22.0 cm

Min point movement: 3.9 / 4.3 px (actual / forecast)

CloseDoubleDoor / demo_2316 / s192 / R



Query now Actual future +11 steps

Object: actual 47.8 cm; forecast 47.3 cm

Min point movement: 22.9 / 27.4 px (actual / forecast)

OpenSingleDoor / demo_2711 / s240 / L

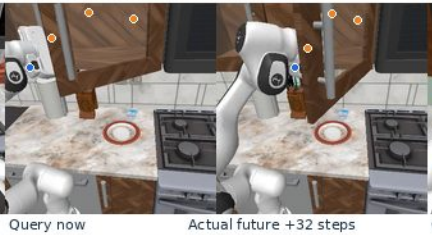


Query now Actual future +9 steps

Object: actual 8.9 cm; forecast 14.3 cm

Min point movement: 28.1 / 46.0 px (actual / forecast)

OpenDoubleDoor / demo_1262 / s352 / R



Query now Actual future +32 steps

Object: actual 24.8 cm; forecast 24.8 cm

Min point movement: 7.7 / 8.7 px (actual / forecast)

OpenSingleDoor / demo_271 / s160 / R

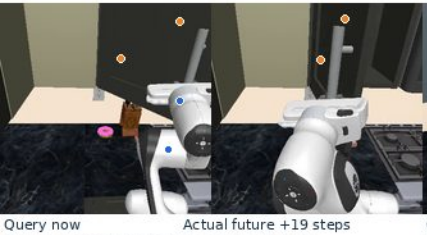


Query now Actual future +32 steps

Object: actual 20.7 cm; forecast 21.5 cm

Min point movement: 15.7 / 13.0 px (actual / forecast)

OpenSingleDoor / demo_2533 / s224 / L

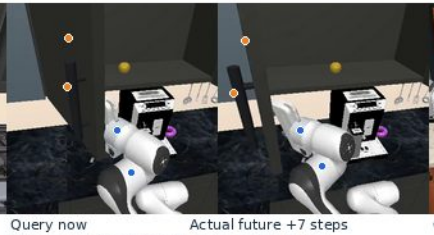


Query now Actual future +19 steps

Object: actual 18.6 cm; forecast 22.6 cm

Min point movement: 7.0 / 9.8 px (actual / forecast)

OpenDoubleDoor / demo_1176 / s864 / L



Query now Actual future +7 steps

Object: actual 10.4 cm; forecast 24.3 cm

Min point movement: 13.7 / 35.9 px (actual / forecast)

CloseDoubleDoor / demo_1617 / s192 / L

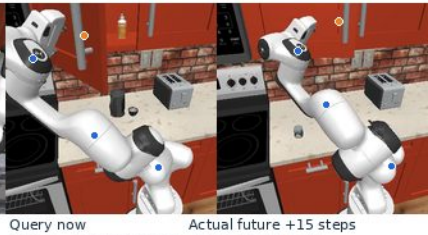


Query now Actual future +22 steps

Object: actual 25.7 cm; forecast 25.8 cm

Min point movement: 5.0 / 3.7 px (actual / forecast)

CloseDoubleDoor / demo_279 / s352 / L



Query now Actual future +15 steps

Object: actual 41.4 cm; forecast 43.1 cm

Min point movement: 15.6 / 21.0 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDoubleDoor / demo_2220 / s352 / L



Query now Actual future +26 steps

Object: actual 41.0 cm; forecast 45.9 cm

Min point movement: 27.5 / 27.5 px (actual / forecast)

CloseDoubleDoor / demo_876 / s416 / L



Query now Actual future +32 steps

Object: actual 25.6 cm; forecast 11.7 cm

Min point movement: 25.2 / 15.1 px (actual / forecast)

CloseDoubleDoor / demo_1617 / s160 / R



Query now Actual future +32 steps

Object: actual 20.6 cm; forecast 28.5 cm

Min point movement: 24.1 / 29.4 px (actual / forecast)

CloseSingleDoor / demo_288 / s208 / L



Query now Actual future +27 steps

Object: actual 40.0 cm; forecast 39.4 cm

Min point movement: 21.5 / 21.7 px (actual / forecast)

OpenSingleDoor / demo_255 / s256 / L



Query now Actual future +17 steps

Object: actual 15.9 cm; forecast 12.8 cm

Min point movement: 9.9 / 8.2 px (actual / forecast)

OpenSingleDoor / demo_2760 / s160 / R



Query now Actual future +32 steps

Object: actual 19.1 cm; forecast 17.9 cm

Min point movement: 16.5 / 18.6 px (actual / forecast)

CloseSingleDoor / demo_1789 / s208 / L

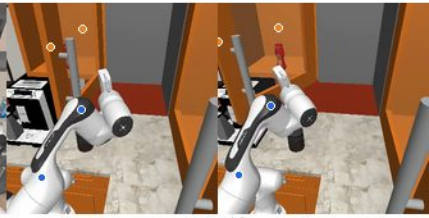


Query now Actual future +32 steps

Object: actual 27.4 cm; forecast 22.3 cm

Min point movement: 8.8 / 4.5 px (actual / forecast)

CloseSingleDoor / demo_2462 / s160 / R



Query now Actual future +16 steps

Object: actual 46.4 cm; forecast 43.2 cm

Min point movement: 15.6 / 13.7 px (actual / forecast)

OpenSingleDoor / demo_641 / s144 / R

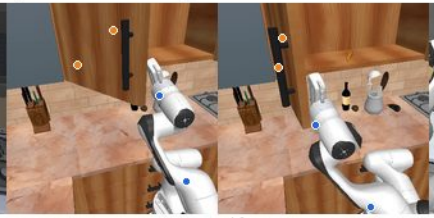


Query now Actual future +32 steps

Object: actual 24.1 cm; forecast 21.9 cm

Min point movement: 10.0 / 8.9 px (actual / forecast)

OpenDoubleDoor / demo_1013 / s752 / L



Query now Actual future +21 steps

Object: actual 16.9 cm; forecast 15.5 cm

Min point movement: 6.5 / 4.5 px (actual / forecast)

OpenDoubleDoor / demo_1225 / s368 / R



Query now Actual future +32 steps

Object: actual 18.1 cm; forecast 17.1 cm

Min point movement: 11.2 / 11.5 px (actual / forecast)

OpenSingleDoor / demo_2989 / s128 / R

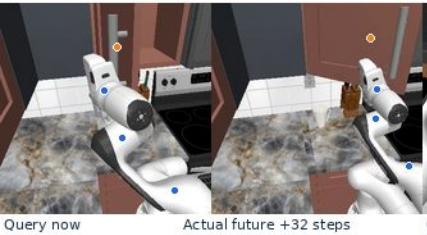


Query now Actual future +32 steps

Object: actual 22.3 cm; forecast 21.2 cm

Min point movement: 25.9 / 23.8 px (actual / forecast)

CloseSingleDoor / demo_64 / s192 / L

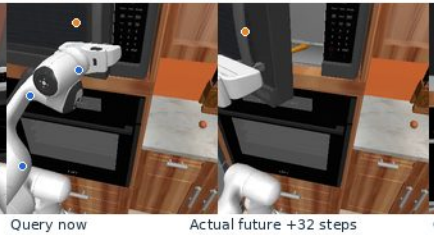


Query now Actual future +32 steps

Object: actual 22.1 cm; forecast 21.3 cm

Min point movement: 20.4 / 25.4 px (actual / forecast)

OpenSingleDoor / demo_156 / s224 / R



Query now Actual future +32 steps

Object: actual 18.5 cm; forecast 21.2 cm

Min point movement: 16.2 / 12.4 px (actual / forecast)

CloseSingleDoor / demo_2758 / s192 / L

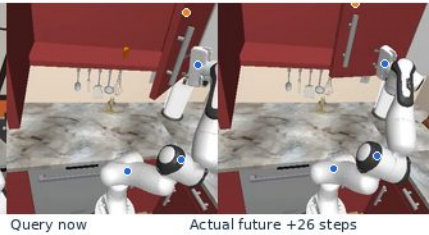


Query now Actual future +32 steps

Object: actual 31.9 cm; forecast 33.0 cm

Min point movement: 26.1 / 19.3 px (actual / forecast)

CloseDoubleDoor / demo_1683 / s192 / L



Query now Actual future +26 steps

Object: actual 29.6 cm; forecast 26.8 cm

Min point movement: 3.8 / 3.6 px (actual / forecast)

OpenSingleDoor / demo_959 / s176 / L

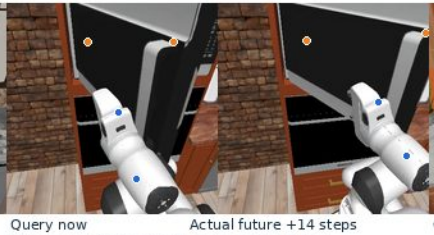


Query now Actual future +23 steps

Object: actual 21.8 cm; forecast 28.4 cm

Min point movement: 3.3 / 6.0 px (actual / forecast)

CloseSingleDoor / demo_609 / s208 / L

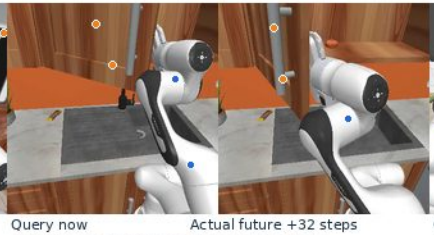


Query now Actual future +14 steps

Object: actual 33.9 cm; forecast 33.2 cm

Min point movement: 10.6 / 12.2 px (actual / forecast)

OpenDoubleDoor / demo_562 / s960 / L



Query now Actual future +32 steps

Object: actual 18.2 cm; forecast 13.8 cm

Min point movement: 20.2 / 14.2 px (actual / forecast)

OpenSingleDoor / demo_1845 / s192 / R



Query now Actual future +32 steps

Object: actual 21.3 cm; forecast 22.7 cm

Min point movement: 22.9 / 15.6 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseSingleDoor / demo_129 / s208 / L

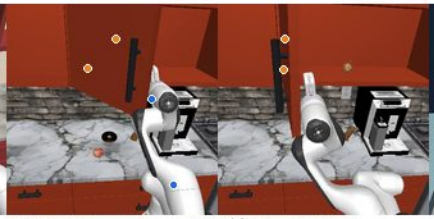


Query now Actual future +28 steps

Object: actual 44.4 cm; forecast 45.1 cm

Min point movement: 9.6 / 9.3 px (actual / forecast)

OpenDoubleDoor / demo_737 / s784 / L

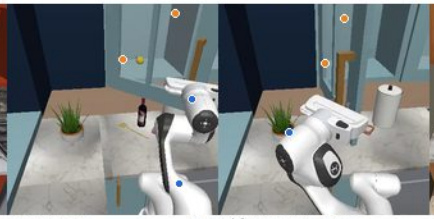


Query now Actual future +26 steps

Object: actual 17.6 cm; forecast 21.2 cm

Min point movement: 9.3 / 13.1 px (actual / forecast)

OpenSingleDoor / demo_191 / s160 / L



Query now Actual future +25 steps

Object: actual 21.4 cm; forecast 21.6 cm

Min point movement: 5.2 / 5.6 px (actual / forecast)

OpenDrawer / demo_2006 / s208 / L

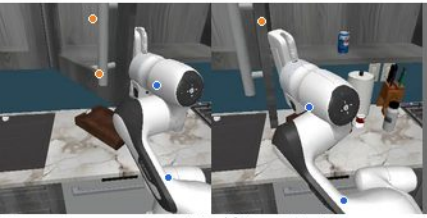


Query now Actual future +31 steps

Object: actual 26.8 cm; forecast 23.4 cm

Min point movement: 4.0 / 4.6 px (actual / forecast)

OpenDoubleDoor / demo_296 / s912 / L

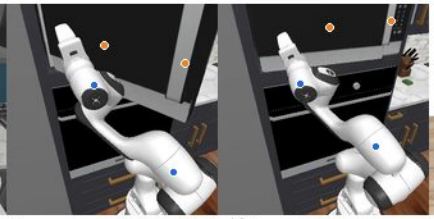


Query now Actual future +15 steps

Object: actual 17.1 cm; forecast 31.4 cm

Min point movement: 21.6 / 48.6 px (actual / forecast)

CloseSingleDoor / demo_1932 / s192 / L



Query now Actual future +28 steps

Object: actual 51.1 cm; forecast 29.5 cm

Min point movement: 12.8 / 10.4 px (actual / forecast)

CloseSingleDoor / demo_2591 / s176 / L



Query now Actual future +32 steps

Object: actual 40.9 cm; forecast 45.3 cm

Min point movement: 6.4 / 6.7 px (actual / forecast)

OpenSingleDoor / demo_83 / s256 / L



Query now Actual future +8 steps

Object: actual 8.8 cm; forecast 15.6 cm

Min point movement: 6.9 / 12.6 px (actual / forecast)

OpenSingleDoor / demo_967 / s272 / L

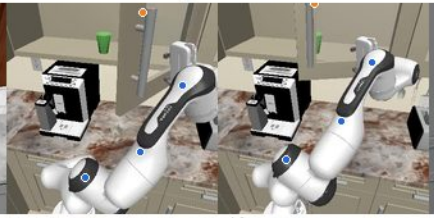


Query now Actual future +27 steps

Object: actual 20.0 cm; forecast 15.9 cm

Min point movement: 8.2 / 6.7 px (actual / forecast)

CloseDoubleDoor / demo_203 / s192 / R

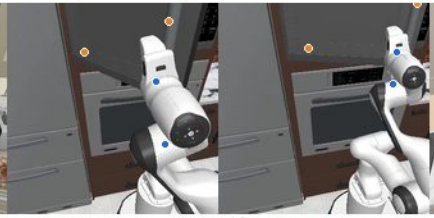


Query now Actual future +9 steps

Object: actual 40.2 cm; forecast 41.2 cm

Min point movement: 25.2 / 24.7 px (actual / forecast)

CloseSingleDoor / demo_1859 / s224 / L

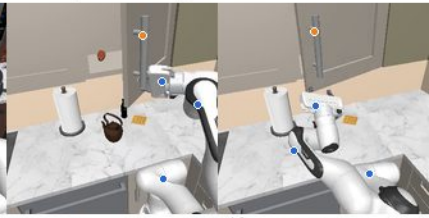


Query now Actual future +30 steps

Object: actual 33.9 cm; forecast 38.3 cm

Min point movement: 8.9 / 9.0 px (actual / forecast)

CloseSingleDoor / demo_2186 / s192 / L



Query now Actual future +32 steps

Object: actual 17.5 cm; forecast 24.3 cm

Min point movement: 4.7 / 6.1 px (actual / forecast)

OpenSingleDoor / demo_1623 / s160 / R



Query now Actual future +32 steps

Object: actual 21.6 cm; forecast 23.7 cm

Min point movement: 20.7 / 21.6 px (actual / forecast)

CloseDoubleDoor / demo_1606 / s352 / L

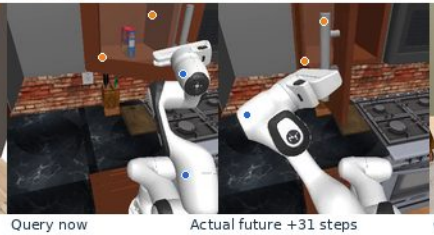


Query now Actual future +32 steps

Object: actual 26.8 cm; forecast 36.9 cm

Min point movement: 24.5 / 20.8 px (actual / forecast)

OpenSingleDoor / demo_1419 / s176 / L

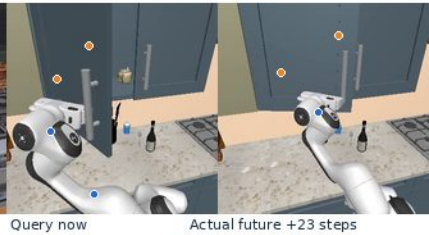


Query now Actual future +31 steps

Object: actual 24.4 cm; forecast 21.2 cm

Min point movement: 5.9 / 5.3 px (actual / forecast)

CloseDoubleDoor / demo_2470 / s352 / L



Query now Actual future +23 steps

Object: actual 41.4 cm; forecast 40.0 cm

Min point movement: 9.2 / 8.4 px (actual / forecast)

CloseDoubleDoor / demo_2316 / s368 / L

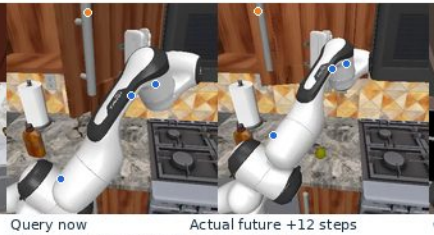


Query now Actual future +25 steps

Object: actual 35.5 cm; forecast 36.9 cm

Min point movement: 24.4 / 24.5 px (actual / forecast)

CloseDoubleDoor / demo_1399 / s192 / R

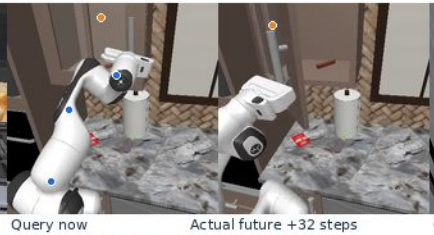


Query now Actual future +12 steps

Object: actual 42.0 cm; forecast 42.6 cm

Min point movement: 24.2 / 24.9 px (actual / forecast)

OpenSingleDoor / demo_959 / s144 / R

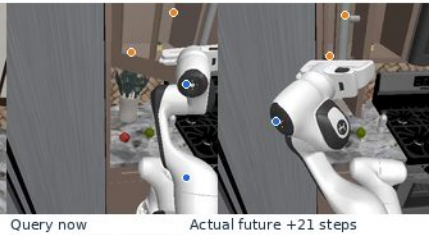


Query now Actual future +32 steps

Object: actual 21.7 cm; forecast 19.1 cm

Min point movement: 18.8 / 18.6 px (actual / forecast)

OpenSingleDoor / demo_1427 / s208 / L



Query now Actual future +21 steps

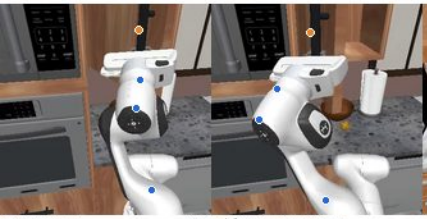
Object: actual 19.9 cm; forecast 24.8 cm

Min point movement: 8.6 / 10.3 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

OpenSingleDoor / demo_498 / s224 / L

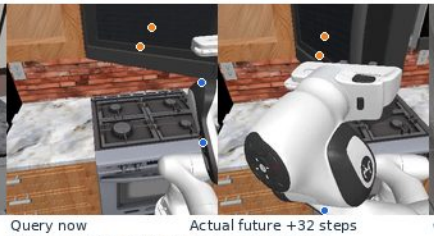


Query now Actual future +9 steps

Object: actual 11.4 cm; forecast 20.3 cm

Min point movement: 23.3 / 44.8 px (actual / forecast)

OpenSingleDoor / demo_2711 / s208 / L



Query now Actual future +32 steps

Object: actual 19.0 cm; forecast 17.3 cm

Min point movement: 20.2 / 18.5 px (actual / forecast)

CloseSingleDoor / demo_24 / s240 / L



Query now Actual future +25 steps

Object: actual 40.3 cm; forecast 35.2 cm

Min point movement: 8.8 / 8.8 px (actual / forecast)

CloseSingleDoor / demo_1680 / s208 / L

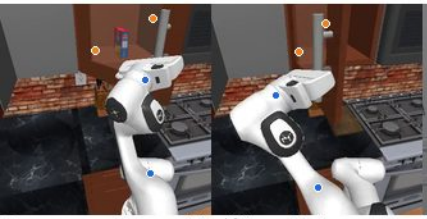


Query now Actual future +26 steps

Object: actual 44.8 cm; forecast 42.8 cm

Min point movement: 9.8 / 11.3 px (actual / forecast)

OpenSingleDoor / demo_1419 / s192 / L



Query now Actual future +15 steps

Object: actual 16.4 cm; forecast 23.3 cm

Min point movement: 4.7 / 7.2 px (actual / forecast)

CloseSingleDoor / demo_258 / s224 / L



Query now Actual future +15 steps

Object: actual 42.3 cm; forecast 46.3 cm

Min point movement: 12.5 / 10.4 px (actual / forecast)

OpenDoubleDoor / demo_562 / s304 / R

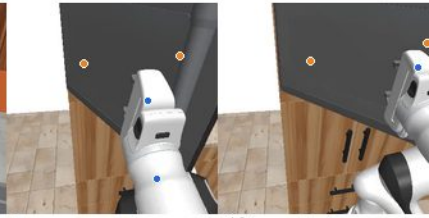


Query now Actual future +32 steps

Object: actual 20.7 cm; forecast 25.7 cm

Min point movement: 8.2 / 8.4 px (actual / forecast)

CloseSingleDoor / demo_103 / s224 / L



Query now Actual future +20 steps

Object: actual 29.4 cm; forecast 27.5 cm

Min point movement: 15.1 / 12.6 px (actual / forecast)

OpenDrawer / demo_1032 / s176 / R

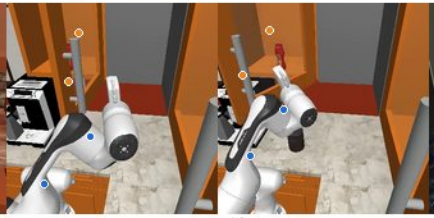


Query now Actual future +26 steps

Object: actual 29.4 cm; forecast 30.5 cm

Min point movement: 5.0 / 4.8 px (actual / forecast)

CloseSingleDoor / demo_2462 / s144 / R

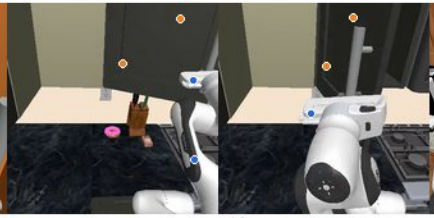


Query now Actual future +32 steps

Object: actual 16.0 cm; forecast 19.0 cm

Min point movement: 12.9 / 13.0 px (actual / forecast)

OpenSingleDoor / demo_2533 / s208 / L



Query now Actual future +32 steps

Object: actual 25.5 cm; forecast 21.7 cm

Min point movement: 5.1 / 4.6 px (actual / forecast)

OpenDoubleDoor / demo_456 / s896 / R

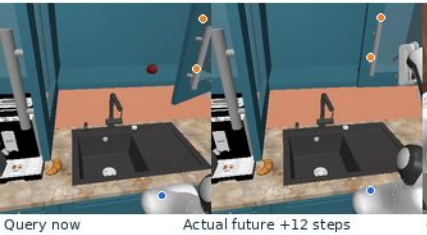


Query now Actual future +32 steps

Object: actual 16.5 cm; forecast 12.6 cm

Min point movement: 20.0 / 16.8 px (actual / forecast)

CloseDoubleDoor / demo_1112 / s192 / L

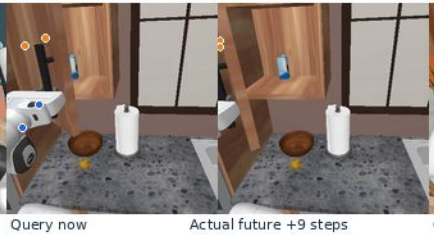


Query now Actual future +12 steps

Object: actual 21.2 cm; forecast 24.2 cm

Min point movement: 4.4 / 4.6 px (actual / forecast)

OpenSingleDoor / demo_498 / s208 / R

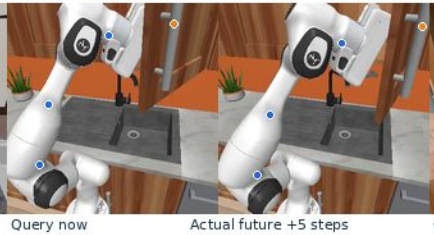


Query now Actual future +9 steps

Object: actual 29.8 cm; forecast 32.4 cm

Min point movement: 8.6 / 7.8 px (actual / forecast)

OpenDoubleDoor / demo_562 / s336 / R

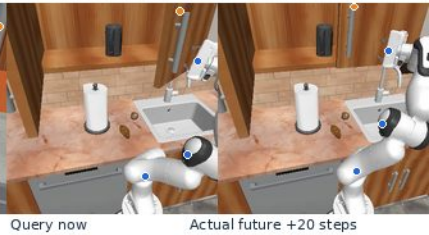


Query now Actual future +5 steps

Object: actual 16.0 cm; forecast 20.4 cm

Min point movement: 12.8 / 11.8 px (actual / forecast)

CloseDoubleDoor / demo_1499 / s192 / L



Query now Actual future +20 steps

Object: actual 24.0 cm; forecast 21.4 cm

Min point movement: 4.0 / 4.9 px (actual / forecast)

OpenDrawer / demo_2308 / s160 / L

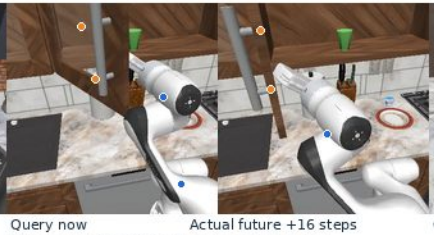


Query now Actual future +32 steps

Object: actual 25.2 cm; forecast 24.5 cm

Min point movement: 4.5 / 4.7 px (actual / forecast)

OpenDoubleDoor / demo_1262 / s752 / L

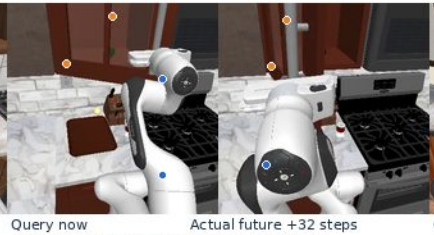


Query now Actual future +16 steps

Object: actual 12.6 cm; forecast 13.1 cm

Min point movement: 19.5 / 17.7 px (actual / forecast)

OpenSingleDoor / demo_2760 / s176 / L



Query now Actual future +32 steps

Object: actual 19.7 cm; forecast 19.2 cm

Min point movement: 3.9 / 4.7 px (actual / forecast)

CloseSingleDoor / demo_1387 / s176 / R



Query now Actual future +20 steps

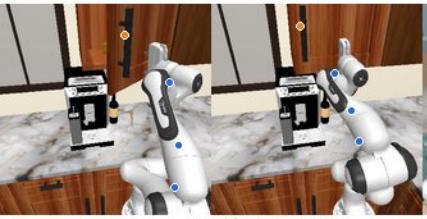
Object: actual 45.9 cm; forecast 35.8 cm

Min point movement: 17.0 / 15.2 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseSingleDoor / demo_1387 / s192 / L

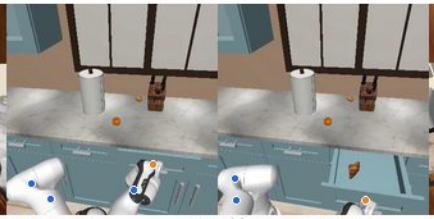


Query now Actual future +19 steps

Object: actual 25.2 cm; forecast 26.3 cm

Min point movement: 22.6 / 23.5 px (actual / forecast)

OpenDrawer / demo_1299 / s160 / R

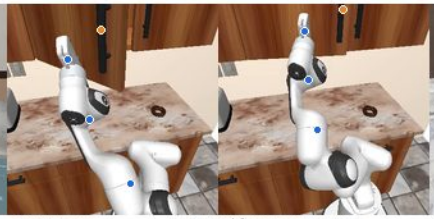


Query now Actual future +29 steps

Object: actual 24.8 cm; forecast 27.8 cm

Min point movement: 5.5 / 4.3 px (actual / forecast)

CloseDoubleDoor / demo_2731 / s368 / L



Query now Actual future +27 steps

Object: actual 34.8 cm; forecast 40.3 cm

Min point movement: 22.3 / 21.3 px (actual / forecast)

OpenSingleDoor / demo_2760 / s192 / L

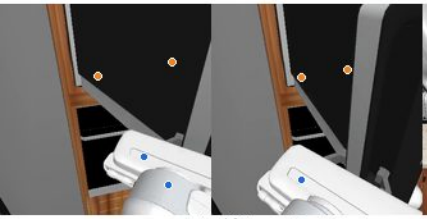


Query now Actual future +22 steps

Object: actual 13.5 cm; forecast 13.0 cm

Min point movement: 19.2 / 17.6 px (actual / forecast)

OpenSingleDoor / demo_1033 / s208 / L

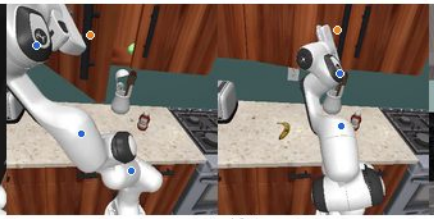


Query now Actual future +13 steps

Object: actual 7.5 cm; forecast 9.0 cm

Min point movement: 4.2 / 5.1 px (actual / forecast)

CloseDoubleDoor / demo_1617 / s368 / L

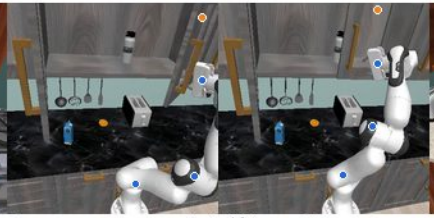


Query now Actual future +32 steps

Object: actual 29.2 cm; forecast 40.2 cm

Min point movement: 8.0 / 21.2 px (actual / forecast)

CloseDoubleDoor / demo_1729 / s176 / L



Query now Actual future +24 steps

Object: actual 22.7 cm; forecast 23.7 cm

Min point movement: 7.1 / 4.3 px (actual / forecast)

CloseDoubleDoor / demo_1617 / s176 / R



Query now Actual future +26 steps

Object: actual 35.7 cm; forecast 41.9 cm

Min point movement: 12.7 / 12.8 px (actual / forecast)

CloseDoubleDoor / demo_1729 / s336 / L

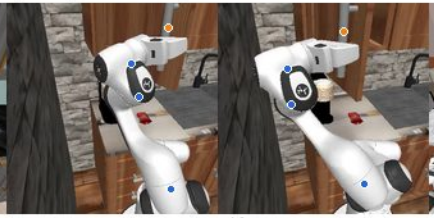


Query now Actual future +27 steps

Object: actual 34.8 cm; forecast 33.4 cm

Min point movement: 21.8 / 20.5 px (actual / forecast)

OpenSingleDoor / demo_1467 / s256 / L



Query now Actual future +10 steps

Object: actual 12.7 cm; forecast 19.7 cm

Min point movement: 11.4 / 23.1 px (actual / forecast)

OpenDrawer / demo_2398 / s160 / R



Query now Actual future +28 steps

Object: actual 25.3 cm; forecast 28.5 cm

Min point movement: 5.8 / 7.0 px (actual / forecast)

CloseDoubleDoor / demo_1717 / s320 / L

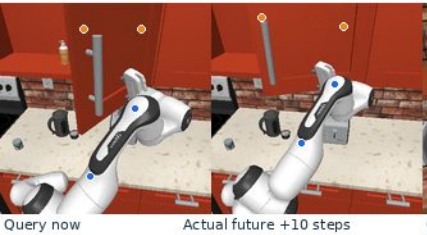


Query now Actual future +27 steps

Object: actual 37.1 cm; forecast 41.7 cm

Min point movement: 21.8 / 24.6 px (actual / forecast)

CloseDoubleDoor / demo_279 / s176 / R

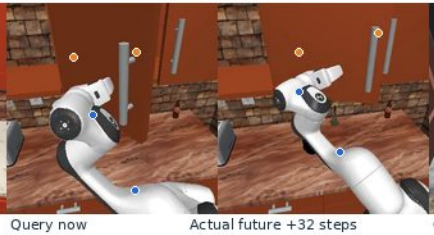


Query now Actual future +10 steps

Object: actual 43.8 cm; forecast 44.8 cm

Min point movement: 6.5 / 7.3 px (actual / forecast)

CloseDoubleDoor / demo_1930 / s352 / L

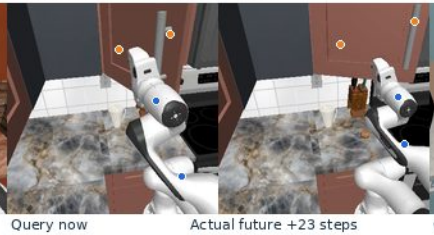


Query now Actual future +32 steps

Object: actual 26.1 cm; forecast 44.3 cm

Min point movement: 8.7 / 8.0 px (actual / forecast)

CloseSingleDoor / demo_64 / s208 / L

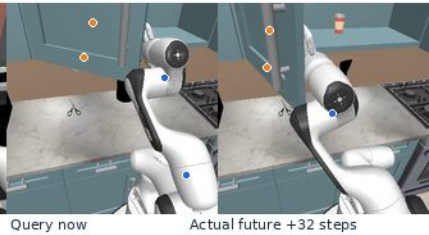


Query now Actual future +23 steps

Object: actual 39.9 cm; forecast 38.0 cm

Min point movement: 7.3 / 7.1 px (actual / forecast)

OpenDoubleDoor / demo_1189 / s832 / L

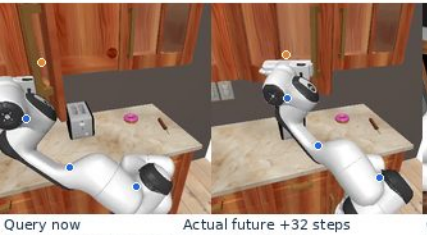


Query now Actual future +32 steps

Object: actual 15.3 cm; forecast 10.1 cm

Min point movement: 18.2 / 8.7 px (actual / forecast)

CloseDoubleDoor / demo_1606 / s336 / L

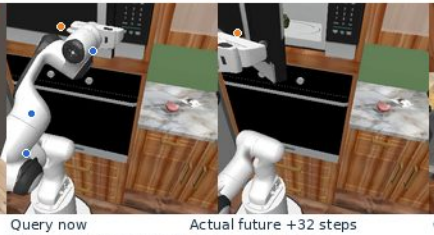


Query now Actual future +32 steps

Object: actual 16.4 cm; forecast 14.6 cm

Min point movement: 19.9 / 15.8 px (actual / forecast)

OpenSingleDoor / demo_1033 / s144 / R



Query now Actual future +32 steps

Object: actual 17.1 cm; forecast 20.0 cm

Min point movement: 19.6 / 17.6 px (actual / forecast)

CloseDoubleDoor / demo_1399 / s368 / L



Query now Actual future +21 steps

Object: actual 37.4 cm; forecast 40.5 cm

Min point movement: 21.1 / 21.5 px (actual / forecast)

CloseSingleDoor / demo_1647 / s224 / L



Query now Actual future +32 steps

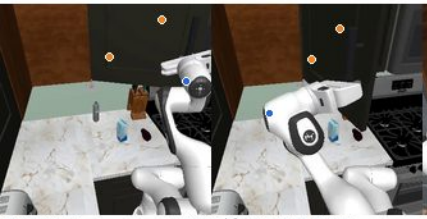
Object: actual 45.2 cm; forecast 47.1 cm

Min point movement: 21.1 / 17.6 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

OpenSingleDoor / demo_271 / s176 / L



Query now Actual future +32 steps

Object: actual 25.7 cm; forecast 25.4 cm

Min point movement: 5.7 / 5.3 px (actual / forecast)

OpenSingleDoor / demo_468 / s192 / R



Query now Actual future +32 steps

Object: actual 17.8 cm; forecast 19.0 cm

Min point movement: 20.9 / 22.1 px (actual / forecast)

OpenSingleDoor / demo_32 / s176 / R



Query now Actual future +32 steps

Object: actual 18.0 cm; forecast 21.5 cm

Min point movement: 18.2 / 18.3 px (actual / forecast)

CloseSingleDoor / demo_2758 / s224 / R

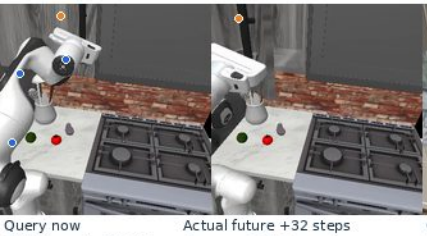


Query now Actual future +14 steps

Object: actual 21.5 cm; forecast 24.5 cm

Min point movement: 16.1 / 17.8 px (actual / forecast)

OpenSingleDoor / demo_122 / s176 / R

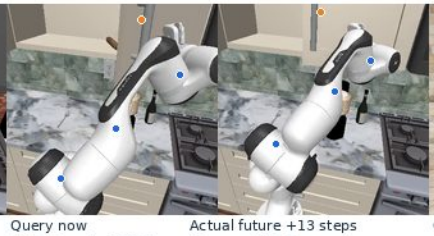


Query now Actual future +32 steps

Object: actual 18.7 cm; forecast 25.0 cm

Min point movement: 20.7 / 15.7 px (actual / forecast)

CloseDoubleDoor / demo_2121 / s176 / R

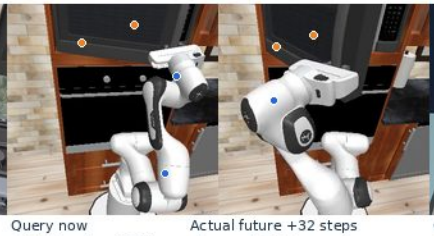


Query now Actual future +13 steps

Object: actual 37.5 cm; forecast 40.3 cm

Min point movement: 20.6 / 21.8 px (actual / forecast)

OpenSingleDoor / demo_2753 / s160 / L

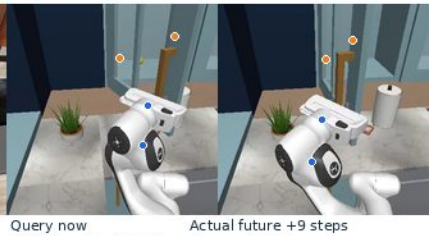


Query now Actual future +32 steps

Object: actual 24.4 cm; forecast 24.1 cm

Min point movement: 9.8 / 8.5 px (actual / forecast)

OpenSingleDoor / demo_191 / s176 / L



Query now Actual future +9 steps

Object: actual 11.2 cm; forecast 22.3 cm

Min point movement: 3.3 / 7.3 px (actual / forecast)

CloseDoubleDoor / demo_2128 / s400 / L



Query now Actual future +22 steps

Object: actual 41.9 cm; forecast 42.6 cm

Min point movement: 10.5 / 9.2 px (actual / forecast)

CloseDoubleDoor / demo_2963 / s192 / L

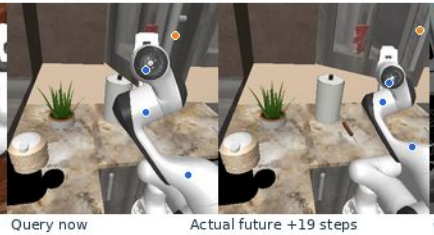


Query now Actual future +21 steps

Object: actual 20.9 cm; forecast 20.7 cm

Min point movement: 4.0 / 3.0 px (actual / forecast)

CloseSingleDoor / demo_1130 / s192 / L

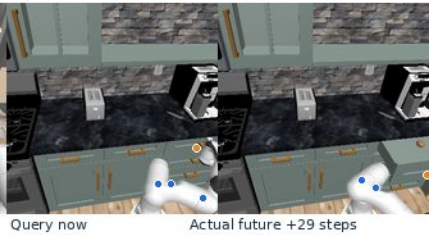


Query now Actual future +19 steps

Object: actual 44.0 cm; forecast 42.3 cm

Min point movement: 20.0 / 17.5 px (actual / forecast)

OpenDrawer / demo_2805 / s176 / L



Query now Actual future +29 steps

Object: actual 29.4 cm; forecast 28.3 cm

Min point movement: 4.9 / 5.6 px (actual / forecast)

OpenDrawer / demo_1684 / s160 / L

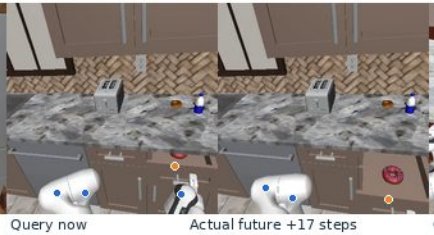


Query now Actual future +26 steps

Object: actual 28.7 cm; forecast 26.1 cm

Min point movement: 4.8 / 3.6 px (actual / forecast)

OpenDrawer / demo_886 / s192 / R



Query now Actual future +17 steps

Object: actual 18.8 cm; forecast 23.9 cm

Min point movement: 3.7 / 4.7 px (actual / forecast)

OpenDrawer / demo_151 / s176 / R



Query now Actual future +32 steps

Object: actual 24.5 cm; forecast 25.1 cm

Min point movement: 4.5 / 5.0 px (actual / forecast)

OpenSingleDoor / demo_935 / s176 / R

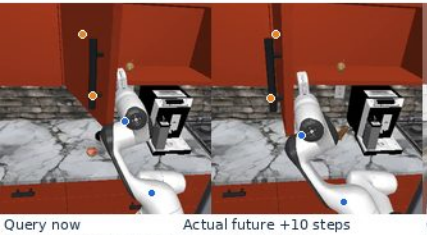


Query now Actual future +32 steps

Object: actual 20.4 cm; forecast 20.4 cm

Min point movement: 4.0 / 9.5 px (actual / forecast)

OpenDoubleDoor / demo_737 / s800 / L

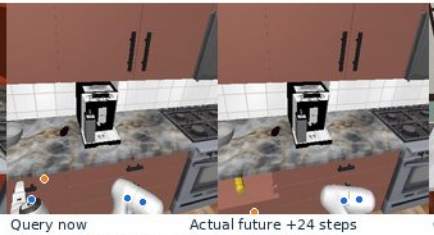


Query now Actual future +10 steps

Object: actual 11.0 cm; forecast 21.0 cm

Min point movement: 10.5 / 19.8 px (actual / forecast)

OpenDrawer / demo_2411 / s208 / L

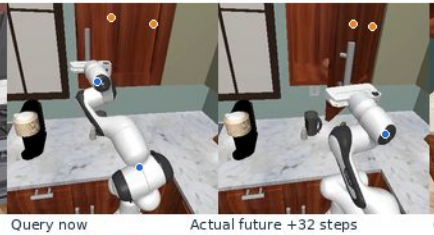


Query now Actual future +24 steps

Object: actual 30.6 cm; forecast 27.6 cm

Min point movement: 4.2 / 3.9 px (actual / forecast)

OpenSingleDoor / demo_255 / s224 / L

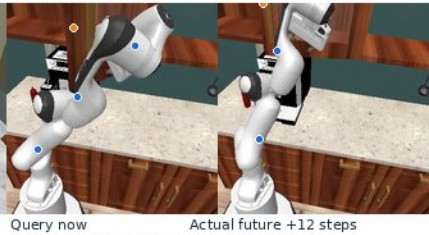


Query now Actual future +32 steps

Object: actual 24.2 cm; forecast 22.9 cm

Min point movement: 4.6 / 5.8 px (actual / forecast)

CloseDoubleDoor / demo_341 / s160 / R



Query now Actual future +12 steps

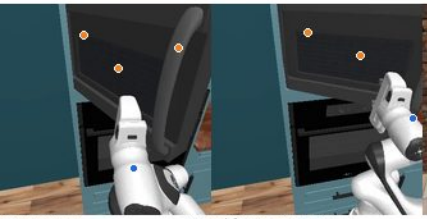
Object: actual 44.5 cm; forecast 42.4 cm

Min point movement: 11.3 / 8.8 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseSingleDoor / demo_1838 / s224 / L

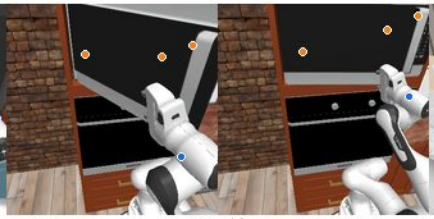


Query now Actual future +32 steps

Object: actual 29.8 cm; forecast 30.5 cm

Min point movement: 8.0 / 9.6 px (actual / forecast)

CloseSingleDoor / demo_609 / s224 / L



Query now Actual future +32 steps

Object: actual 40.5 cm; forecast 39.2 cm

Min point movement: 4.3 / 4.1 px (actual / forecast)

CloseSingleDoor / demo_2186 / s144 / R



Query now Actual future +32 steps

Object: actual 16.0 cm; forecast 11.2 cm

Min point movement: 19.4 / 15.8 px (actual / forecast)

OpenDoubleDoor / demo_1130 / s704 / L



Query now Actual future +22 steps

Object: actual 13.5 cm; forecast 19.7 cm

Min point movement: 13.5 / 22.3 px (actual / forecast)

OpenDrawer / demo_2814 / s208 / L

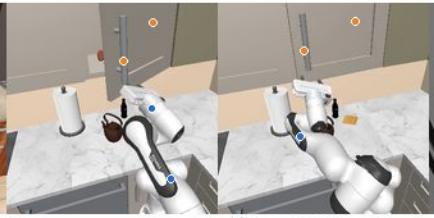


Query now Actual future +32 steps

Object: actual 23.3 cm; forecast 19.2 cm

Min point movement: 4.3 / 4.0 px (actual / forecast)

CloseSingleDoor / demo_2186 / s208 / L

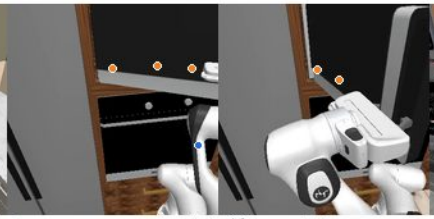


Query now Actual future +29 steps

Object: actual 20.3 cm; forecast 21.2 cm

Min point movement: 5.9 / 6.7 px (actual / forecast)

OpenSingleDoor / demo_1105 / s192 / L

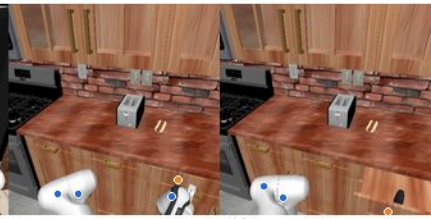


Query now Actual future +32 steps

Object: actual 26.4 cm; forecast 27.1 cm

Min point movement: 3.8 / 4.3 px (actual / forecast)

OpenDrawer / demo_559 / s208 / R



Query now Actual future +22 steps

Object: actual 25.9 cm; forecast 28.5 cm

Min point movement: 4.9 / 6.1 px (actual / forecast)

CloseSingleDoor / demo_695 / s208 / L

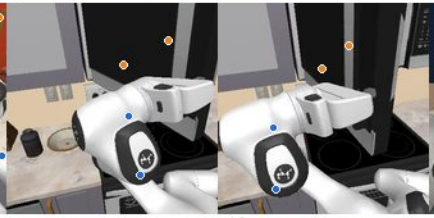


Query now Actual future +22 steps

Object: actual 37.8 cm; forecast 34.3 cm

Min point movement: 19.2 / 17.7 px (actual / forecast)

OpenSingleDoor / demo_935 / s240 / L

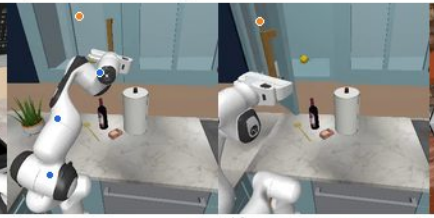


Query now Actual future +10 steps

Object: actual 8.9 cm; forecast 14.6 cm

Min point movement: 8.2 / 13.5 px (actual / forecast)

OpenSingleDoor / demo_191 / s128 / R

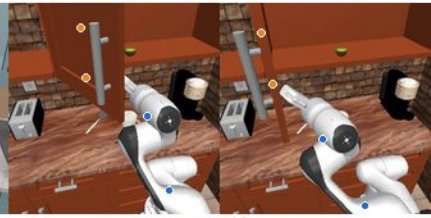


Query now Actual future +32 steps

Object: actual 20.4 cm; forecast 17.3 cm

Min point movement: 19.1 / 15.2 px (actual / forecast)

OpenDoubleDoor / demo_1088 / s736 / L

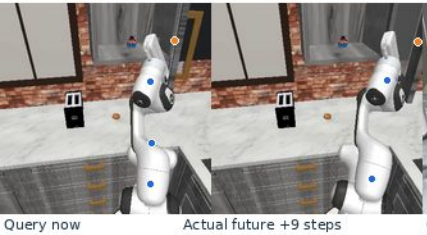


Query now Actual future +13 steps

Object: actual 10.2 cm; forecast 13.0 cm

Min point movement: 13.2 / 14.8 px (actual / forecast)

OpenSingleDoor / demo_770 / s432 / L

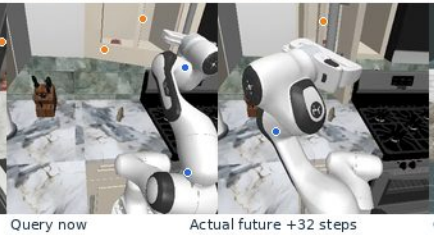


Query now Actual future +9 steps

Object: actual 13.2 cm; forecast 26.7 cm

Min point movement: 5.4 / 19.5 px (actual / forecast)

OpenSingleDoor / demo_1227 / s160 / L

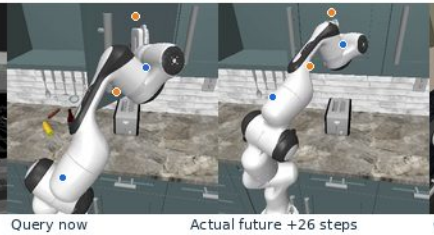


Query now Actual future +32 steps

Object: actual 23.3 cm; forecast 20.6 cm

Min point movement: 5.0 / 5.7 px (actual / forecast)

CloseDoubleDoor / demo_254 / s160 / R

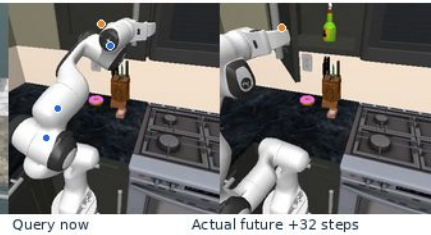


Query now Actual future +26 steps

Object: actual 33.1 cm; forecast 34.2 cm

Min point movement: 10.5 / 11.5 px (actual / forecast)

OpenSingleDoor / demo_2533 / s192 / R

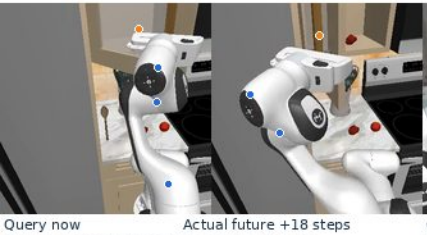


Query now Actual future +32 steps

Object: actual 18.2 cm; forecast 18.7 cm

Min point movement: 18.7 / 19.3 px (actual / forecast)

OpenSingleDoor / demo_1623 / s192 / L

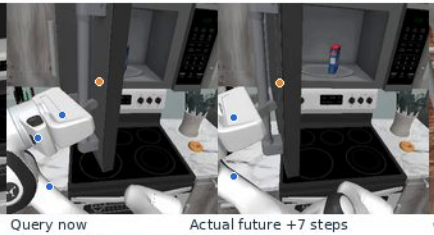


Query now Actual future +18 steps

Object: actual 13.7 cm; forecast 15.5 cm

Min point movement: 18.7 / 22.1 px (actual / forecast)

OpenSingleDoor / demo_32 / s240 / L



Query now Actual future +7 steps

Object: actual 6.6 cm; forecast 10.1 cm

Min point movement: 17.1 / 18.1 px (actual / forecast)

CloseDoubleDoor / demo_1995 / s176 / L

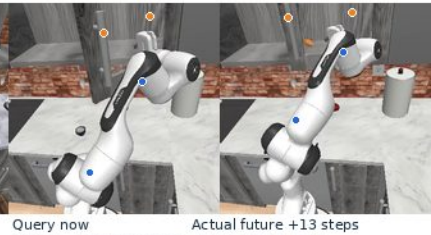


Query now Actual future +14 steps

Object: actual 19.4 cm; forecast 18.7 cm

Min point movement: 6.6 / 7.8 px (actual / forecast)

CloseDoubleDoor / demo_1895 / s176 / R



Query now Actual future +13 steps

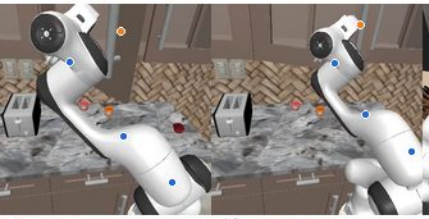
Object: actual 36.4 cm; forecast 36.1 cm

Min point movement: 6.3 / 6.4 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDoubleDoor / demo_2238 / s368 / L

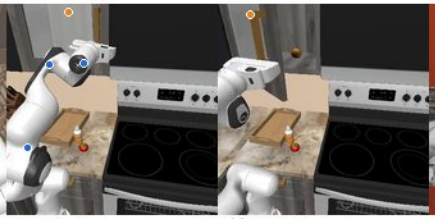


Query now Actual future +21 steps

Object: actual 39.2 cm; forecast 39.3 cm

Min point movement: 18.3 / 18.3 px (actual / forecast)

OpenSingleDoor / demo_843 / s128 / R

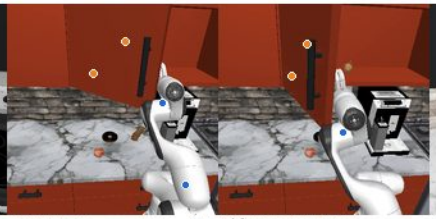


Query now Actual future +32 steps

Object: actual 19.3 cm; forecast 17.8 cm

Min point movement: 18.2 / 16.8 px (actual / forecast)

OpenDoubleDoor / demo_737 / s768 / L

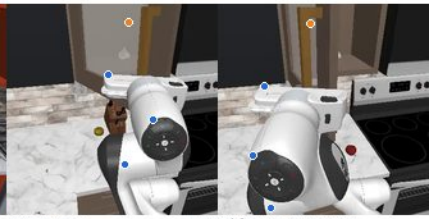


Query now Actual future +32 steps

Object: actual 16.1 cm; forecast 15.4 cm

Min point movement: 7.5 / 7.2 px (actual / forecast)

OpenSingleDoor / demo_2213 / s208 / L



Query now Actual future +14 steps

Object: actual 10.0 cm; forecast 10.9 cm

Min point movement: 18.1 / 20.4 px (actual / forecast)

OpenSingleDoor / demo_1467 / s224 / R

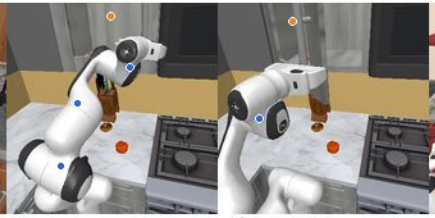


Query now Actual future +32 steps

Object: actual 18.4 cm; forecast 17.1 cm

Min point movement: 18.1 / 15.4 px (actual / forecast)

OpenSingleDoor / demo_1049 / s144 / R



Query now Actual future +32 steps

Object: actual 19.3 cm; forecast 20.0 cm

Min point movement: 18.0 / 19.5 px (actual / forecast)

OpenDrawer / demo_2030 / s208 / R



Query now Actual future +23 steps

Object: actual 20.4 cm; forecast 27.3 cm

Min point movement: 6.1 / 7.7 px (actual / forecast)

OpenSingleDoor / demo_1427 / s192 / L

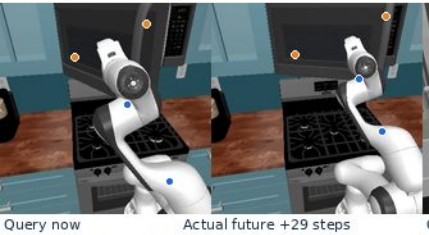


Query now Actual future +32 steps

Object: actual 23.8 cm; forecast 19.5 cm

Min point movement: 4.3 / 11.0 px (actual / forecast)

CloseSingleDoor / demo_2334 / s208 / L



Query now Actual future +29 steps

Object: actual 34.4 cm; forecast 36.3 cm

Min point movement: 6.3 / 5.9 px (actual / forecast)

CloseDoubleDoor / demo_876 / s256 / L



Query now Actual future +16 steps

Object: actual 19.0 cm; forecast 17.0 cm

Min point movement: 11.3 / 8.2 px (actual / forecast)

CloseSingleDoor / demo_2758 / s208 / L

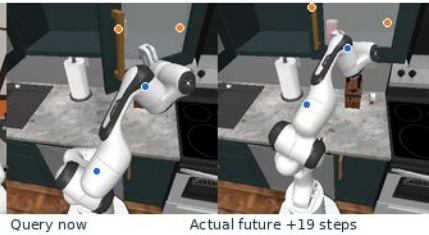


Query now Actual future +18 steps

Object: actual 44.8 cm; forecast 38.5 cm

Min point movement: 7.3 / 7.8 px (actual / forecast)

CloseDoubleDoor / demo_275 / s176 / R

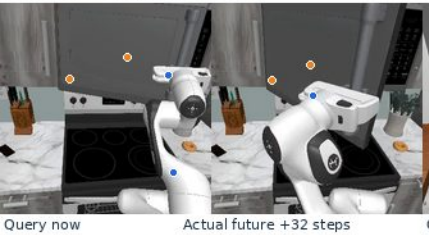


Query now Actual future +19 steps

Object: actual 30.9 cm; forecast 30.9 cm

Min point movement: 4.2 / 5.1 px (actual / forecast)

OpenSingleDoor / demo_32 / s192 / L



Query now Actual future +32 steps

Object: actual 19.6 cm; forecast 20.4 cm

Min point movement: 5.4 / 4.3 px (actual / forecast)

OpenSingleDoor / demo_967 / s256 / L

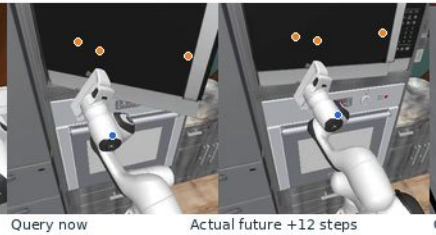


Query now Actual future +32 steps

Object: actual 31.4 cm; forecast 27.8 cm

Min point movement: 4.2 / 3.9 px (actual / forecast)

CloseSingleDoor / demo_2519 / s208 / L

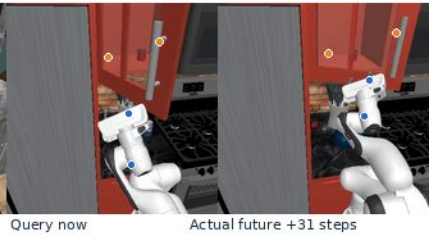


Query now Actual future +12 steps

Object: actual 30.9 cm; forecast 30.3 cm

Min point movement: 5.1 / 5.9 px (actual / forecast)

CloseSingleDoor / demo_2451 / s352 / L

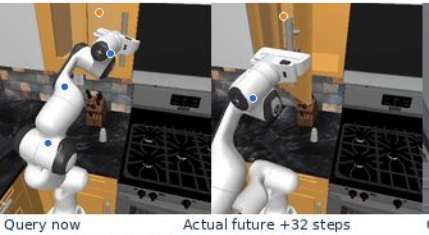


Query now Actual future +31 steps

Object: actual 28.0 cm; forecast 30.5 cm

Min point movement: 5.5 / 3.8 px (actual / forecast)

OpenSingleDoor / demo_1381 / s272 / R



Query now Actual future +32 steps

Object: actual 20.6 cm; forecast 19.8 cm

Min point movement: 16.7 / 17.0 px (actual / forecast)

OpenSingleDoor / demo_122 / s368 / L



Query now Actual future +12 steps

Object: actual 9.7 cm; forecast 15.6 cm

Min point movement: 12.7 / 21.3 px (actual / forecast)

OpenDrawer / demo_2398 / s144 / R



Query now Actual future +32 steps

Object: actual 20.2 cm; forecast 23.6 cm

Min point movement: 5.5 / 4.0 px (actual / forecast)

CloseDoubleDoor / demo_2128 / s336 / L



Query now Actual future +27 steps

Object: actual 8.5 cm; forecast 6.7 cm

Min point movement: 5.6 / 9.0 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

OpenSingleDoor / demo_498 / s192 / L



Query now Actual future +32 steps

Object: actual 21.0 cm; forecast 21.1 cm

Min point movement: 3.7 / 4.0 px (actual / forecast)

CloseDoubleDoor / demo_2872 / s336 / L

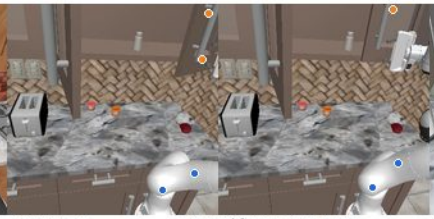


Query now Actual future +16 steps

Object: actual 35.2 cm; forecast 41.6 cm

Min point movement: 5.8 / 6.0 px (actual / forecast)

CloseDoubleDoor / demo_2238 / s192 / L



Query now Actual future +11 steps

Object: actual 16.6 cm; forecast 15.1 cm

Min point movement: 7.6 / 6.0 px (actual / forecast)

OpenSingleDoor / demo_1227 / s176 / L



Query now Actual future +19 steps

Object: actual 12.5 cm; forecast 12.1 cm

Min point movement: 16.3 / 16.0 px (actual / forecast)

OpenSingleDoor / demo_49 / s160 / R

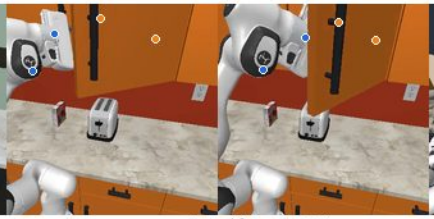


Query now Actual future +32 steps

Object: actual 14.6 cm; forecast 16.2 cm

Min point movement: 16.1 / 17.5 px (actual / forecast)

OpenDoubleDoor / demo_971 / s368 / R



Query now Actual future +32 steps

Object: actual 12.7 cm; forecast 22.3 cm

Min point movement: 6.1 / 12.3 px (actual / forecast)

OpenDrawer / demo_1397 / s192 / R



Query now Actual future +24 steps

Object: actual 26.4 cm; forecast 29.0 cm

Min point movement: 4.5 / 4.7 px (actual / forecast)

CloseSingleDoor / demo_869 / s240 / R



Query now Actual future +20 steps

Object: actual 18.2 cm; forecast 20.4 cm

Min point movement: 5.0 / 5.8 px (actual / forecast)

CloseSingleDoor / demo_1829 / s256 / L



Query now Actual future +28 steps

Object: actual 39.2 cm; forecast 43.2 cm

Min point movement: 6.1 / 6.6 px (actual / forecast)

OpenSingleDoor / demo_468 / s208 / L



Query now Actual future +32 steps

Object: actual 26.2 cm; forecast 25.6 cm

Min point movement: 6.8 / 9.2 px (actual / forecast)

OpenDoubleDoor / demo_456 / s384 / R

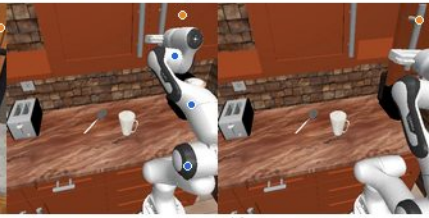


Query now Actual future +5 steps

Object: actual 17.0 cm; forecast 19.1 cm

Min point movement: 15.7 / 15.7 px (actual / forecast)

OpenDoubleDoor / demo_1088 / s160 / L

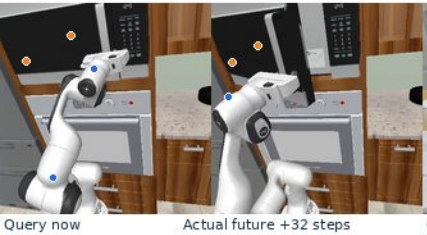


Query now Actual future +32 steps

Object: actual 16.0 cm; forecast 15.6 cm

Min point movement: 15.7 / 16.0 px (actual / forecast)

OpenSingleDoor / demo_1115 / s160 / R

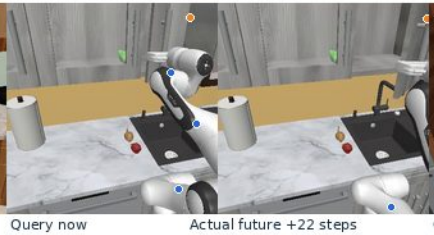


Query now Actual future +32 steps

Object: actual 16.1 cm; forecast 15.9 cm

Min point movement: 3.2 / 3.5 px (actual / forecast)

OpenDoubleDoor / demo_1225 / s224 / L

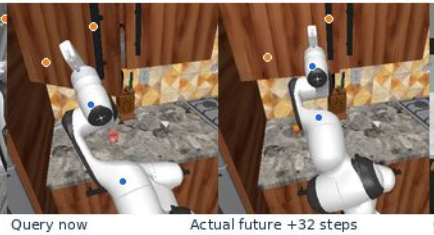


Query now Actual future +22 steps

Object: actual 15.1 cm; forecast 13.7 cm

Min point movement: 11.7 / 11.2 px (actual / forecast)

CloseDoubleDoor / demo_1029 / s416 / L



Query now Actual future +32 steps

Object: actual 25.4 cm; forecast 29.6 cm

Min point movement: 6.2 / 5.1 px (actual / forecast)

CloseSingleDoor / demo_972 / s224 / L

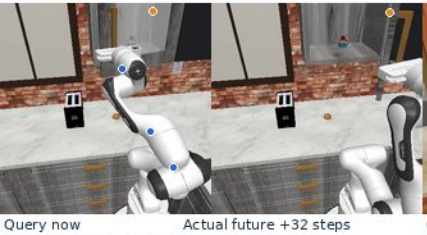


Query now Actual future +32 steps

Object: actual 27.4 cm; forecast 22.9 cm

Min point movement: 13.9 / 7.2 px (actual / forecast)

OpenSingleDoor / demo_770 / s288 / L

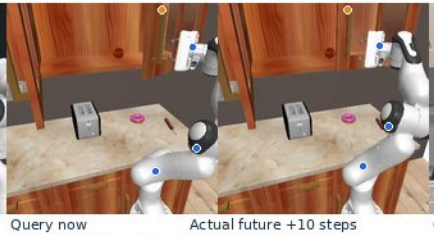


Query now Actual future +32 steps

Object: actual 13.6 cm; forecast 12.5 cm

Min point movement: 15.4 / 13.7 px (actual / forecast)

CloseDoubleDoor / demo_1606 / s176 / L



Query now Actual future +10 steps

Object: actual 17.4 cm; forecast 20.4 cm

Min point movement: 9.5 / 5.7 px (actual / forecast)

CloseDoubleDoor / demo_1175 / s368 / L



Query now Actual future +23 steps

Object: actual 30.1 cm; forecast 30.8 cm

Min point movement: 15.3 / 14.2 px (actual / forecast)

OpenDoubleDoor / demo_1130 / s912 / L



Query now Actual future +32 steps

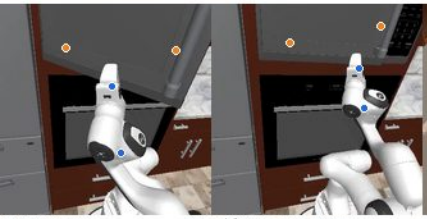
Object: actual 8.5 cm; forecast 4.9 cm

Min point movement: 8.9 / 4.5 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseSingleDoor / demo_821 / s240 / L

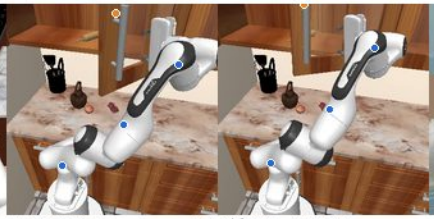


Query now Actual future +22 steps

Object: actual 33.3 cm; forecast 39.3 cm

Min point movement: 8.6 / 6.7 px (actual / forecast)

CloseDoubleDoor / demo_1087 / s192 / R

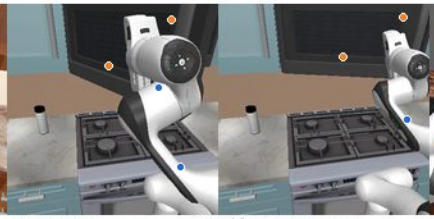


Query now Actual future +7 steps

Object: actual 36.3 cm; forecast 36.3 cm

Min point movement: 8.1 / 10.3 px (actual / forecast)

CloseSingleDoor / demo_535 / s208 / L



Query now Actual future +32 steps

Object: actual 35.3 cm; forecast 35.3 cm

Min point movement: 14.6 / 10.6 px (actual / forecast)

OpenSingleDoor / demo_1419 / s144 / R



Query now Actual future +32 steps

Object: actual 15.1 cm; forecast 14.5 cm

Min point movement: 15.0 / 14.3 px (actual / forecast)

OpenDoubleDoor / demo_1088 / s608 / R

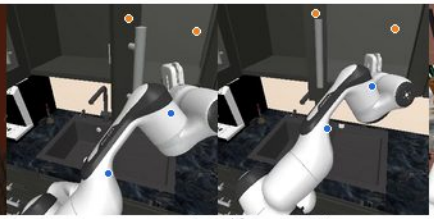


Query now Actual future +32 steps

Object: actual 13.9 cm; forecast 11.9 cm

Min point movement: 15.0 / 12.2 px (actual / forecast)

CloseDoubleDoor / demo_1584 / s192 / R



Query now Actual future +19 steps

Object: actual 38.3 cm; forecast 29.7 cm

Min point movement: 8.6 / 9.5 px (actual / forecast)

CloseSingleDoor / demo_2151 / s224 / R



Query now Actual future +10 steps

Object: actual 16.7 cm; forecast 22.2 cm

Min point movement: 5.1 / 6.4 px (actual / forecast)

CloseDoubleDoor / demo_443 / s368 / L

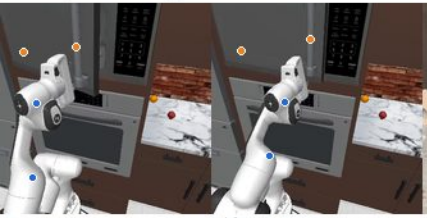


Query now Actual future +29 steps

Object: actual 15.5 cm; forecast 30.8 cm

Min point movement: 6.1 / 5.1 px (actual / forecast)

CloseSingleDoor / demo_1859 / s256 / R

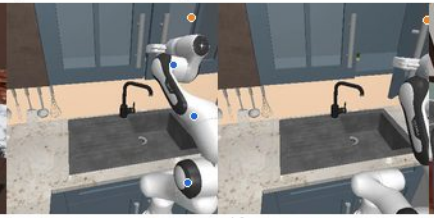


Query now Actual future +10 steps

Object: actual 13.6 cm; forecast 17.3 cm

Min point movement: 4.2 / 4.9 px (actual / forecast)

OpenDoubleDoor / demo_1392 / s240 / L



Query now Actual future +26 steps

Object: actual 15.7 cm; forecast 14.9 cm

Min point movement: 14.7 / 14.9 px (actual / forecast)

OpenSingleDoor / demo_255 / s256 / R

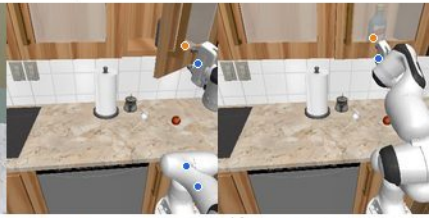


Query now Actual future +17 steps

Object: actual 13.5 cm; forecast 10.3 cm

Min point movement: 5.2 / 3.9 px (actual / forecast)

CloseSingleDoor / demo_611 / s192 / L

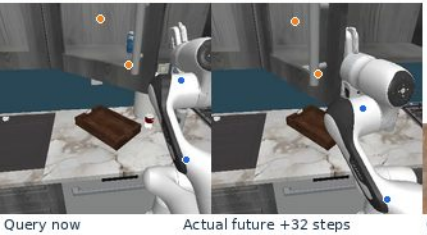


Query now Actual future +24 steps

Object: actual 16.7 cm; forecast 15.1 cm

Min point movement: 7.3 / 4.1 px (actual / forecast)

OpenDoubleDoor / demo_296 / s880 / L

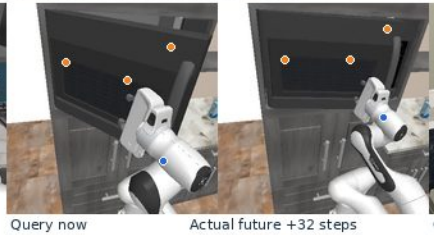


Query now Actual future +32 steps

Object: actual 12.0 cm; forecast 10.5 cm

Min point movement: 10.5 / 9.5 px (actual / forecast)

CloseSingleDoor / demo_1789 / s224 / L



Query now Actual future +32 steps

Object: actual 32.7 cm; forecast 28.6 cm

Min point movement: 5.2 / 3.3 px (actual / forecast)

OpenDrawer / demo_2082 / s224 / L

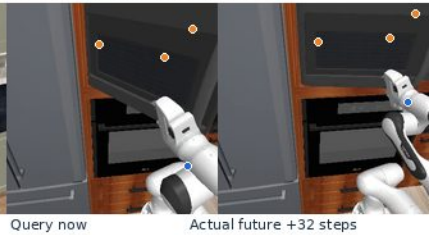


Query now Actual future +20 steps

Object: actual 26.0 cm; forecast 24.6 cm

Min point movement: 4.7 / 3.2 px (actual / forecast)

CloseSingleDoor / demo_1172 / s208 / L



Query now Actual future +32 steps

Object: actual 33.1 cm; forecast 33.9 cm

Min point movement: 4.8 / 3.7 px (actual / forecast)

OpenSingleDoor / demo_122 / s208 / L

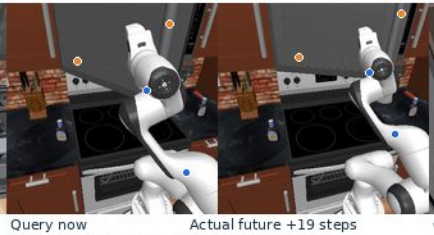


Query now Actual future +26 steps

Object: actual 15.4 cm; forecast 20.7 cm

Min point movement: 6.2 / 9.0 px (actual / forecast)

CloseSingleDoor / demo_651 / s224 / L



Query now Actual future +19 steps

Object: actual 45.7 cm; forecast 46.8 cm

Min point movement: 9.4 / 8.8 px (actual / forecast)

OpenSingleDoor / demo_1427 / s176 / R



Query now Actual future +32 steps

Object: actual 15.2 cm; forecast 15.8 cm

Min point movement: 14.0 / 14.8 px (actual / forecast)

OpenDoubleDoor / demo_1392 / s432 / R



Query now Actual future +32 steps

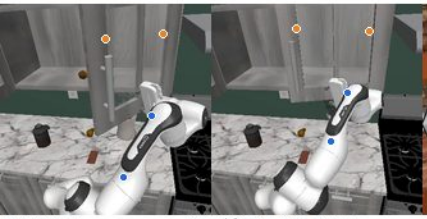
Object: actual 12.9 cm; forecast 18.3 cm

Min point movement: 4.7 / 5.2 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDoubleDoor / demo_1032 / s176 / R

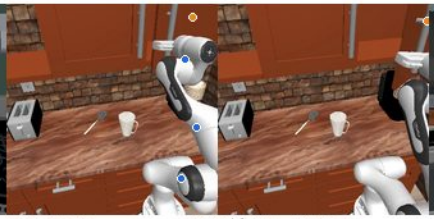


Query now Actual future +12 steps

Object: actual 36.0 cm; forecast 37.4 cm

Min point movement: 3.3 / 3.9 px (actual / forecast)

OpenDoubleDoor / demo_1088 / s176 / L



Query now Actual future +19 steps

Object: actual 16.2 cm; forecast 14.6 cm

Min point movement: 13.8 / 14.1 px (actual / forecast)

OpenDrawer / demo_186 / s192 / L

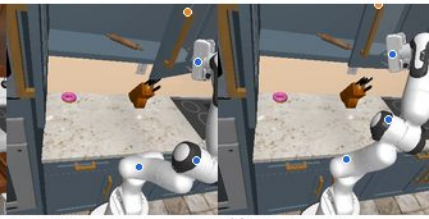


Query now Actual future +14 steps

Object: actual 14.2 cm; forecast 19.6 cm

Min point movement: 3.3 / 3.9 px (actual / forecast)

CloseDoubleDoor / demo_2220 / s192 / L

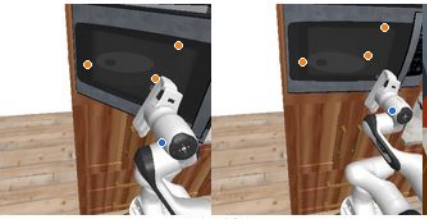


Query now Actual future +23 steps

Object: actual 16.6 cm; forecast 18.1 cm

Min point movement: 5.0 / 5.1 px (actual / forecast)

CloseSingleDoor / demo_2591 / s192 / L

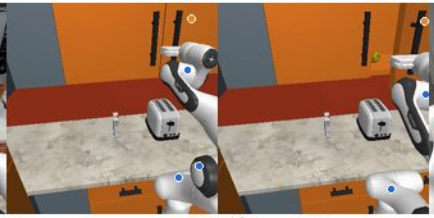


Query now Actual future +26 steps

Object: actual 29.8 cm; forecast 33.5 cm

Min point movement: 3.2 / 4.0 px (actual / forecast)

OpenDoubleDoor / demo_971 / s192 / L



Query now Actual future +32 steps

Object: actual 13.6 cm; forecast 17.9 cm

Min point movement: 6.9 / 7.5 px (actual / forecast)

CloseDoubleDoor / demo_876 / s448 / R

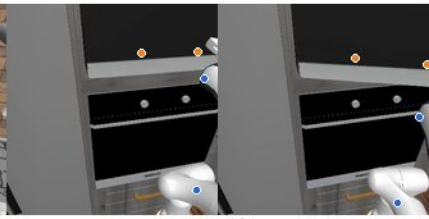


Query now Actual future +15 steps

Object: actual 13.6 cm; forecast 15.2 cm

Min point movement: 4.7 / 4.1 px (actual / forecast)

OpenSingleDoor / demo_2989 / s128 / L



Query now Actual future +20 steps

Object: actual 27.5 cm; forecast 27.2 cm

Min point movement: 5.5 / 4.7 px (actual / forecast)

CloseDoubleDoor / demo_2827 / s176 / R



Query now Actual future +8 steps

Object: actual 30.4 cm; forecast 30.9 cm

Min point movement: 13.3 / 13.1 px (actual / forecast)

CloseSingleDoor / demo_2592 / s224 / L

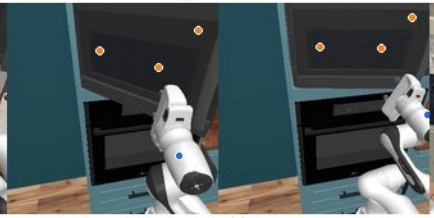


Query now Actual future +32 steps

Object: actual 13.4 cm; forecast 15.3 cm

Min point movement: 12.6 / 14.0 px (actual / forecast)

CloseSingleDoor / demo_1838 / s240 / L



Query now Actual future +32 steps

Object: actual 30.5 cm; forecast 24.2 cm

Min point movement: 6.4 / 6.4 px (actual / forecast)

CloseDrawer / demo_1360 / s144 / R



Query now Actual future +32 steps

Object: actual 13.5 cm; forecast 10.4 cm

Min point movement: 7.5 / 7.2 px (actual / forecast)

CloseDoubleDoor / demo_2827 / s496 / L



Query now Actual future +11 steps

Object: actual 33.7 cm; forecast 37.9 cm

Min point movement: 4.1 / 6.0 px (actual / forecast)

CloseDoubleDoor / demo_2963 / s176 / R

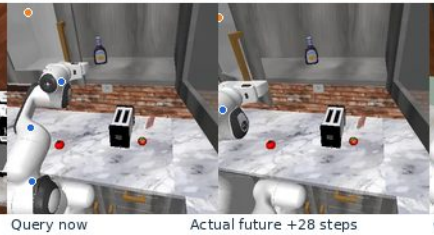


Query now Actual future +26 steps

Object: actual 30.5 cm; forecast 33.8 cm

Min point movement: 7.5 / 6.4 px (actual / forecast)

OpenDoubleDoor / demo_293 / s656 / R

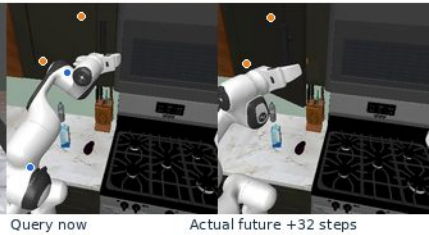


Query now Actual future +28 steps

Object: actual 14.0 cm; forecast 12.8 cm

Min point movement: 12.9 / 13.1 px (actual / forecast)

OpenSingleDoor / demo_271 / s144 / R

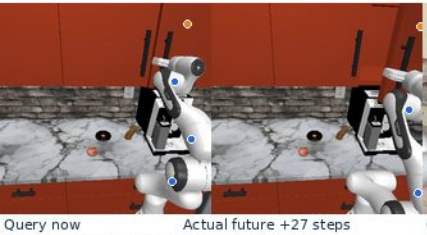


Query now Actual future +32 steps

Object: actual 15.5 cm; forecast 17.6 cm

Min point movement: 4.7 / 5.5 px (actual / forecast)

OpenDoubleDoor / demo_737 / s176 / L



Query now Actual future +27 steps

Object: actual 15.8 cm; forecast 17.9 cm

Min point movement: 12.8 / 12.7 px (actual / forecast)

CloseDrawer / demo_1347 / s208 / R



Query now Actual future +32 steps

Object: actual 16.3 cm; forecast 12.6 cm

Min point movement: 6.0 / 4.5 px (actual / forecast)

CloseDrawer / demo_649 / s192 / L

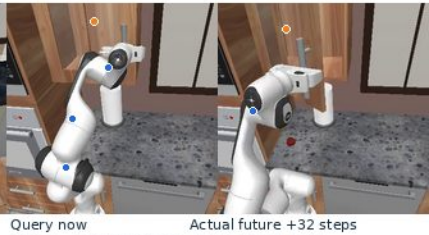


Query now Actual future +32 steps

Object: actual 17.1 cm; forecast 18.4 cm

Min point movement: 5.6 / 6.4 px (actual / forecast)

OpenSingleDoor / demo_1017 / s128 / R



Query now Actual future +32 steps

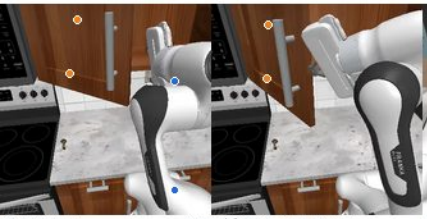
Object: actual 16.3 cm; forecast 14.2 cm

Min point movement: 12.6 / 10.4 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

OpenDoubleDoor / demo_1130 / s688 / L

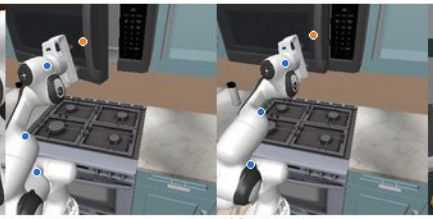


Query now Actual future +32 steps

Object: actual 8.9 cm; forecast 8.7 cm

Min point movement: 8.5 / 8.5 px (actual / forecast)

CloseSingleDoor / demo_535 / s240 / R

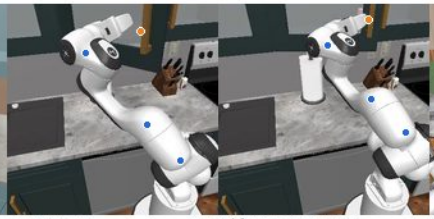


Query now Actual future +9 steps

Object: actual 13.2 cm; forecast 18.4 cm

Min point movement: 5.1 / 7.8 px (actual / forecast)

CloseDoubleDoor / demo_275 / s352 / L



Query now Actual future +11 steps

Object: actual 31.2 cm; forecast 27.1 cm

Min point movement: 12.5 / 11.2 px (actual / forecast)

OpenDrawer / demo_623 / s208 / R

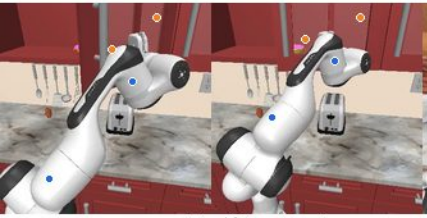


Query now Actual future +16 steps

Object: actual 23.0 cm; forecast 31.4 cm

Min point movement: 3.7 / 3.9 px (actual / forecast)

CloseDoubleDoor / demo_2123 / s160 / R

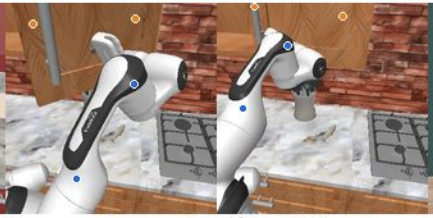


Query now Actual future +12 steps

Object: actual 30.4 cm; forecast 31.2 cm

Min point movement: 4.5 / 5.0 px (actual / forecast)

CloseDoubleDoor / demo_2872 / s176 / R



Query now Actual future +19 steps

Object: actual 37.4 cm; forecast 33.6 cm

Min point movement: 3.3 / 3.9 px (actual / forecast)

CloseSingleDoor / demo_2592 / s192 / L

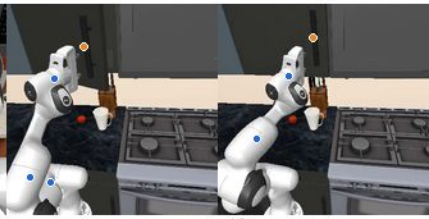


Query now Actual future +16 steps

Object: actual 11.4 cm; forecast 16.9 cm

Min point movement: 12.3 / 22.0 px (actual / forecast)

CloseSingleDoor / demo_466 / s272 / R

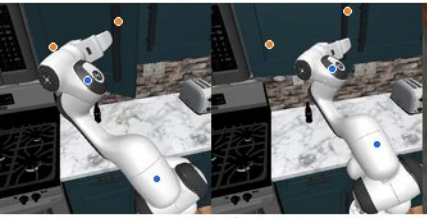


Query now Actual future +13 steps

Object: actual 14.0 cm; forecast 18.7 cm

Min point movement: 5.9 / 9.2 px (actual / forecast)

CloseDoubleDoor / demo_2593 / s384 / L



Query now Actual future +13 steps

Object: actual 29.9 cm; forecast 32.1 cm

Min point movement: 4.5 / 3.5 px (actual / forecast)

CloseDoubleDoor / demo_2963 / s400 / L

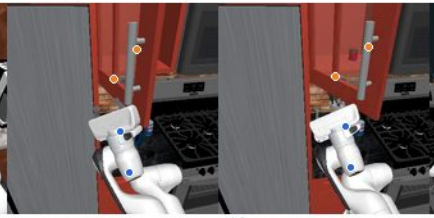


Query now Actual future +32 steps

Object: actual 18.9 cm; forecast 21.5 cm

Min point movement: 9.6 / 7.7 px (actual / forecast)

CloseSingleDoor / demo_2451 / s320 / L

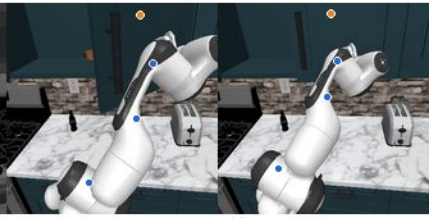


Query now Actual future +32 steps

Object: actual 7.8 cm; forecast 16.9 cm

Min point movement: 5.8 / 4.7 px (actual / forecast)

CloseDoubleDoor / demo_2593 / s192 / R



Query now Actual future +12 steps

Object: actual 23.6 cm; forecast 25.0 cm

Min point movement: 12.1 / 12.6 px (actual / forecast)

OpenDoubleDoor / demo_503 / s176 / L

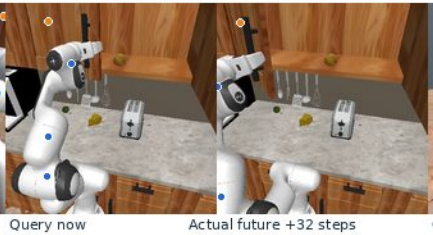


Query now Actual future +32 steps

Object: actual 11.8 cm; forecast 11.6 cm

Min point movement: 12.0 / 12.0 px (actual / forecast)

OpenDoubleDoor / demo_456 / s880 / R

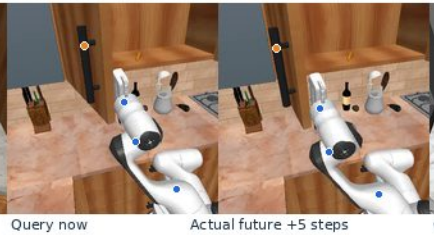


Query now Actual future +32 steps

Object: actual 10.7 cm; forecast 8.6 cm

Min point movement: 12.0 / 8.5 px (actual / forecast)

OpenDoubleDoor / demo_1013 / s768 / L

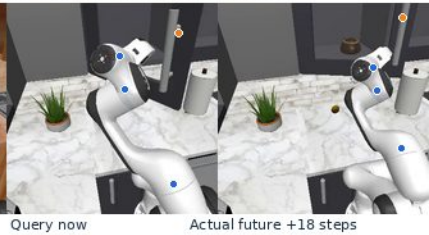


Query now Actual future +5 steps

Object: actual 5.5 cm; forecast 12.4 cm

Min point movement: 6.5 / 14.9 px (actual / forecast)

CloseSingleDoor / demo_1892 / s208 / L

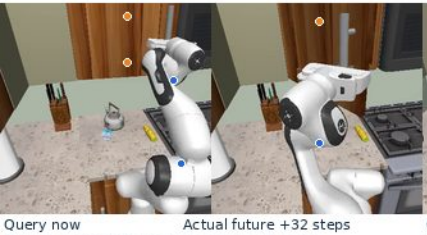


Query now Actual future +18 steps

Object: actual 39.6 cm; forecast 42.3 cm

Min point movement: 11.9 / 12.7 px (actual / forecast)

OpenSingleDoor / demo_2131 / s160 / L

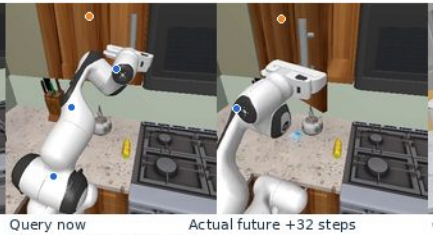


Query now Actual future +32 steps

Object: actual 23.4 cm; forecast 22.2 cm

Min point movement: 9.2 / 9.1 px (actual / forecast)

OpenSingleDoor / demo_2131 / s144 / R

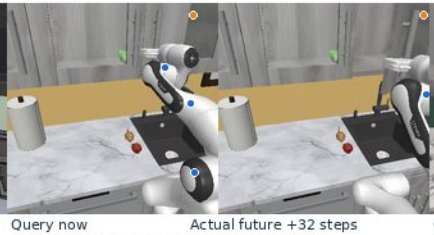


Query now Actual future +32 steps

Object: actual 14.0 cm; forecast 9.8 cm

Min point movement: 11.8 / 7.8 px (actual / forecast)

OpenDoubleDoor / demo_1225 / s208 / L

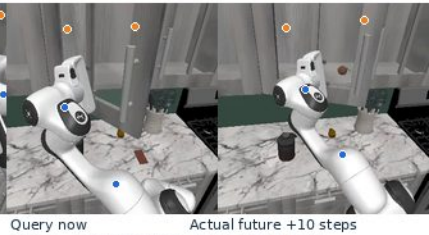


Query now Actual future +32 steps

Object: actual 11.4 cm; forecast 10.5 cm

Min point movement: 11.8 / 11.2 px (actual / forecast)

CloseDoubleDoor / demo_1032 / s352 / L



Query now Actual future +10 steps

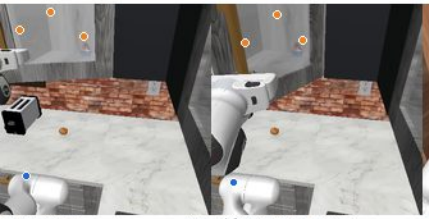
Object: actual 34.5 cm; forecast 36.7 cm

Min point movement: 5.2 / 5.5 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

OpenSingleDoor / demo_770 / s288 / R



Query now Actual future +32 steps

Object: actual 28.5 cm; forecast 24.6 cm

Min point movement: 4.7 / 4.1 px (actual / forecast)

OpenSingleDoor / demo_498 / s160 / R



Query now Actual future +32 steps

Object: actual 14.4 cm; forecast 17.8 cm

Min point movement: 3.3 / 4.2 px (actual / forecast)

CloseSingleDoor / demo_2186 / s160 / R

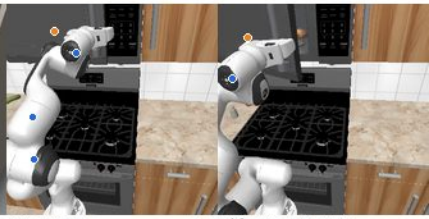


Query now Actual future +31 steps

Object: actual 26.0 cm; forecast 34.8 cm

Min point movement: 10.3 / 11.0 px (actual / forecast)

OpenSingleDoor / demo_1845 / s176 / R

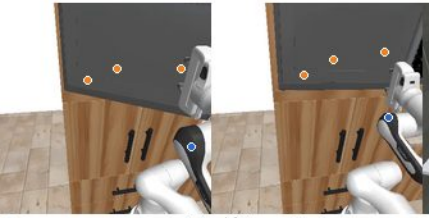


Query now Actual future +32 steps

Object: actual 11.8 cm; forecast 9.1 cm

Min point movement: 11.6 / 8.3 px (actual / forecast)

CloseSingleDoor / demo_103 / s256 / L

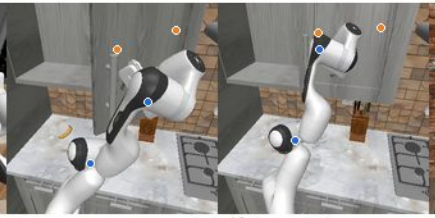


Query now Actual future +15 steps

Object: actual 18.9 cm; forecast 21.6 cm

Min point movement: 4.6 / 4.6 px (actual / forecast)

CloseDoubleDoor / demo_876 / s192 / R



Query now Actual future +26 steps

Object: actual 30.0 cm; forecast 31.9 cm

Min point movement: 4.5 / 6.3 px (actual / forecast)

CloseDoubleDoor / demo_1995 / s368 / L



Query now Actual future +22 steps

Object: actual 27.6 cm; forecast 24.0 cm

Min point movement: 11.3 / 8.4 px (actual / forecast)

CloseDrawer / demo_1886 / s176 / R

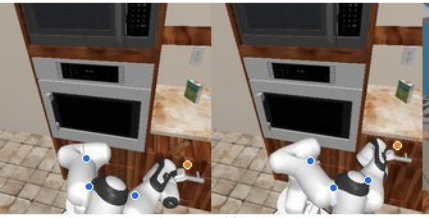


Query now Actual future +29 steps

Object: actual 20.4 cm; forecast 14.1 cm

Min point movement: 3.4 / 3.7 px (actual / forecast)

CloseDrawer / demo_1907 / s176 / R

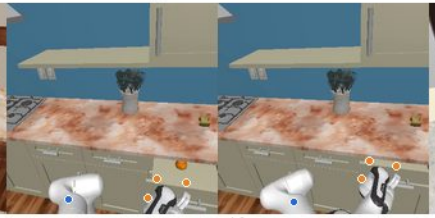


Query now Actual future +32 steps

Object: actual 14.3 cm; forecast 14.6 cm

Min point movement: 7.9 / 10.6 px (actual / forecast)

CloseDrawer / demo_183 / s160 / R

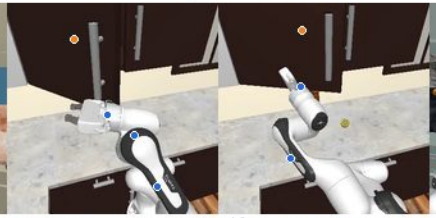


Query now Actual future +32 steps

Object: actual 13.1 cm; forecast 14.0 cm

Min point movement: 8.6 / 7.8 px (actual / forecast)

CloseDoubleDoor / demo_1987 / s384 / L



Query now Actual future +32 steps

Object: actual 12.0 cm; forecast 18.4 cm

Min point movement: 11.2 / 14.2 px (actual / forecast)

CloseDrawer / demo_41 / s160 / R

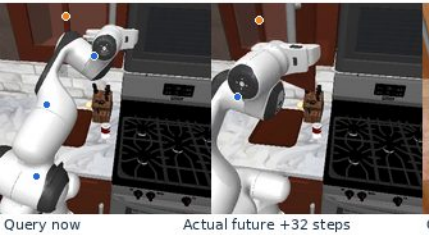


Query now Actual future +32 steps

Object: actual 15.0 cm; forecast 10.0 cm

Min point movement: 6.6 / 7.6 px (actual / forecast)

OpenSingleDoor / demo_2760 / s144 / R

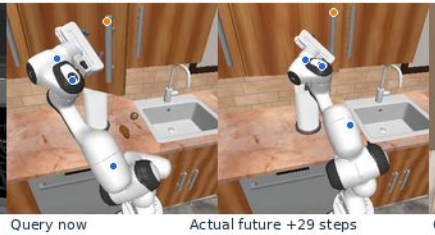


Query now Actual future +32 steps

Object: actual 12.1 cm; forecast 11.3 cm

Min point movement: 11.1 / 10.3 px (actual / forecast)

CloseDoubleDoor / demo_1499 / s384 / L

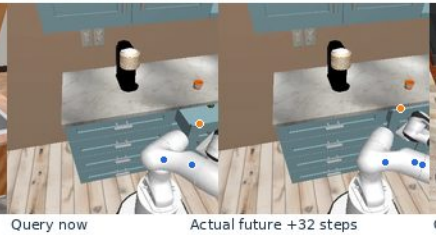


Query now Actual future +29 steps

Object: actual 25.5 cm; forecast 26.7 cm

Min point movement: 11.0 / 9.0 px (actual / forecast)

CloseDrawer / demo_329 / s208 / L

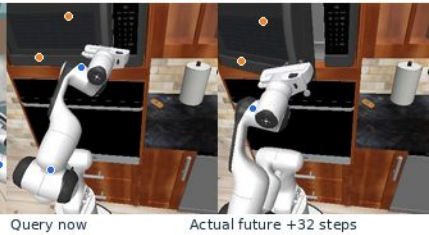


Query now Actual future +32 steps

Object: actual 13.9 cm; forecast 14.6 cm

Min point movement: 6.2 / 6.5 px (actual / forecast)

OpenSingleDoor / demo_2753 / s128 / R

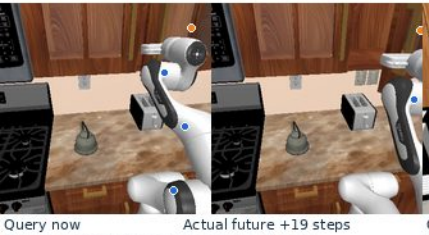


Query now Actual future +32 steps

Object: actual 13.1 cm; forecast 12.5 cm

Min point movement: 6.5 / 5.6 px (actual / forecast)

OpenDoubleDoor / demo_503 / s192 / L



Query now Actual future +19 steps

Object: actual 14.1 cm; forecast 13.3 cm

Min point movement: 11.0 / 10.7 px (actual / forecast)

OpenDoubleDoor / demo_456 / s192 / L

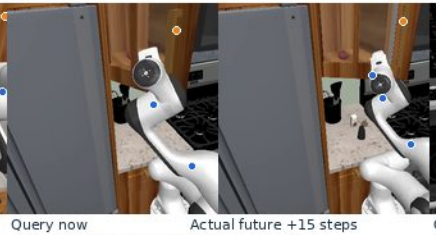


Query now Actual future +27 steps

Object: actual 12.8 cm; forecast 11.3 cm

Min point movement: 10.9 / 10.3 px (actual / forecast)

CloseSingleDoor / demo_1182 / s240 / L

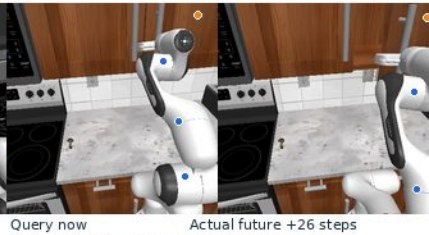


Query now Actual future +15 steps

Object: actual 43.2 cm; forecast 46.3 cm

Min point movement: 10.9 / 11.0 px (actual / forecast)

OpenDoubleDoor / demo_1130 / s176 / L



Query now Actual future +26 steps

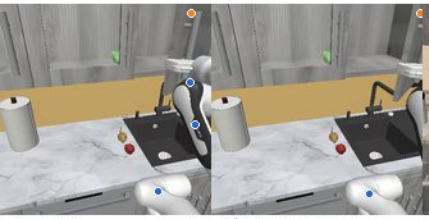
Object: actual 13.6 cm; forecast 11.6 cm

Min point movement: 10.9 / 11.1 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

OpenDoubleDoor / demo_1225 / s240 / L

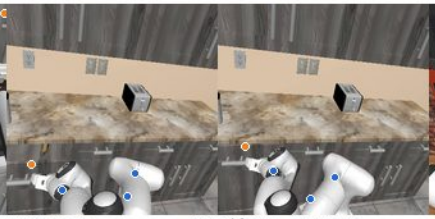


Query now Actual future +10 steps

Object: actual 8.5 cm; forecast 6.4 cm

Min point movement: 3.9 / 3.5 px (actual / forecast)

CloseDrawer / demo_2326 / s208 / L

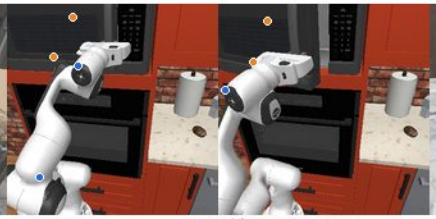


Query now Actual future +32 steps

Object: actual 13.6 cm; forecast 13.7 cm

Min point movement: 7.5 / 7.9 px (actual / forecast)

OpenSingleDoor / demo_1154 / s208 / R

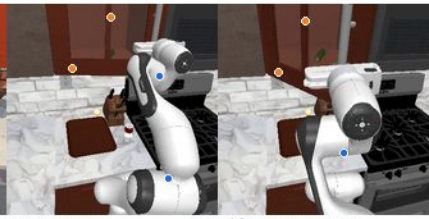


Query now Actual future +32 steps

Object: actual 12.0 cm; forecast 12.6 cm

Min point movement: 7.6 / 8.1 px (actual / forecast)

OpenSingleDoor / demo_2760 / s160 / L

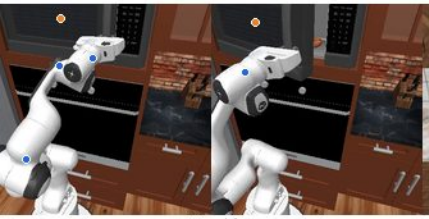


Query now Actual future +32 steps

Object: actual 19.1 cm; forecast 17.9 cm

Min point movement: 4.0 / 3.4 px (actual / forecast)

OpenSingleDoor / demo_1026 / s144 / R



Query now Actual future +32 steps

Object: actual 10.8 cm; forecast 12.2 cm

Min point movement: 10.5 / 12.4 px (actual / forecast)

CloseDrawer / demo_2471 / s160 / L

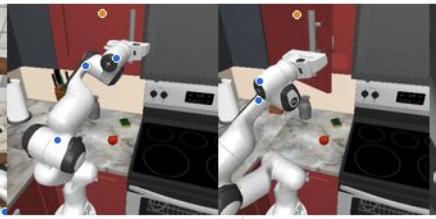


Query now Actual future +32 steps

Object: actual 13.8 cm; forecast 14.7 cm

Min point movement: 8.6 / 7.8 px (actual / forecast)

OpenSingleDoor / demo_461 / s144 / R

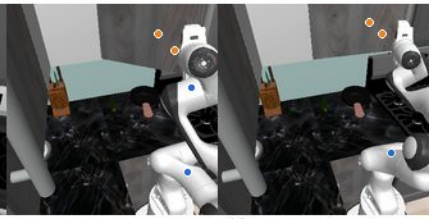


Query now Actual future +32 steps

Object: actual 14.7 cm; forecast 14.3 cm

Min point movement: 10.4 / 10.1 px (actual / forecast)

CloseSingleDoor / demo_1135 / s256 / L

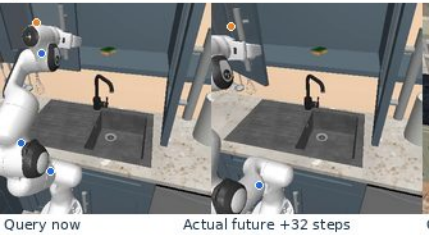


Query now Actual future +32 steps

Object: actual 23.6 cm; forecast 25.3 cm

Min point movement: 6.6 / 6.9 px (actual / forecast)

OpenDoubleDoor / demo_1392 / s672 / R

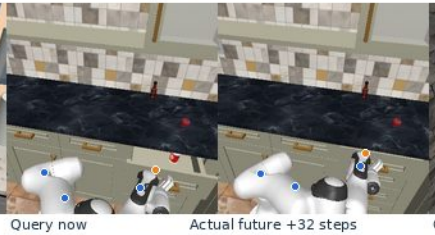


Query now Actual future +32 steps

Object: actual 10.9 cm; forecast 9.6 cm

Min point movement: 9.1 / 8.8 px (actual / forecast)

CloseDrawer / demo_1360 / s176 / R

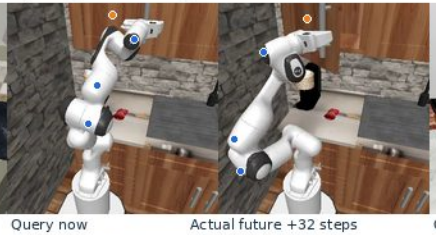


Query now Actual future +32 steps

Object: actual 12.6 cm; forecast 13.4 cm

Min point movement: 3.2 / 5.4 px (actual / forecast)

OpenSingleDoor / demo_1467 / s208 / R

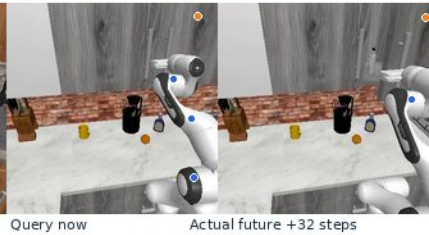


Query now Actual future +32 steps

Object: actual 15.8 cm; forecast 14.3 cm

Min point movement: 10.2 / 9.0 px (actual / forecast)

OpenDoubleDoor / demo_654 / s176 / L

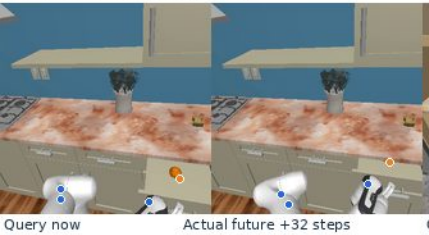


Query now Actual future +32 steps

Object: actual 10.4 cm; forecast 11.5 cm

Min point movement: 10.2 / 10.8 px (actual / forecast)

CloseDrawer / demo_183 / s144 / R



Query now Actual future +32 steps

Object: actual 13.8 cm; forecast 13.5 cm

Min point movement: 6.2 / 8.1 px (actual / forecast)

CloseDrawer / demo_2024 / s176 / R



Query now Actual future +32 steps

Object: actual 15.2 cm; forecast 12.6 cm

Min point movement: 8.0 / 8.0 px (actual / forecast)

CloseDrawer / demo_2464 / s192 / L

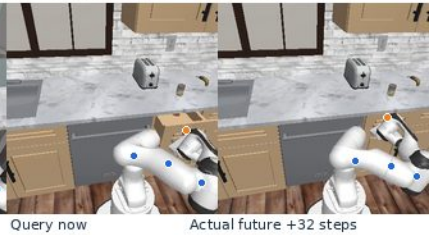


Query now Actual future +32 steps

Object: actual 13.6 cm; forecast 11.2 cm

Min point movement: 4.5 / 7.7 px (actual / forecast)

CloseDrawer / demo_2931 / s192 / L



Query now Actual future +32 steps

Object: actual 15.0 cm; forecast 13.2 cm

Min point movement: 4.9 / 3.3 px (actual / forecast)

CloseSingleDoor / demo_288 / s224 / L

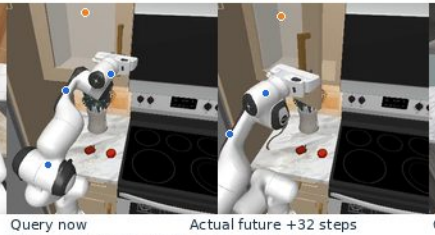


Query now Actual future +15 steps

Object: actual 33.9 cm; forecast 37.1 cm

Min point movement: 6.5 / 7.1 px (actual / forecast)

OpenSingleDoor / demo_1623 / s144 / R

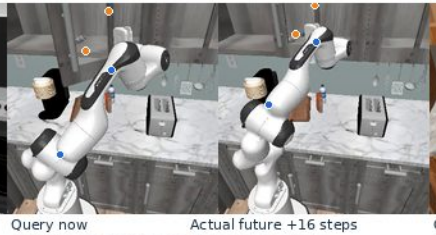


Query now Actual future +32 steps

Object: actual 13.7 cm; forecast 12.1 cm

Min point movement: 10.0 / 8.9 px (actual / forecast)

CloseDoubleDoor / demo_928 / s176 / R

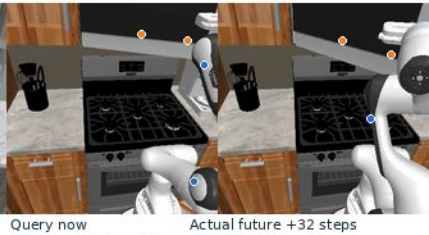


Query now Actual future +16 steps

Object: actual 26.5 cm; forecast 26.2 cm

Min point movement: 5.3 / 5.6 px (actual / forecast)

OpenSingleDoor / demo_321 / s176 / L



Query now Actual future +32 steps

Object: actual 21.1 cm; forecast 23.6 cm

Min point movement: 7.1 / 8.4 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

OpenSingleDoor / demo_32 / s160 / R

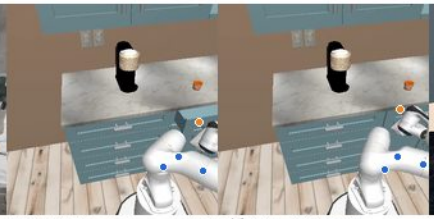


Query now Actual future +32 steps

Object: actual 11.6 cm; forecast 9.1 cm

Min point movement: 10.0 / 6.8 px (actual / forecast)

CloseDrawer / demo_329 / s224 / L

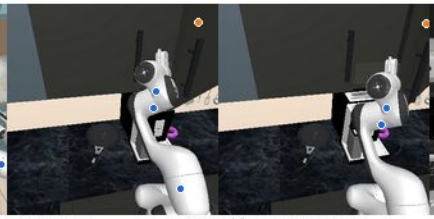


Query now Actual future +28 steps

Object: actual 13.6 cm; forecast 11.4 cm

Min point movement: 4.8 / 3.2 px (actual / forecast)

OpenDoubleDoor / demo_1176 / s400 / L



Query now Actual future +14 steps

Object: actual 23.4 cm; forecast 29.2 cm

Min point movement: 9.9 / 10.9 px (actual / forecast)

CloseDrawer / demo_2142 / s192 / L



Query now Actual future +28 steps

Object: actual 16.0 cm; forecast 14.5 cm

Min point movement: 7.0 / 6.8 px (actual / forecast)

CloseSingleDoor / demo_869 / s224 / L

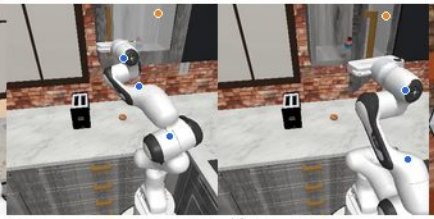


Query now Actual future +30 steps

Object: actual 35.3 cm; forecast 37.9 cm

Min point movement: 5.4 / 3.5 px (actual / forecast)

OpenSingleDoor / demo_770 / s272 / L

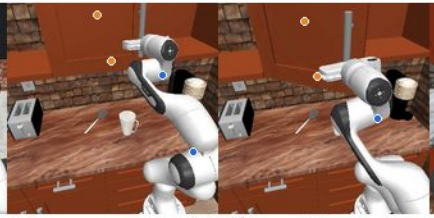


Query now Actual future +32 steps

Object: actual 12.6 cm; forecast 14.0 cm

Min point movement: 9.8 / 10.2 px (actual / forecast)

OpenDoubleDoor / demo_1088 / s608 / L

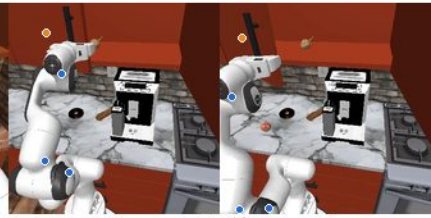


Query now Actual future +32 steps

Object: actual 17.7 cm; forecast 16.0 cm

Min point movement: 5.0 / 4.1 px (actual / forecast)

OpenDoubleDoor / demo_737 / s624 / R

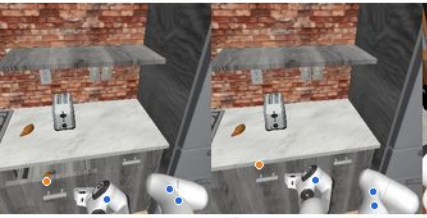


Query now Actual future +32 steps

Object: actual 9.2 cm; forecast 14.1 cm

Min point movement: 9.8 / 15.4 px (actual / forecast)

CloseDrawer / demo_2934 / s208 / L

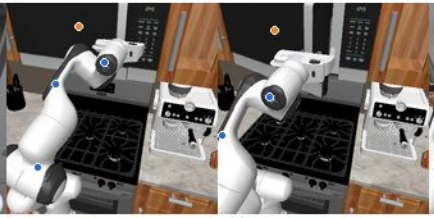


Query now Actual future +26 steps

Object: actual 14.2 cm; forecast 12.6 cm

Min point movement: 5.8 / 5.0 px (actual / forecast)

OpenSingleDoor / demo_321 / s144 / R

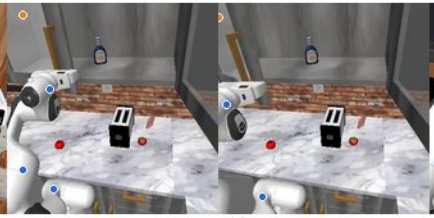


Query now Actual future +32 steps

Object: actual 11.1 cm; forecast 11.9 cm

Min point movement: 9.8 / 10.7 px (actual / forecast)

OpenDoubleDoor / demo_293 / s672 / R

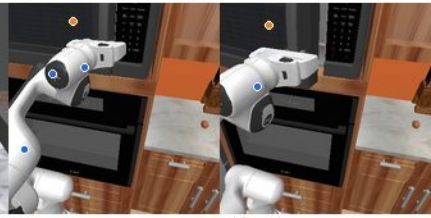


Query now Actual future +13 steps

Object: actual 14.7 cm; forecast 13.0 cm

Min point movement: 6.9 / 5.8 px (actual / forecast)

OpenSingleDoor / demo_156 / s208 / R

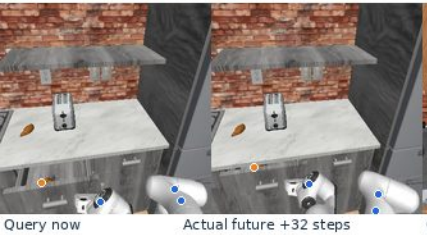


Query now Actual future +32 steps

Object: actual 9.6 cm; forecast 10.9 cm

Min point movement: 9.7 / 11.2 px (actual / forecast)

CloseDrawer / demo_2934 / s192 / L

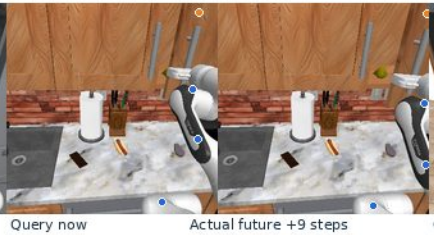


Query now Actual future +32 steps

Object: actual 10.9 cm; forecast 12.7 cm

Min point movement: 6.1 / 5.2 px (actual / forecast)

OpenDoubleDoor / demo_176 / s192 / L

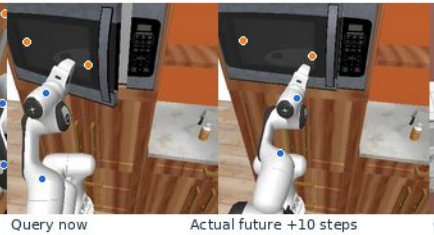


Query now Actual future +9 steps

Object: actual 19.3 cm; forecast 18.3 cm

Min point movement: 4.5 / 3.6 px (actual / forecast)

CloseSingleDoor / demo_2591 / s208 / R

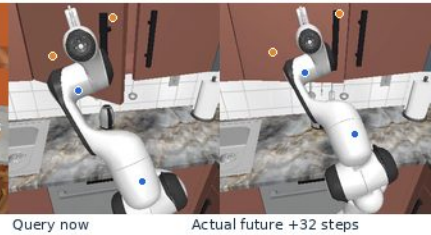


Query now Actual future +10 steps

Object: actual 13.2 cm; forecast 14.4 cm

Min point movement: 3.6 / 3.9 px (actual / forecast)

CloseDoubleDoor / demo_1757 / s448 / L

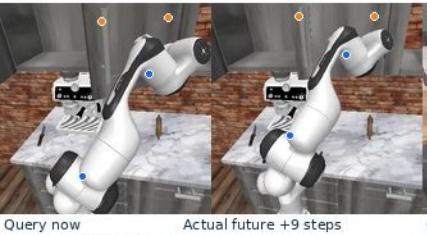


Query now Actual future +32 steps

Object: actual 21.3 cm; forecast 28.4 cm

Min point movement: 6.0 / 5.3 px (actual / forecast)

CloseDoubleDoor / demo_1995 / s176 / R

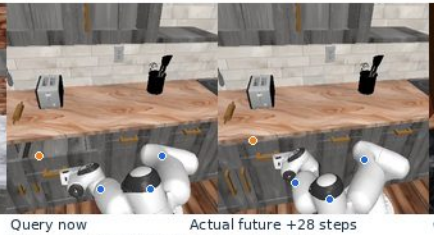


Query now Actual future +9 steps

Object: actual 30.0 cm; forecast 29.2 cm

Min point movement: 3.6 / 4.4 px (actual / forecast)

CloseDrawer / demo_2465 / s160 / L

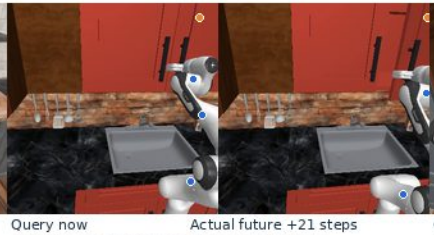


Query now Actual future +28 steps

Object: actual 15.4 cm; forecast 13.9 cm

Min point movement: 6.3 / 6.1 px (actual / forecast)

OpenDoubleDoor / demo_332 / s192 / L

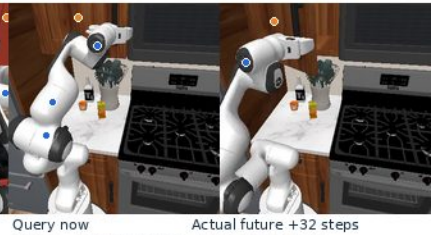


Query now Actual future +21 steps

Object: actual 18.6 cm; forecast 18.8 cm

Min point movement: 9.5 / 9.8 px (actual / forecast)

OpenSingleDoor / demo_468 / s176 / R



Query now Actual future +32 steps

Object: actual 11.8 cm; forecast 10.3 cm

Min point movement: 9.5 / 7.3 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDoubleDoor / demo_1029 / s224 / L

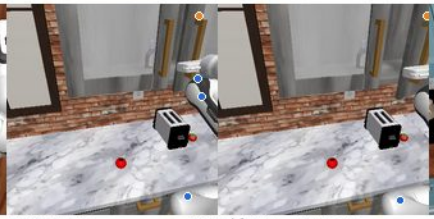


Query now Actual future +9 steps

Object: actual 11.5 cm; forecast 12.3 cm

Min point movement: 4.1 / 6.3 px (actual / forecast)

OpenDoubleDoor / demo_293 / s192 / L



Query now Actual future +10 steps

Object: actual 23.0 cm; forecast 22.4 cm

Min point movement: 4.6 / 4.0 px (actual / forecast)

CloseSingleDoor / demo_2334 / s224 / L

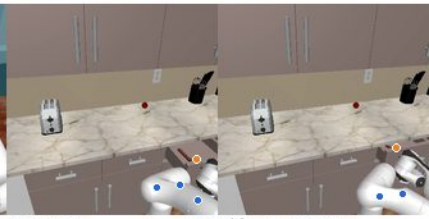


Query now Actual future +16 steps

Object: actual 35.6 cm; forecast 37.4 cm

Min point movement: 3.5 / 3.1 px (actual / forecast)

CloseDrawer / demo_21 / s160 / L



Query now Actual future +32 steps

Object: actual 13.4 cm; forecast 14.4 cm

Min point movement: 8.8 / 8.0 px (actual / forecast)

CloseDrawer / demo_995 / s176 / R



Query now Actual future +32 steps

Object: actual 12.9 cm; forecast 11.6 cm

Min point movement: 8.6 / 8.2 px (actual / forecast)

CloseDrawer / demo_2368 / s128 / R



Query now Actual future +32 steps

Object: actual 12.7 cm; forecast 11.9 cm

Min point movement: 5.8 / 7.8 px (actual / forecast)

CloseSingleDoor / demo_2592 / s240 / L

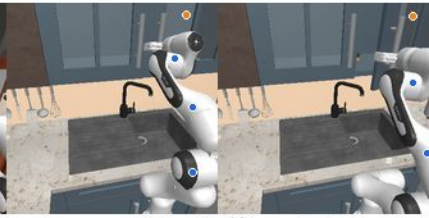


Query now Actual future +25 steps

Object: actual 21.6 cm; forecast 20.2 cm

Min point movement: 5.6 / 7.8 px (actual / forecast)

OpenDoubleDoor / demo_1392 / s224 / L

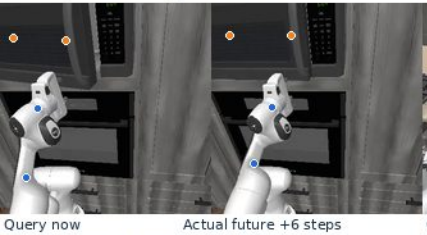


Query now Actual future +32 steps

Object: actual 10.5 cm; forecast 10.6 cm

Min point movement: 9.3 / 10.1 px (actual / forecast)

CloseSingleDoor / demo_24 / s272 / R

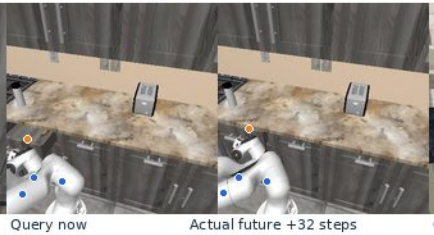


Query now Actual future +6 steps

Object: actual 8.8 cm; forecast 13.5 cm

Min point movement: 3.7 / 5.1 px (actual / forecast)

CloseDrawer / demo_2326 / s208 / R

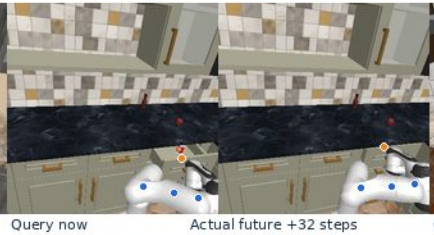


Query now Actual future +32 steps

Object: actual 13.6 cm; forecast 13.7 cm

Min point movement: 3.9 / 3.5 px (actual / forecast)

CloseDrawer / demo_1360 / s176 / L



Query now Actual future +32 steps

Object: actual 12.6 cm; forecast 12.6 cm

Min point movement: 3.9 / 5.2 px (actual / forecast)

CloseDrawer / demo_2471 / s192 / R

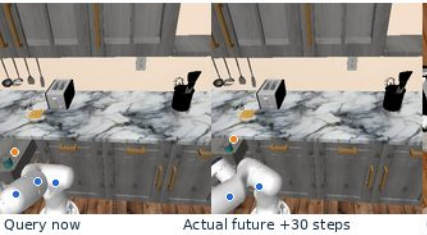


Query now Actual future +25 steps

Object: actual 12.2 cm; forecast 11.8 cm

Min point movement: 4.3 / 4.1 px (actual / forecast)

CloseDrawer / demo_1064 / s176 / R

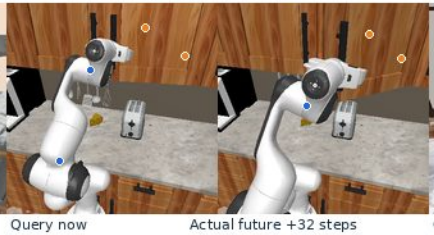


Query now Actual future +30 steps

Object: actual 14.4 cm; forecast 14.0 cm

Min point movement: 5.6 / 3.4 px (actual / forecast)

OpenDoubleDoor / demo_456 / s208 / R

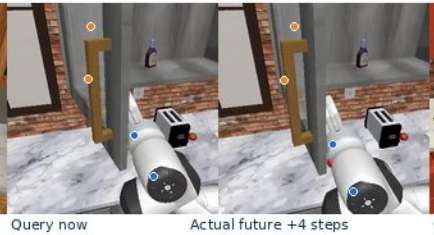


Query now Actual future +32 steps

Object: actual 16.4 cm; forecast 14.1 cm

Min point movement: 3.7 / 3.3 px (actual / forecast)

OpenDoubleDoor / demo_293 / s848 / L

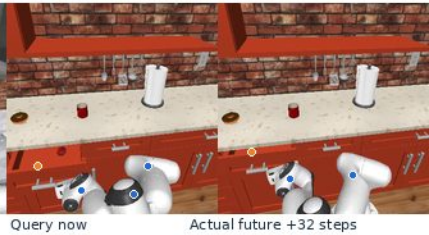


Query now Actual future +4 steps

Object: actual 3.0 cm; forecast 13.5 cm

Min point movement: 4.8 / 28.2 px (actual / forecast)

CloseDrawer / demo_1572 / s160 / L



Query now Actual future +32 steps

Object: actual 11.2 cm; forecast 12.6 cm

Min point movement: 6.8 / 5.8 px (actual / forecast)

CloseDrawer / demo_2931 / s160 / L

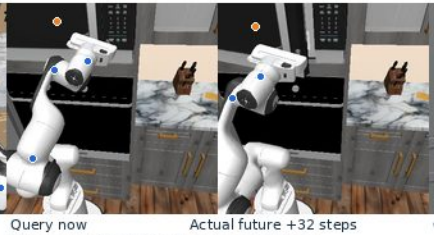


Query now Actual future +32 steps

Object: actual 14.0 cm; forecast 14.8 cm

Min point movement: 5.9 / 9.4 px (actual / forecast)

OpenSingleDoor / demo_2989 / s112 / R

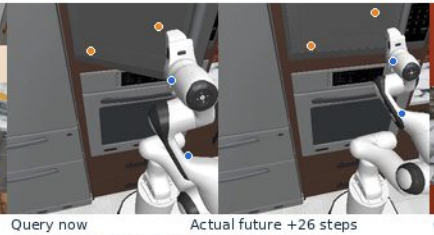


Query now Actual future +32 steps

Object: actual 10.6 cm; forecast 7.9 cm

Min point movement: 8.9 / 6.3 px (actual / forecast)

CloseSingleDoor / demo_1859 / s240 / L

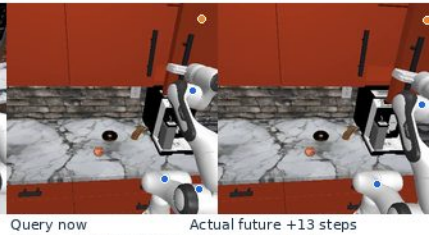


Query now Actual future +26 steps

Object: actual 25.2 cm; forecast 28.2 cm

Min point movement: 6.4 / 6.3 px (actual / forecast)

OpenDoubleDoor / demo_737 / s192 / L



Query now Actual future +13 steps

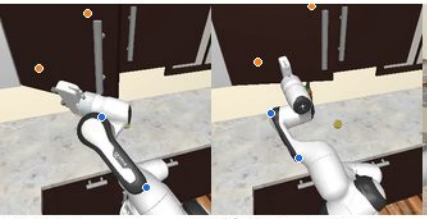
Object: actual 11.1 cm; forecast 14.6 cm

Min point movement: 4.4 / 5.1 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDoubleDoor / demo_1987 / s400 / L



Query now Actual future +32 steps

Object: actual 19.7 cm; forecast 21.6 cm

Min point movement: 5.5 / 4.4 px (actual / forecast)

CloseDrawer / demo_1347 / s192 / R



Query now Actual future +32 steps

Object: actual 11.6 cm; forecast 11.9 cm

Min point movement: 3.5 / 3.5 px (actual / forecast)

CloseDrawer / demo_1572 / s176 / L

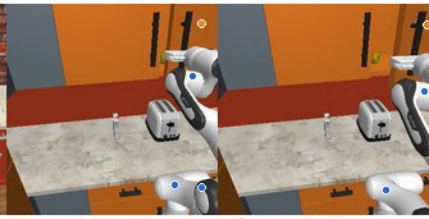


Query now Actual future +24 steps

Object: actual 16.6 cm; forecast 14.6 cm

Min point movement: 8.3 / 8.8 px (actual / forecast)

OpenDoubleDoor / demo_971 / s208 / L

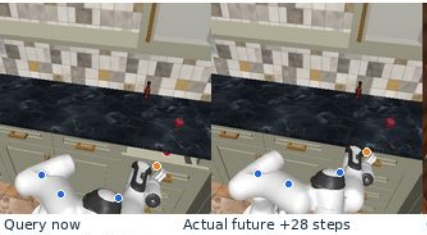


Query now Actual future +16 steps

Object: actual 12.9 cm; forecast 16.0 cm

Min point movement: 5.0 / 5.0 px (actual / forecast)

CloseDrawer / demo_1360 / s192 / R

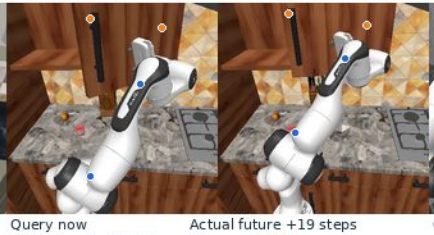


Query now Actual future +28 steps

Object: actual 11.3 cm; forecast 12.1 cm

Min point movement: 3.3 / 4.6 px (actual / forecast)

CloseDoubleDoor / demo_1029 / s208 / R

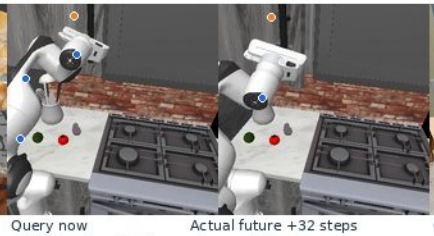


Query now Actual future +19 steps

Object: actual 28.8 cm; forecast 27.7 cm

Min point movement: 4.1 / 3.8 px (actual / forecast)

OpenSingleDoor / demo_122 / s160 / R

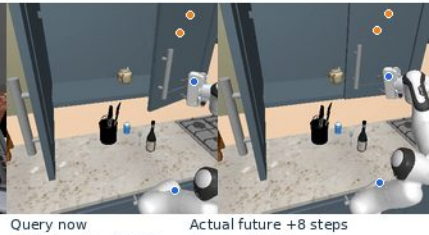


Query now Actual future +32 steps

Object: actual 11.4 cm; forecast 11.5 cm

Min point movement: 8.6 / 9.6 px (actual / forecast)

CloseDoubleDoor / demo_2470 / s192 / L

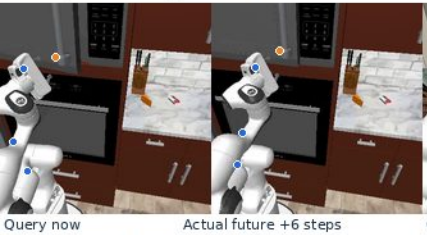


Query now Actual future +8 steps

Object: actual 11.3 cm; forecast 12.1 cm

Min point movement: 7.6 / 8.6 px (actual / forecast)

CloseSingleDoor / demo_821 / s256 / R

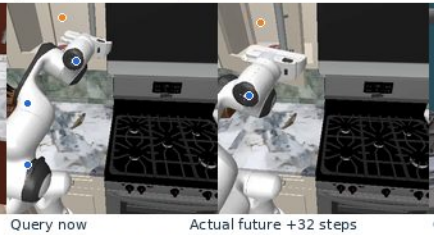


Query now Actual future +6 steps

Object: actual 8.7 cm; forecast 13.3 cm

Min point movement: 3.7 / 7.9 px (actual / forecast)

OpenSingleDoor / demo_1227 / s128 / R

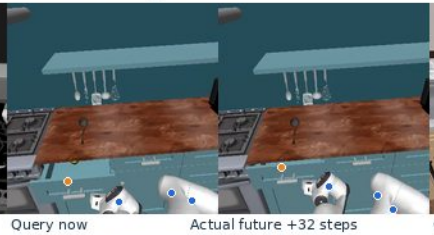


Query now Actual future +32 steps

Object: actual 11.7 cm; forecast 10.0 cm

Min point movement: 8.6 / 7.1 px (actual / forecast)

CloseDrawer / demo_1989 / s192 / L

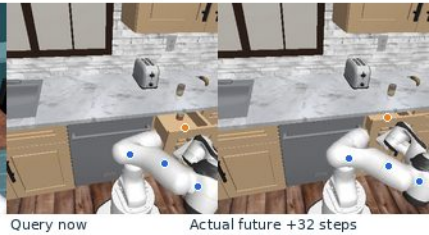


Query now Actual future +32 steps

Object: actual 10.3 cm; forecast 9.8 cm

Min point movement: 4.0 / 4.5 px (actual / forecast)

CloseDrawer / demo_2931 / s176 / L



Query now Actual future +32 steps

Object: actual 11.9 cm; forecast 13.7 cm

Min point movement: 5.2 / 5.5 px (actual / forecast)

CloseDrawer / demo_2877 / s176 / R

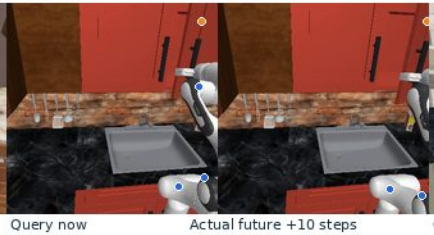


Query now Actual future +32 steps

Object: actual 11.1 cm; forecast 10.3 cm

Min point movement: 7.2 / 7.9 px (actual / forecast)

OpenDoubleDoor / demo_332 / s208 / L



Query now Actual future +10 steps

Object: actual 18.5 cm; forecast 18.2 cm

Min point movement: 3.6 / 4.4 px (actual / forecast)

CloseDrawer / demo_1360 / s144 / L



Query now Actual future +32 steps

Object: actual 13.5 cm; forecast 10.9 cm

Min point movement: 6.7 / 6.9 px (actual / forecast)

CloseDrawer / demo_2142 / s208 / L



Query now Actual future +20 steps

Object: actual 12.5 cm; forecast 12.7 cm

Min point movement: 4.7 / 4.6 px (actual / forecast)

CloseDrawer / demo_479 / s176 / L

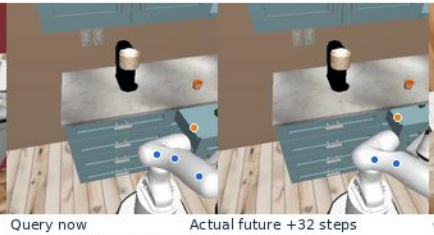


Query now Actual future +32 steps

Object: actual 11.8 cm; forecast 14.3 cm

Min point movement: 5.4 / 5.0 px (actual / forecast)

CloseDrawer / demo_329 / s176 / L

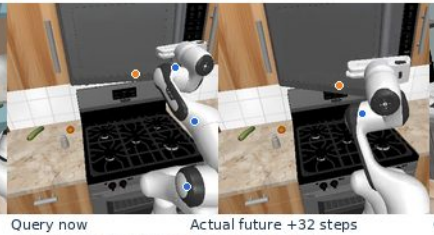


Query now Actual future +32 steps

Object: actual 10.8 cm; forecast 12.4 cm

Min point movement: 5.4 / 7.3 px (actual / forecast)

OpenSingleDoor / demo_1845 / s192 / L

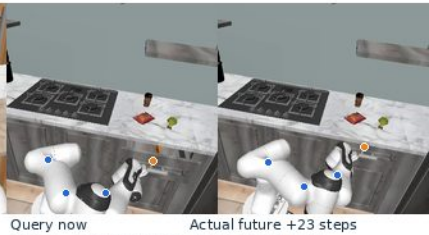


Query now Actual future +32 steps

Object: actual 16.2 cm; forecast 16.8 cm

Min point movement: 8.4 / 8.7 px (actual / forecast)

CloseDrawer / demo_2464 / s208 / R



Query now Actual future +23 steps

Object: actual 12.7 cm; forecast 12.2 cm

Min point movement: 5.6 / 6.1 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDrawer / demo_2464 / s192 / R



Query now Actual future +29 steps

Object: actual 13.6 cm; forecast 9.9 cm

Min point movement: 6.9 / 7.1 px (actual / forecast)

CloseDoubleDoor / demo_876 / s432 / L



Query now Actual future +22 steps

Object: actual 26.8 cm; forecast 28.9 cm

Min point movement: 5.4 / 5.3 px (actual / forecast)

CloseDrawer / demo_1075 / s176 / L

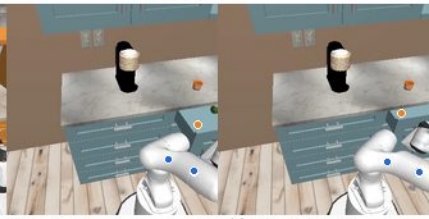


Query now Actual future +23 steps

Object: actual 12.3 cm; forecast 13.8 cm

Min point movement: 4.6 / 5.3 px (actual / forecast)

CloseDrawer / demo_329 / s192 / L



Query now Actual future +32 steps

Object: actual 12.1 cm; forecast 14.5 cm

Min point movement: 5.4 / 8.3 px (actual / forecast)

OpenDoubleDoor / demo_176 / s176 / L

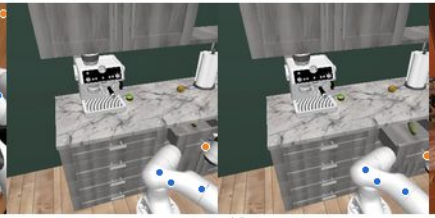


Query now Actual future +19 steps

Object: actual 17.5 cm; forecast 16.8 cm

Min point movement: 8.2 / 8.6 px (actual / forecast)

OpenDrawer / demo_1004 / s224 / L



Query now Actual future +11 steps

Object: actual 16.8 cm; forecast 26.9 cm

Min point movement: 5.4 / 6.9 px (actual / forecast)

CloseDrawer / demo_728 / s192 / L



Query now Actual future +27 steps

Object: actual 14.7 cm; forecast 14.8 cm

Min point movement: 6.0 / 5.9 px (actual / forecast)

OpenSingleDoor / demo_271 / s208 / L



Query now Actual future +4 steps

Object: actual 4.3 cm; forecast 11.5 cm

Min point movement: 7.9 / 23.4 px (actual / forecast)

CloseDrawer / demo_520 / s176 / R

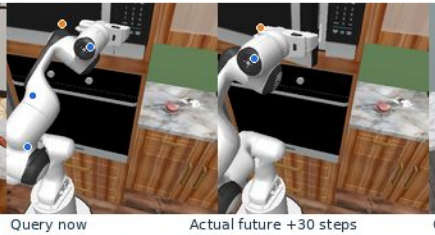


Query now Actual future +29 steps

Object: actual 11.3 cm; forecast 13.3 cm

Min point movement: 4.3 / 4.2 px (actual / forecast)

OpenSingleDoor / demo_1033 / s128 / R

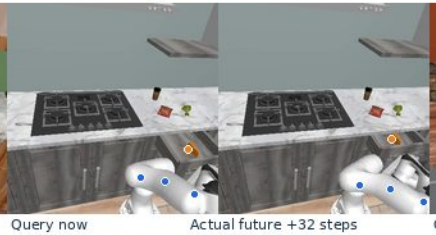


Query now Actual future +30 steps

Object: actual 10.1 cm; forecast 10.6 cm

Min point movement: 8.0 / 9.9 px (actual / forecast)

CloseDrawer / demo_2464 / s160 / L

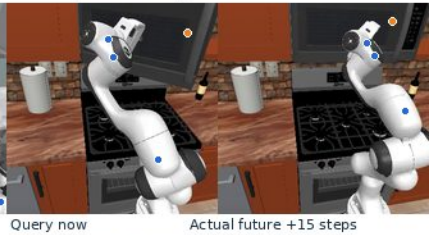


Query now Actual future +32 steps

Object: actual 11.9 cm; forecast 12.2 cm

Min point movement: 6.3 / 7.4 px (actual / forecast)

CloseSingleDoor / demo_1645 / s240 / L

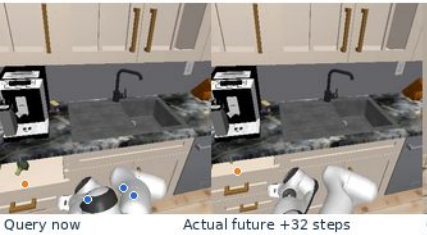


Query now Actual future +15 steps

Object: actual 30.5 cm; forecast 34.1 cm

Min point movement: 7.9 / 8.3 px (actual / forecast)

CloseDrawer / demo_319 / s160 / L

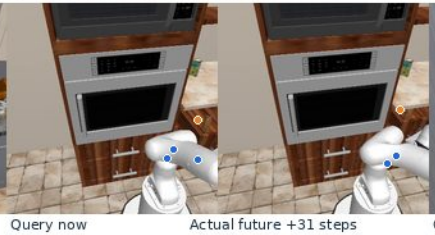


Query now Actual future +32 steps

Object: actual 10.0 cm; forecast 10.2 cm

Min point movement: 7.9 / 7.6 px (actual / forecast)

CloseDrawer / demo_1907 / s160 / L

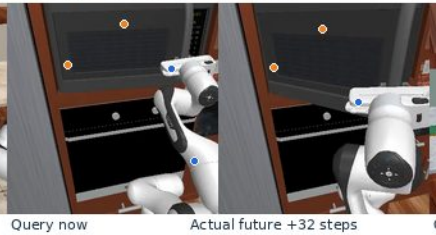


Query now Actual future +31 steps

Object: actual 13.2 cm; forecast 14.5 cm

Min point movement: 6.8 / 7.5 px (actual / forecast)

OpenSingleDoor / demo_1026 / s160 / L



Query now Actual future +32 steps

Object: actual 18.4 cm; forecast 21.3 cm

Min point movement: 3.5 / 3.9 px (actual / forecast)

CloseSingleDoor / demo_1680 / s224 / L

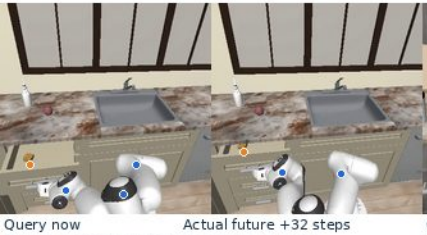


Query now Actual future +21 steps

Object: actual 34.7 cm; forecast 36.6 cm

Min point movement: 3.4 / 4.2 px (actual / forecast)

CloseDrawer / demo_1347 / s192 / L

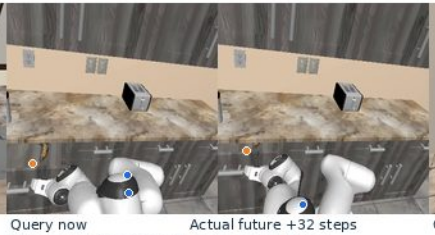


Query now Actual future +32 steps

Object: actual 11.6 cm; forecast 11.9 cm

Min point movement: 6.7 / 4.7 px (actual / forecast)

CloseDrawer / demo_2326 / s192 / L



Query now Actual future +32 steps

Object: actual 11.4 cm; forecast 12.9 cm

Min point movement: 7.8 / 8.7 px (actual / forecast)

CloseDrawer / demo_1907 / s192 / L

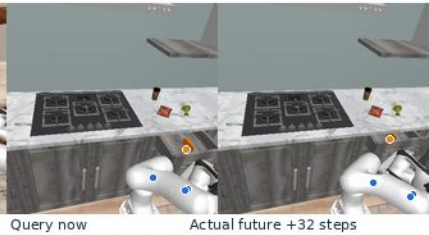


Query now Actual future +19 steps

Object: actual 15.8 cm; forecast 14.8 cm

Min point movement: 6.5 / 5.9 px (actual / forecast)

CloseDrawer / demo_2464 / s176 / L



Query now Actual future +32 steps

Object: actual 11.6 cm; forecast 13.4 cm

Min point movement: 6.1 / 8.9 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDrawer / demo_2142 / s176 / L

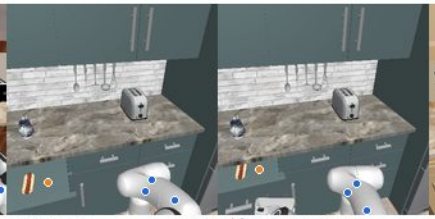


Query now Actual future +32 steps

Object: actual 11.9 cm; forecast 13.9 cm

Min point movement: 5.5 / 8.0 px (actual / forecast)

CloseDrawer / demo_145 / s128 / L



Query now Actual future +32 steps

Object: actual 10.7 cm; forecast 9.2 cm

Min point movement: 4.5 / 4.7 px (actual / forecast)

CloseDrawer / demo_2024 / s160 / L



Query now Actual future +32 steps

Object: actual 11.1 cm; forecast 12.3 cm

Min point movement: 5.2 / 8.6 px (actual / forecast)

CloseDoubleDoor / demo_341 / s128 / R



Query now Actual future +17 steps

Object: actual 14.1 cm; forecast 13.6 cm

Min point movement: 7.7 / 10.0 px (actual / forecast)

CloseDrawer / demo_728 / s176 / L

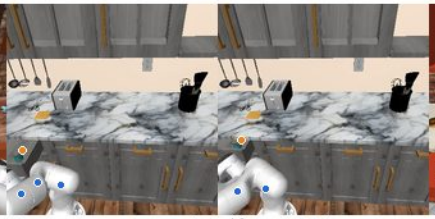


Query now Actual future +32 steps

Object: actual 11.8 cm; forecast 13.0 cm

Min point movement: 6.6 / 8.4 px (actual / forecast)

CloseDrawer / demo_1064 / s192 / R



Query now Actual future +22 steps

Object: actual 15.1 cm; forecast 13.9 cm

Min point movement: 4.9 / 4.5 px (actual / forecast)

CloseDrawer / demo_1572 / s128 / R

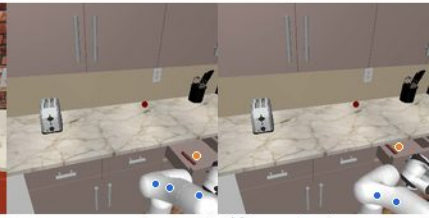


Query now Actual future +32 steps

Object: actual 9.5 cm; forecast 8.4 cm

Min point movement: 5.2 / 3.6 px (actual / forecast)

CloseDrawer / demo_21 / s144 / L



Query now Actual future +32 steps

Object: actual 11.0 cm; forecast 11.1 cm

Min point movement: 7.6 / 7.9 px (actual / forecast)

CloseDrawer / demo_2201 / s208 / R



Query now Actual future +32 steps

Object: actual 9.8 cm; forecast 11.8 cm

Min point movement: 3.4 / 3.9 px (actual / forecast)

CloseDrawer / demo_2471 / s144 / L



Query now Actual future +32 steps

Object: actual 10.7 cm; forecast 10.6 cm

Min point movement: 7.3 / 7.7 px (actual / forecast)

CloseDrawer / demo_728 / s160 / L



Query now Actual future +32 steps

Object: actual 11.3 cm; forecast 11.9 cm

Min point movement: 7.1 / 7.9 px (actual / forecast)

CloseDrawer / demo_2415 / s176 / L

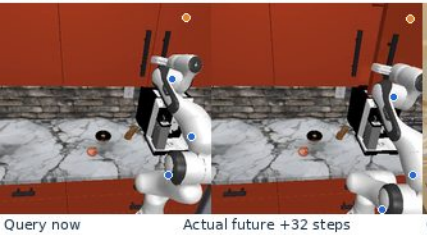


Query now Actual future +32 steps

Object: actual 10.7 cm; forecast 12.7 cm

Min point movement: 6.3 / 8.4 px (actual / forecast)

OpenDoubleDoor / demo_737 / s160 / L



Query now Actual future +32 steps

Object: actual 9.5 cm; forecast 13.1 cm

Min point movement: 7.5 / 11.7 px (actual / forecast)

CloseDrawer / demo_2024 / s144 / L

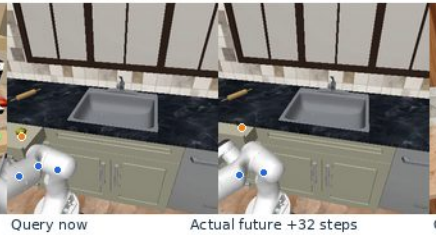


Query now Actual future +32 steps

Object: actual 10.9 cm; forecast 11.7 cm

Min point movement: 6.6 / 8.0 px (actual / forecast)

CloseDrawer / demo_649 / s176 / R

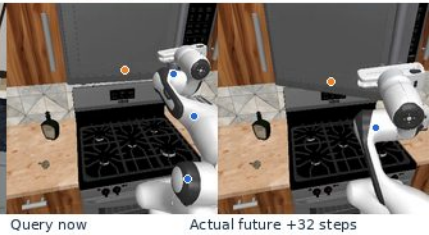


Query now Actual future +32 steps

Object: actual 11.4 cm; forecast 12.3 cm

Min point movement: 4.1 / 3.9 px (actual / forecast)

OpenSingleDoor / demo_83 / s192 / L

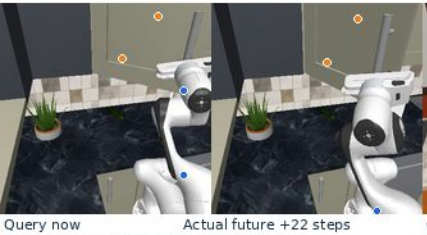


Query now Actual future +32 steps

Object: actual 15.0 cm; forecast 16.0 cm

Min point movement: 7.4 / 8.0 px (actual / forecast)

OpenSingleDoor / demo_1549 / s192 / L

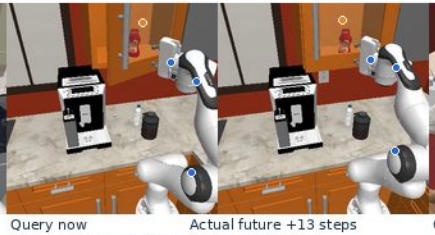


Query now Actual future +22 steps

Object: actual 10.3 cm; forecast 8.9 cm

Min point movement: 3.9 / 3.5 px (actual / forecast)

CloseSingleDoor / demo_2462 / s192 / L

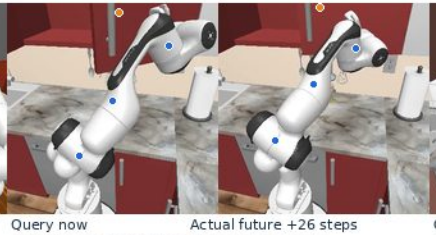


Query now Actual future +13 steps

Object: actual 10.5 cm; forecast 9.3 cm

Min point movement: 7.2 / 6.6 px (actual / forecast)

CloseDoubleDoor / demo_1683 / s192 / R

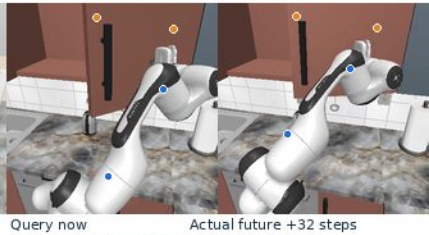


Query now Actual future +26 steps

Object: actual 26.7 cm; forecast 24.6 cm

Min point movement: 7.1 / 6.6 px (actual / forecast)

CloseDoubleDoor / demo_1757 / s176 / R



Query now Actual future +32 steps

Object: actual 16.4 cm; forecast 31.7 cm

Min point movement: 4.6 / 5.4 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDrawer / demo_41 / s176 / R

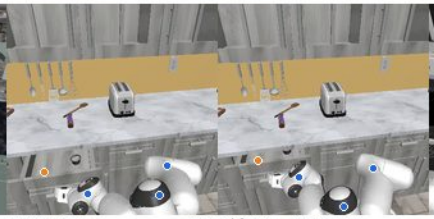


Query now Actual future +19 steps

Object: actual 9.9 cm; forecast 12.2 cm

Min point movement: 4.0 / 3.6 px (actual / forecast)

CloseDrawer / demo_1605 / s192 / L



Query now Actual future +32 steps

Object: actual 10.1 cm; forecast 12.2 cm

Min point movement: 3.5 / 4.4 px (actual / forecast)

CloseDrawer / demo_2142 / s160 / R



Query now Actual future +32 steps

Object: actual 10.4 cm; forecast 13.3 cm

Min point movement: 5.6 / 6.4 px (actual / forecast)

OpenDoubleDoor / demo_332 / s176 / L

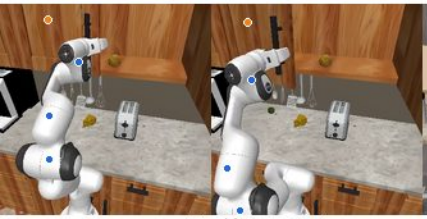


Query now Actual future +32 steps

Object: actual 9.0 cm; forecast 9.5 cm

Min point movement: 6.9 / 8.4 px (actual / forecast)

OpenDoubleDoor / demo_456 / s864 / R

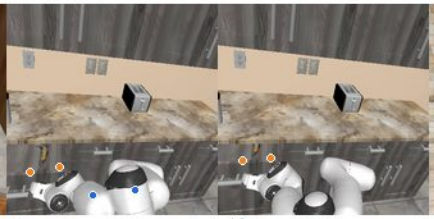


Query now Actual future +32 steps

Object: actual 8.1 cm; forecast 8.9 cm

Min point movement: 6.9 / 7.1 px (actual / forecast)

CloseDrawer / demo_2326 / s176 / L

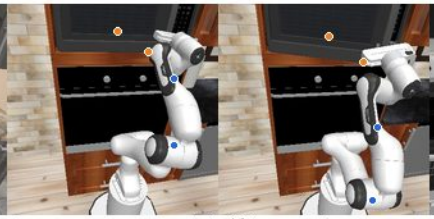


Query now Actual future +32 steps

Object: actual 7.4 cm; forecast 10.9 cm

Min point movement: 5.8 / 8.0 px (actual / forecast)

OpenSingleDoor / demo_2753 / s128 / L



Query now Actual future +30 steps

Object: actual 17.0 cm; forecast 15.8 cm

Min point movement: 3.4 / 3.1 px (actual / forecast)

CloseDrawer / demo_2142 / s160 / L

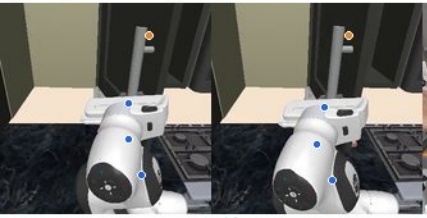


Query now Actual future +32 steps

Object: actual 10.4 cm; forecast 13.3 cm

Min point movement: 5.5 / 7.8 px (actual / forecast)

OpenSingleDoor / demo_2533 / s240 / L



Query now Actual future +3 steps

Object: actual 3.0 cm; forecast 6.9 cm

Min point movement: 6.7 / 16.2 px (actual / forecast)

CloseDrawer / demo_2465 / s128 / R



Query now Actual future +32 steps

Object: actual 10.6 cm; forecast 13.1 cm

Min point movement: 3.5 / 3.2 px (actual / forecast)

CloseDrawer / demo_2201 / s208 / L



Query now Actual future +32 steps

Object: actual 9.8 cm; forecast 11.2 cm

Min point movement: 6.7 / 7.4 px (actual / forecast)

CloseDrawer / demo_2363 / s160 / L

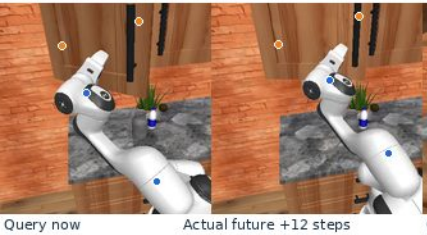


Query now Actual future +24 steps

Object: actual 12.7 cm; forecast 13.7 cm

Min point movement: 3.0 / 3.5 px (actual / forecast)

CloseDoubleDoor / demo_1569 / s368 / L

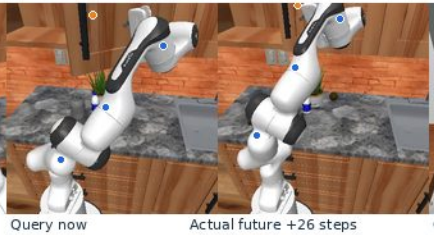


Query now Actual future +12 steps

Object: actual 29.4 cm; forecast 32.3 cm

Min point movement: 3.4 / 3.1 px (actual / forecast)

CloseDoubleDoor / demo_1569 / s192 / R



Query now Actual future +26 steps

Object: actual 25.7 cm; forecast 26.4 cm

Min point movement: 6.7 / 7.7 px (actual / forecast)

OpenSingleDoor / demo_967 / s240 / L

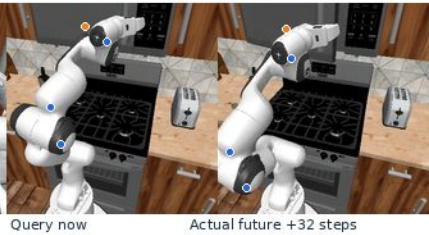


Query now Actual future +32 steps

Object: actual 16.3 cm; forecast 16.8 cm

Min point movement: 3.3 / 3.4 px (actual / forecast)

OpenSingleDoor / demo_83 / s176 / R

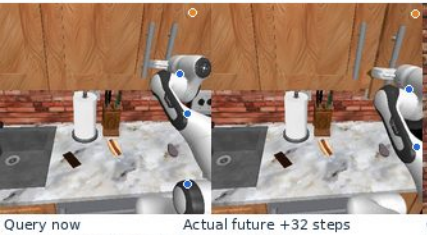


Query now Actual future +32 steps

Object: actual 8.5 cm; forecast 9.9 cm

Min point movement: 6.6 / 7.7 px (actual / forecast)

OpenDoubleDoor / demo_176 / s160 / L

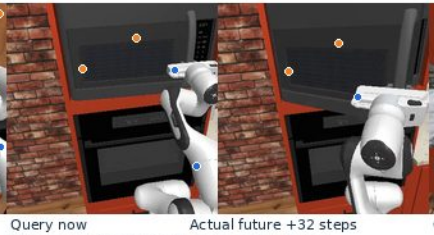


Query now Actual future +32 steps

Object: actual 7.8 cm; forecast 9.1 cm

Min point movement: 6.6 / 8.5 px (actual / forecast)

OpenSingleDoor / demo_1154 / s224 / L

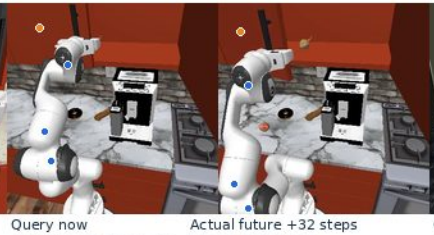


Query now Actual future +32 steps

Object: actual 18.6 cm; forecast 20.8 cm

Min point movement: 3.1 / 3.2 px (actual / forecast)

OpenDoubleDoor / demo_737 / s608 / R



Query now Actual future +32 steps

Object: actual 7.1 cm; forecast 10.0 cm

Min point movement: 6.5 / 9.2 px (actual / forecast)

OpenSingleDoor / demo_2533 / s192 / L



Query now Actual future +32 steps

Object: actual 21.5 cm; forecast 22.7 cm

Min point movement: 6.4 / 7.2 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDrawer / demo_2415 / s160 / L

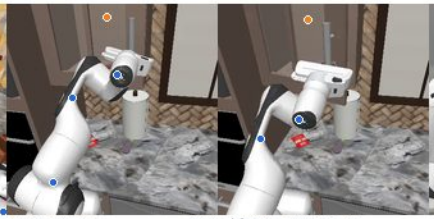


Query now Actual future +32 steps

Object: actual 8.5 cm; forecast 9.6 cm

Min point movement: 6.4 / 5.6 px (actual / forecast)

OpenSingleDoor / demo_959 / s128 / R

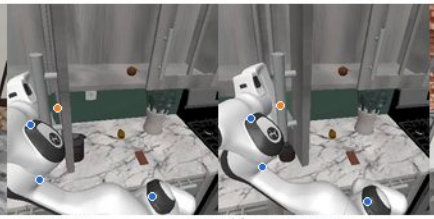


Query now Actual future +32 steps

Object: actual 10.4 cm; forecast 11.6 cm

Min point movement: 6.4 / 8.3 px (actual / forecast)

CloseDoubleDoor / demo_1032 / s336 / L

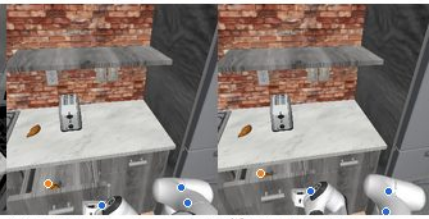


Query now Actual future +7 steps

Object: actual 32.1 cm; forecast 28.1 cm

Min point movement: 6.3 / 14.2 px (actual / forecast)

CloseDrawer / demo_2934 / s176 / L

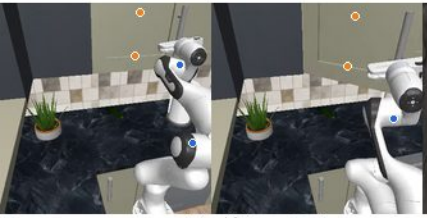


Query now Actual future +32 steps

Object: actual 8.1 cm; forecast 11.9 cm

Min point movement: 3.6 / 3.7 px (actual / forecast)

OpenSingleDoor / demo_1549 / s160 / L

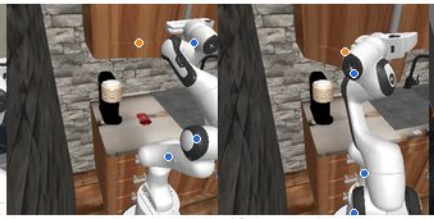


Query now Actual future +32 steps

Object: actual 16.9 cm; forecast 18.1 cm

Min point movement: 3.8 / 3.7 px (actual / forecast)

OpenSingleDoor / demo_1467 / s224 / L

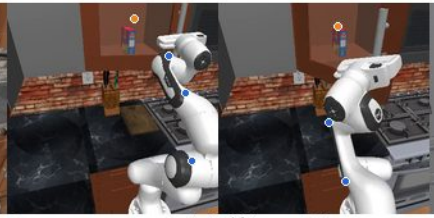


Query now Actual future +25 steps

Object: actual 20.4 cm; forecast 19.8 cm

Min point movement: 6.2 / 7.5 px (actual / forecast)

OpenSingleDoor / demo_1419 / s160 / L

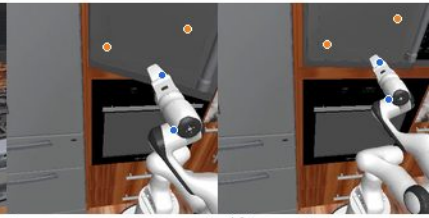


Query now Actual future +32 steps

Object: actual 18.0 cm; forecast 19.2 cm

Min point movement: 6.1 / 6.7 px (actual / forecast)

CloseSingleDoor / demo_258 / s240 / L



Query now Actual future +21 steps

Object: actual 26.1 cm; forecast 28.9 cm

Min point movement: 5.7 / 4.3 px (actual / forecast)

CloseDoubleDoor / demo_1617 / s384 / L

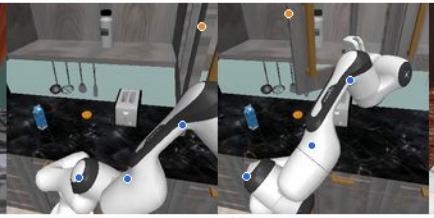


Query now Actual future +19 steps

Object: actual 22.4 cm; forecast 24.4 cm

Min point movement: 6.0 / 6.1 px (actual / forecast)

CloseDoubleDoor / demo_1729 / s144 / R

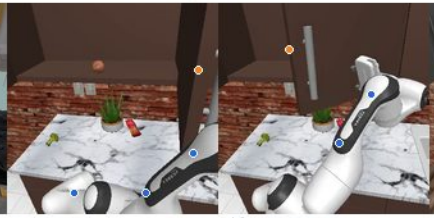


Query now Actual future +32 steps

Object: actual 36.9 cm; forecast 22.3 cm

Min point movement: 26.1 / 13.9 px (actual / forecast)

CloseDoubleDoor / demo_842 / s144 / R

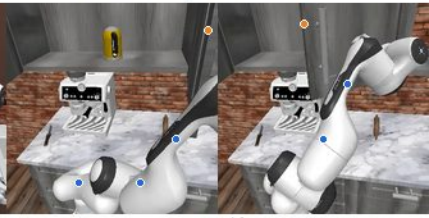


Query now Actual future +32 steps

Object: actual 39.3 cm; forecast 35.7 cm

Min point movement: 4.9 / 4.2 px (actual / forecast)

CloseDoubleDoor / demo_1995 / s144 / R

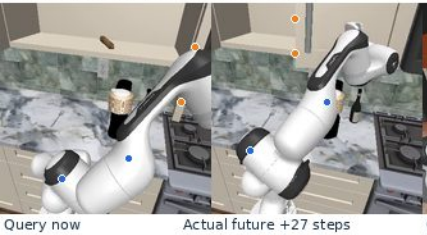


Query now Actual future +32 steps

Object: actual 36.5 cm; forecast 21.2 cm

Min point movement: 5.4 / 4.6 px (actual / forecast)

CloseDoubleDoor / demo_2121 / s160 / R

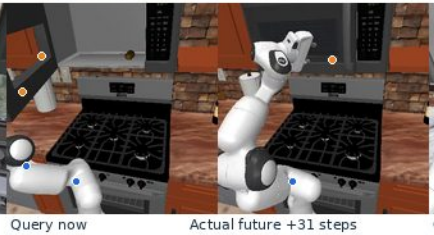


Query now Actual future +27 steps

Object: actual 58.5 cm; forecast 53.1 cm

Min point movement: 30.9 / 32.0 px (actual / forecast)

CloseSingleDoor / demo_1645 / s224 / R

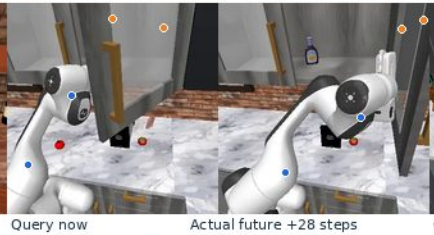


Query now Actual future +31 steps

Object: actual 50.2 cm; forecast 51.4 cm

Min point movement: 3.2 / 4.9 px (actual / forecast)

OpenDoubleDoor / demo_293 / s416 / R

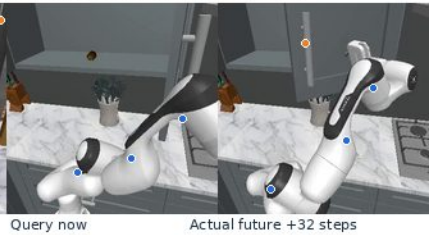


Query now Actual future +28 steps

Object: actual 27.8 cm; forecast 29.5 cm

Min point movement: 19.7 / 23.2 px (actual / forecast)

CloseDoubleDoor / demo_1480 / s144 / R

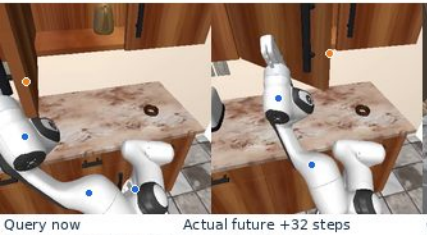


Query now Actual future +32 steps

Object: actual 31.9 cm; forecast 11.0 cm

Min point movement: 10.4 / 9.1 px (actual / forecast)

CloseDoubleDoor / demo_2731 / s336 / L

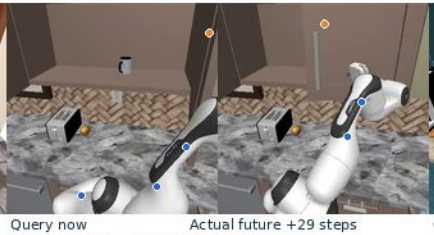


Query now Actual future +32 steps

Object: actual 30.6 cm; forecast 34.2 cm

Min point movement: 4.5 / 22.5 px (actual / forecast)

CloseDoubleDoor / demo_1717 / s144 / R

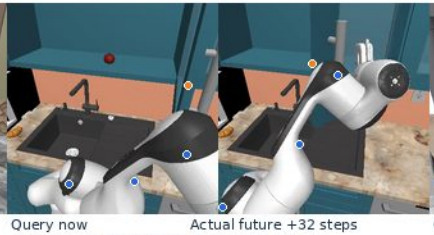


Query now Actual future +29 steps

Object: actual 48.4 cm; forecast 43.8 cm

Min point movement: 3.4 / 6.4 px (actual / forecast)

CloseDoubleDoor / demo_1112 / s144 / R

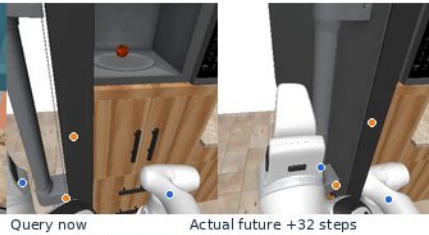


Query now Actual future +32 steps

Object: actual 23.0 cm; forecast 15.5 cm

Min point movement: 35.8 / 7.7 px (actual / forecast)

CloseSingleDoor / demo_103 / s176 / L



Query now Actual future +32 steps

Object: actual 11.7 cm; forecast 6.9 cm

Min point movement: 5.4 / 4.0 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

OpenDoubleDoor / demo_654 / s384 / R

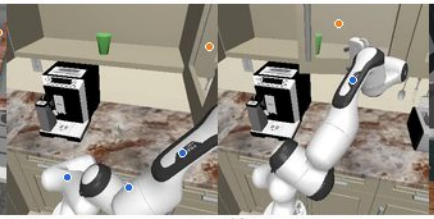


Query now Actual future +18 steps

Object: actual 33.3 cm; forecast 17.3 cm

Min point movement: 23.4 / 10.5 px (actual / forecast)

CloseDoubleDoor / demo_203 / s176 / R



Query now Actual future +28 steps

Object: actual 43.0 cm; forecast 41.1 cm

Min point movement: 4.6 / 7.4 px (actual / forecast)

CloseDoubleDoor / demo_275 / s144 / R



Query now Actual future +32 steps

Object: actual 26.6 cm; forecast 20.4 cm

Min point movement: 5.8 / 3.8 px (actual / forecast)

CloseSingleDoor / demo_610 / s192 / L

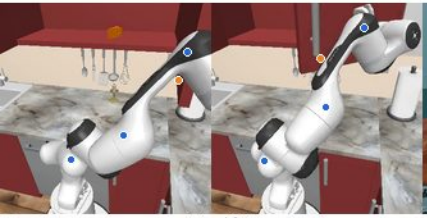


Query now Actual future +25 steps

Object: actual 55.4 cm; forecast 51.9 cm

Min point movement: 44.9 / 40.8 px (actual / forecast)

CloseDoubleDoor / demo_1683 / s160 / R



Query now Actual future +32 steps

Object: actual 30.8 cm; forecast 26.5 cm

Min point movement: 10.8 / 7.8 px (actual / forecast)

CloseSingleDoor / demo_1829 / s224 / L



Query now Actual future +32 steps

Object: actual 26.7 cm; forecast 29.5 cm

Min point movement: 15.1 / 15.4 px (actual / forecast)

CloseSingleDoor / demo_1060 / s224 / R



Query now Actual future +32 steps

Object: actual 29.3 cm; forecast 32.8 cm

Min point movement: 7.1 / 4.2 px (actual / forecast)

CloseDoubleDoor / demo_2128 / s176 / R



Query now Actual future +32 steps

Object: actual 21.5 cm; forecast 33.9 cm

Min point movement: 5.6 / 20.4 px (actual / forecast)

CloseDoubleDoor / demo_1029 / s160 / R



Query now Actual future +32 steps

Object: actual 16.2 cm; forecast 9.6 cm

Min point movement: 9.7 / 11.0 px (actual / forecast)

CloseDoubleDoor / demo_2470 / s320 / L

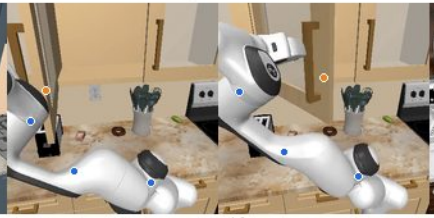


Query now Actual future +32 steps

Object: actual 19.6 cm; forecast 9.7 cm

Min point movement: 9.5 / 8.1 px (actual / forecast)

CloseDoubleDoor / demo_2827 / s320 / L

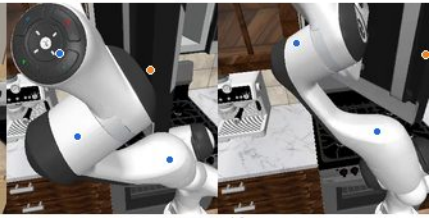


Query now Actual future +32 steps

Object: actual 21.3 cm; forecast 16.1 cm

Min point movement: 6.1 / 7.6 px (actual / forecast)

CloseSingleDoor / demo_1060 / s208 / L

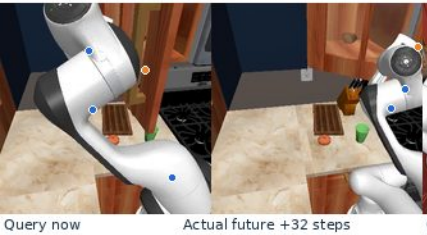


Query now Actual future +32 steps

Object: actual 16.0 cm; forecast 16.9 cm

Min point movement: 17.4 / 10.7 px (actual / forecast)

CloseSingleDoor / demo_1647 / s208 / L

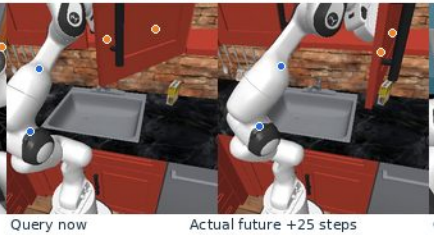


Query now Actual future +32 steps

Object: actual 33.0 cm; forecast 28.0 cm

Min point movement: 25.8 / 23.9 px (actual / forecast)

OpenDoubleDoor / demo_332 / s400 / R

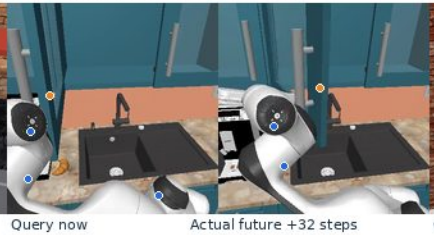


Query now Actual future +25 steps

Object: actual 20.8 cm; forecast 23.6 cm

Min point movement: 11.8 / 12.9 px (actual / forecast)

CloseDoubleDoor / demo_1112 / s320 / L

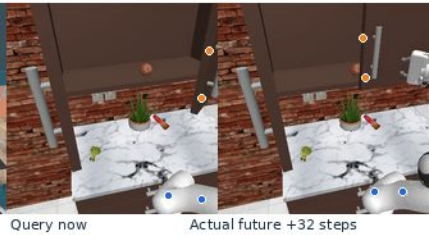


Query now Actual future +32 steps

Object: actual 15.3 cm; forecast 7.3 cm

Min point movement: 13.8 / 7.2 px (actual / forecast)

CloseDoubleDoor / demo_842 / s176 / L

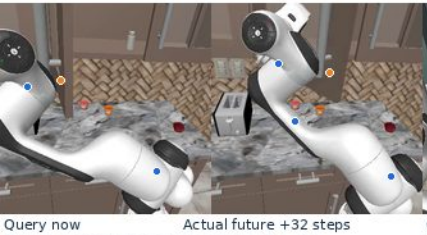


Query now Actual future +32 steps

Object: actual 29.2 cm; forecast 30.2 cm

Min point movement: 3.4 / 3.6 px (actual / forecast)

CloseDoubleDoor / demo_2238 / s336 / L

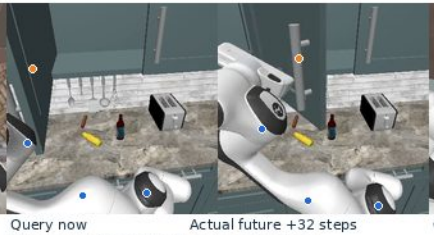


Query now Actual future +32 steps

Object: actual 17.3 cm; forecast 18.1 cm

Min point movement: 14.2 / 12.4 px (actual / forecast)

CloseDoubleDoor / demo_254 / s304 / L

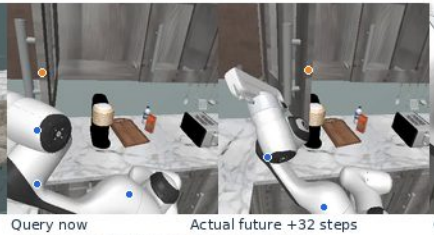


Query now Actual future +32 steps

Object: actual 16.8 cm; forecast 6.7 cm

Min point movement: 9.6 / 6.6 px (actual / forecast)

CloseDoubleDoor / demo_928 / s304 / L



Query now Actual future +32 steps

Object: actual 15.1 cm; forecast 5.2 cm

Min point movement: 12.6 / 12.1 px (actual / forecast)

CloseSingleDoor / demo_379 / s192 / L



Query now Actual future +23 steps

Object: actual 27.3 cm; forecast 26.8 cm

Min point movement: 32.4 / 37.5 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDoubleDoor / demo_1995 / s320 / L

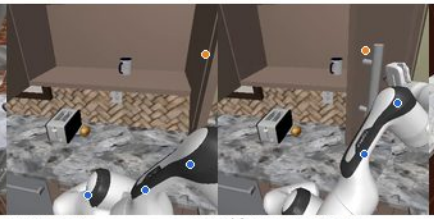


Query now Actual future +32 steps

Object: actual 14.6 cm; forecast 6.5 cm

Min point movement: 11.9 / 5.7 px (actual / forecast)

CloseDoubleDoor / demo_1717 / s128 / R

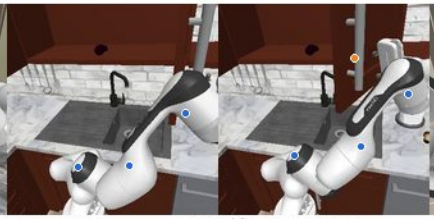


Query now Actual future +32 steps

Object: actual 14.0 cm; forecast 5.1 cm

Min point movement: 11.5 / 9.7 px (actual / forecast)

CloseDoubleDoor / demo_2316 / s160 / R



Query now Actual future +32 steps

Object: actual 13.4 cm; forecast 22.0 cm

Min point movement: 7.7 / 6.3 px (actual / forecast)

CloseDoubleDoor / demo_1175 / s320 / L

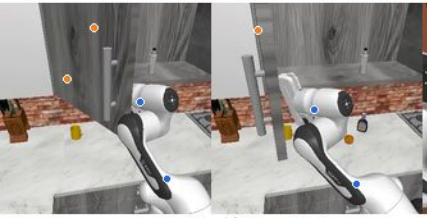


Query now Actual future +32 steps

Object: actual 14.2 cm; forecast 5.1 cm

Min point movement: 9.0 / 10.9 px (actual / forecast)

OpenDoubleDoor / demo_654 / s752 / L

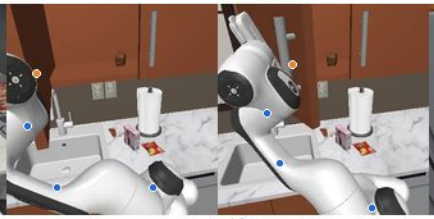


Query now Actual future +16 steps

Object: actual 14.8 cm; forecast 28.1 cm

Min point movement: 7.5 / 17.5 px (actual / forecast)

CloseDoubleDoor / demo_1020 / s320 / L



Query now Actual future +32 steps

Object: actual 14.3 cm; forecast 5.5 cm

Min point movement: 13.2 / 11.2 px (actual / forecast)

CloseSingleDoor / demo_1182 / s208 / L



Query now Actual future +32 steps

Object: actual 18.4 cm; forecast 29.8 cm

Min point movement: 27.8 / 31.4 px (actual / forecast)

CloseSingleDoor / demo_972 / s192 / L



Query now Actual future +16 steps

Object: actual 27.7 cm; forecast 36.5 cm

Min point movement: 24.7 / 44.2 px (actual / forecast)

CloseDoubleDoor / demo_1399 / s320 / L



Query now Actual future +32 steps

Object: actual 11.0 cm; forecast 3.0 cm

Min point movement: 7.7 / 7.6 px (actual / forecast)

CloseSingleDoor / demo_1829 / s208 / L

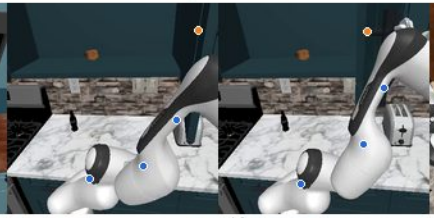


Query now Actual future +32 steps

Object: actual 12.4 cm; forecast 30.9 cm

Min point movement: 3.5 / 28.6 px (actual / forecast)

CloseDoubleDoor / demo_2593 / s144 / R



Query now Actual future +32 steps

Object: actual 10.1 cm; forecast 25.5 cm

Min point movement: 5.1 / 6.6 px (actual / forecast)

CloseDoubleDoor / demo_1906 / s336 / L



Query now Actual future +32 steps

Object: actual 12.0 cm; forecast 31.4 cm

Min point movement: 12.1 / 11.8 px (actual / forecast)

CloseSingleDoor / demo_198 / s224 / L

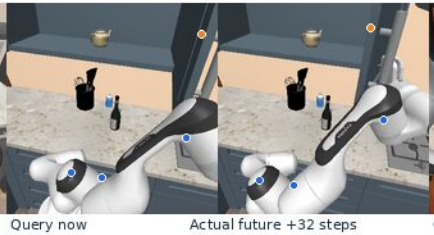


Query now Actual future +25 steps

Object: actual 48.7 cm; forecast 51.8 cm

Min point movement: 25.7 / 25.5 px (actual / forecast)

CloseDoubleDoor / demo_2470 / s144 / R

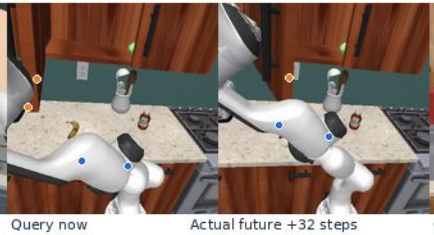


Query now Actual future +32 steps

Object: actual 9.4 cm; forecast 13.4 cm

Min point movement: 12.3 / 8.8 px (actual / forecast)

CloseDoubleDoor / demo_1617 / s320 / L

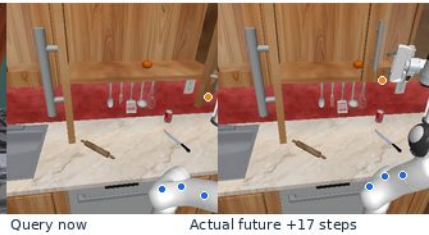


Query now Actual future +32 steps

Object: actual 12.5 cm; forecast 4.5 cm

Min point movement: 18.8 / 8.2 px (actual / forecast)

CloseDoubleDoor / demo_99 / s176 / L



Query now Actual future +17 steps

Object: actual 25.1 cm; forecast 25.3 cm

Min point movement: 6.3 / 4.9 px (actual / forecast)

CloseSingleDoor / demo_1892 / s176 / R

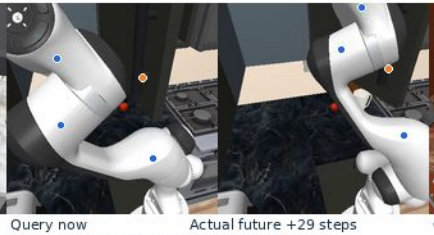


Query now Actual future +32 steps

Object: actual 14.8 cm; forecast 20.1 cm

Min point movement: 4.9 / 6.1 px (actual / forecast)

CloseSingleDoor / demo_466 / s208 / L

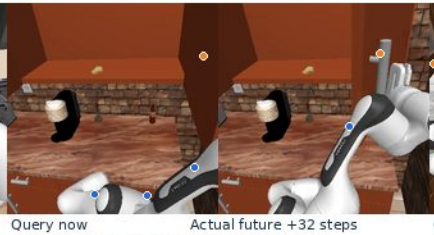


Query now Actual future +29 steps

Object: actual 15.4 cm; forecast 25.2 cm

Min point movement: 21.6 / 37.2 px (actual / forecast)

CloseDoubleDoor / demo_1930 / s128 / R



Query now Actual future +32 steps

Object: actual 10.7 cm; forecast 12.7 cm

Min point movement: 4.0 / 5.0 px (actual / forecast)

CloseDoubleDoor / demo_126 / s352 / R



Query now Actual future +26 steps

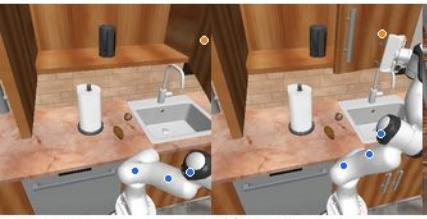
Object: actual 20.1 cm; forecast 21.1 cm

Min point movement: 8.4 / 5.3 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseDoubleDoor / demo_1499 / s176 / L



Query now Actual future +32 steps

Object: actual 18.4 cm; forecast 18.7 cm

Min point movement: 3.6 / 3.2 px (actual / forecast)

CloseDoubleDoor / demo_1995 / s160 / L

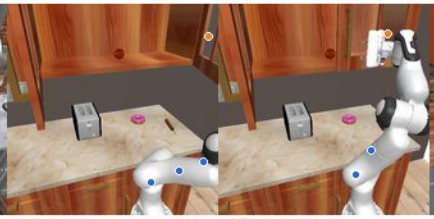


Query now Actual future +28 steps

Object: actual 18.4 cm; forecast 15.7 cm

Min point movement: 6.7 / 3.9 px (actual / forecast)

CloseDoubleDoor / demo_1606 / s160 / L



Query now Actual future +30 steps

Object: actual 18.6 cm; forecast 18.0 cm

Min point movement: 6.4 / 3.2 px (actual / forecast)

CloseSingleDoor / demo_1645 / s192 / L

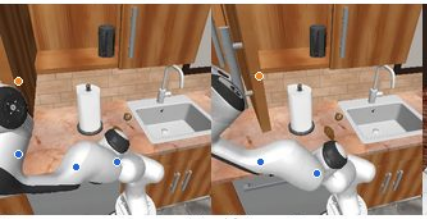


Query now Actual future +32 steps

Object: actual 9.4 cm; forecast 15.4 cm

Min point movement: 3.9 / 10.8 px (actual / forecast)

CloseDoubleDoor / demo_1499 / s320 / L

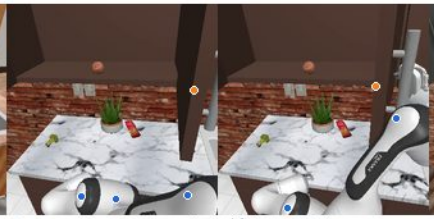


Query now Actual future +32 steps

Object: actual 8.3 cm; forecast 5.2 cm

Min point movement: 9.6 / 8.2 px (actual / forecast)

CloseDoubleDoor / demo_842 / s128 / R

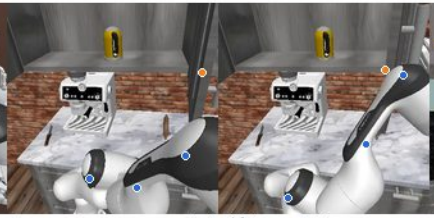


Query now Actual future +32 steps

Object: actual 6.2 cm; forecast 5.5 cm

Min point movement: 4.2 / 8.4 px (actual / forecast)

CloseDoubleDoor / demo_1995 / s128 / R

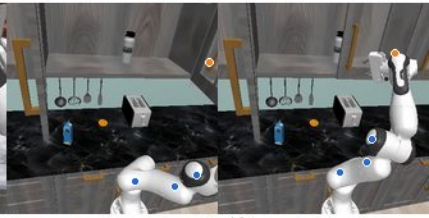


Query now Actual future +32 steps

Object: actual 6.1 cm; forecast 2.4 cm

Min point movement: 14.0 / 6.8 px (actual / forecast)

CloseDoubleDoor / demo_1729 / s160 / L



Query now Actual future +32 steps

Object: actual 17.0 cm; forecast 17.2 cm

Min point movement: 6.5 / 3.9 px (actual / forecast)

CloseDoubleDoor / demo_1029 / s192 / L

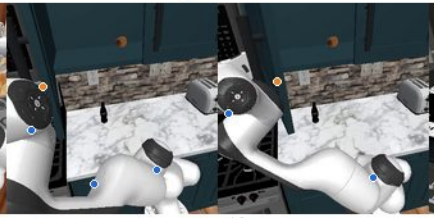


Query now Actual future +28 steps

Object: actual 14.0 cm; forecast 15.9 cm

Min point movement: 5.8 / 3.9 px (actual / forecast)

CloseDoubleDoor / demo_2593 / s320 / L

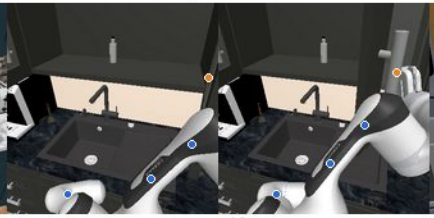


Query now Actual future +32 steps

Object: actual 5.5 cm; forecast 4.2 cm

Min point movement: 5.9 / 10.8 px (actual / forecast)

CloseDoubleDoor / demo_1584 / s160 / R



Query now Actual future +22 steps

Object: actual 13.2 cm; forecast 25.5 cm

Min point movement: 3.0 / 4.3 px (actual / forecast)

CloseDoubleDoor / demo_2220 / s176 / L

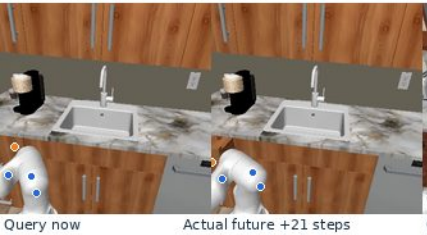


Query now Actual future +14 steps

Object: actual 19.1 cm; forecast 18.4 cm

Min point movement: 6.1 / 4.1 px (actual / forecast)

OpenDrawer / demo_2006 / s208 / R



Query now Actual future +21 steps

Object: actual 26.8 cm; forecast 22.9 cm

Min point movement: 4.7 / 5.9 px (actual / forecast)

OpenDrawer / demo_918 / s160 / R

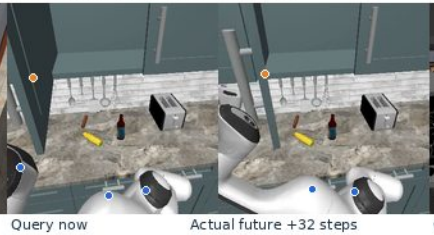


Query now Actual future +16 steps

Object: actual 26.3 cm; forecast 29.5 cm

Min point movement: 3.6 / 4.8 px (actual / forecast)

CloseDoubleDoor / demo_254 / s288 / L



Query now Actual future +32 steps

Object: actual 5.5 cm; forecast 2.9 cm

Min point movement: 6.1 / 6.7 px (actual / forecast)

CloseDrawer / demo_1438 / s144 / L

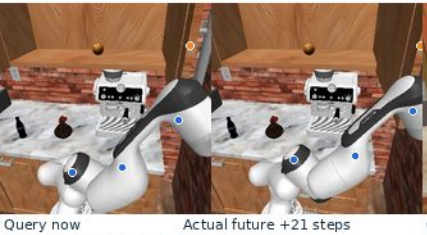


Query now Actual future +32 steps

Object: actual 14.6 cm; forecast 8.7 cm

Min point movement: 6.5 / 7.6 px (actual / forecast)

CloseDoubleDoor / demo_2128 / s144 / R

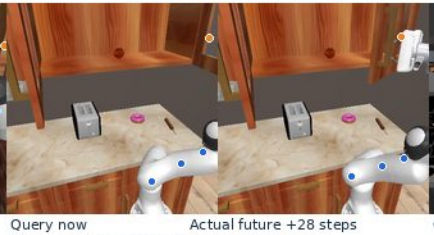


Query now Actual future +21 steps

Object: actual 6.4 cm; forecast 3.7 cm

Min point movement: 11.9 / 8.9 px (actual / forecast)

CloseDoubleDoor / demo_1606 / s144 / L



Query now Actual future +28 steps

Object: actual 11.6 cm; forecast 11.1 cm

Min point movement: 7.9 / 5.4 px (actual / forecast)

CloseDrawer / demo_1438 / s128 / L

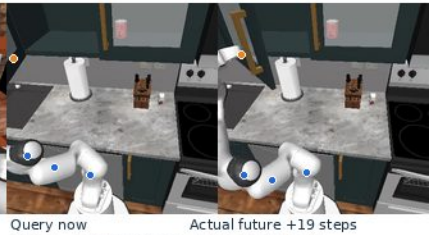


Query now Actual future +26 steps

Object: actual 18.4 cm; forecast 9.7 cm

Min point movement: 9.4 / 7.0 px (actual / forecast)

CloseDoubleDoor / demo_275 / s336 / R



Query now Actual future +19 steps

Object: actual 16.3 cm; forecast 16.6 cm

Min point movement: 3.9 / 3.8 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.

K=4 · all four points visibly move · ours succeeds / baseline fails · 717 views · all 1,200 episodes audited

CloseSingleDoor / demo_1647 / s192 / R

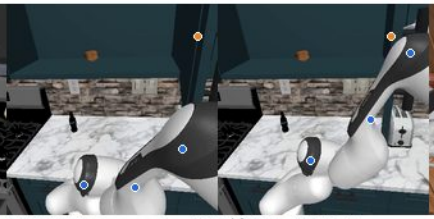


Query now Actual future +32 steps

Object: actual 7.4 cm; forecast 9.6 cm

Min point movement: 4.9 / 3.8 px (actual / forecast)

CloseDoubleDoor / demo_2593 / s128 / R

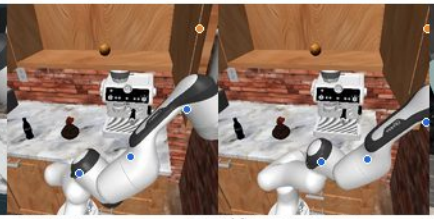


Query now Actual future +32 steps

Object: actual 3.8 cm; forecast 3.6 cm

Min point movement: 10.5 / 9.8 px (actual / forecast)

CloseDoubleDoor / demo_2128 / s160 / R

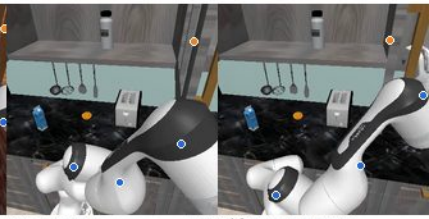


Query now Actual future +24 steps

Object: actual 4.3 cm; forecast 6.3 cm

Min point movement: 9.9 / 8.6 px (actual / forecast)

CloseDoubleDoor / demo_1729 / s128 / R



Query now Actual future +31 steps

Object: actual 3.8 cm; forecast 2.3 cm

Min point movement: 9.7 / 6.5 px (actual / forecast)

CloseDrawer / demo_995 / s192 / L



Query now Actual future +32 steps

Object: actual 14.3 cm; forecast 12.9 cm

Min point movement: 6.2 / 7.4 px (actual / forecast)

OpenSingleDoor / demo_49 / s208 / R



Query now Actual future +3 steps

Object: actual 34.3 cm; forecast 36.0 cm

Min point movement: 4.2 / 4.2 px (actual / forecast)

CloseDrawer / demo_1075 / s144 / L

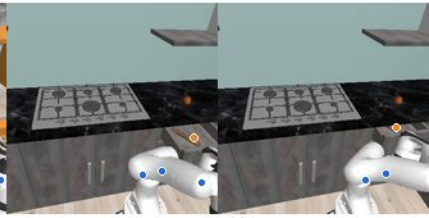


Query now Actual future +32 steps

Object: actual 15.3 cm; forecast 15.6 cm

Min point movement: 5.6 / 9.7 px (actual / forecast)

CloseDrawer / demo_1116 / s176 / L

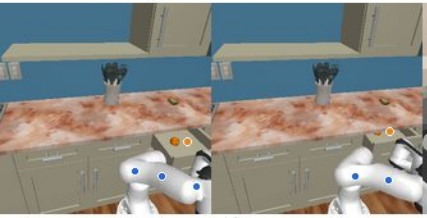


Query now Actual future +32 steps

Object: actual 13.6 cm; forecast 15.4 cm

Min point movement: 7.2 / 8.4 px (actual / forecast)

CloseDrawer / demo_183 / s144 / L



Query now Actual future +32 steps

Object: actual 13.8 cm; forecast 13.4 cm

Min point movement: 6.6 / 8.1 px (actual / forecast)

CloseSingleDoor / demo_1881 / s192 / L

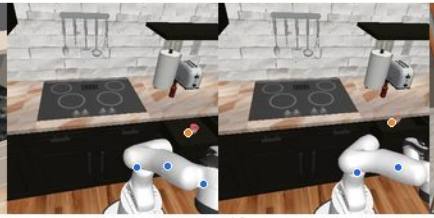


Query now Actual future +12 steps

Object: actual 18.4 cm; forecast 22.5 cm

Min point movement: 8.1 / 24.0 px (actual / forecast)

CloseDrawer / demo_2368 / s128 / L



Query now Actual future +32 steps

Object: actual 12.7 cm; forecast 11.9 cm

Min point movement: 5.6 / 7.2 px (actual / forecast)

CloseDrawer / demo_995 / s208 / L

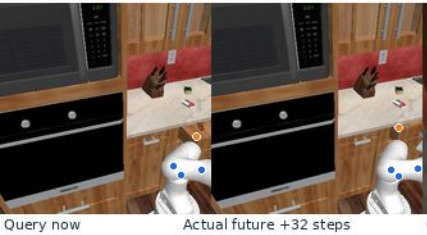


Query now Actual future +19 steps

Object: actual 14.8 cm; forecast 9.3 cm

Min point movement: 4.4 / 5.1 px (actual / forecast)

CloseDrawer / demo_1902 / s160 / L



Query now Actual future +32 steps

Object: actual 11.2 cm; forecast 13.8 cm

Min point movement: 3.5 / 5.2 px (actual / forecast)

CloseDoubleDoor / demo_2963 / s160 / L



Query now Actual future +30 steps

Object: actual 6.2 cm; forecast 7.1 cm

Min point movement: 4.0 / 3.1 px (actual / forecast)

CloseDrawer / demo_995 / s160 / L

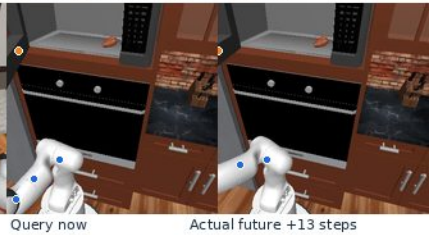


Query now Actual future +32 steps

Object: actual 10.2 cm; forecast 9.5 cm

Min point movement: 6.9 / 6.5 px (actual / forecast)

OpenSingleDoor / demo_1026 / s208 / R

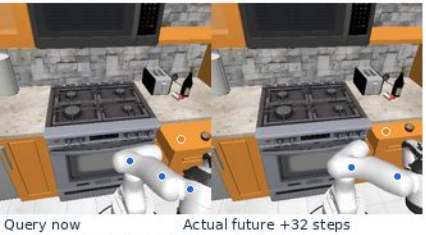


Query now Actual future +13 steps

Object: actual 5.5 cm; forecast 6.5 cm

Min point movement: 3.1 / 3.5 px (actual / forecast)

CloseDrawer / demo_1075 / s128 / L



Query now Actual future +32 steps

Object: actual 10.2 cm; forecast 11.2 cm

Min point movement: 6.6 / 7.2 px (actual / forecast)

Orange object · blue robot. Future frame shows a peak of visible object motion. Complete playback and per-point metrics are on the website.